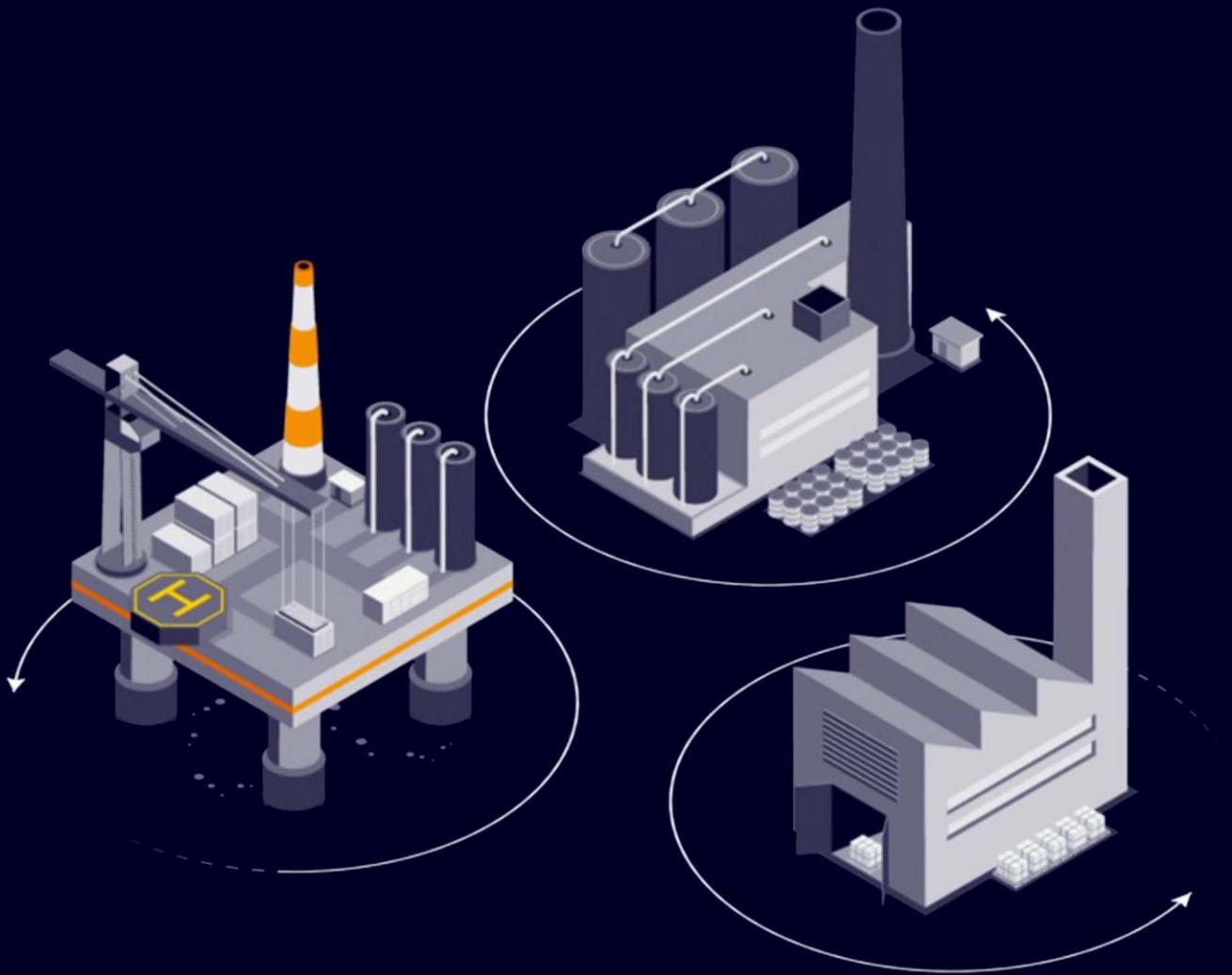


Tutorial – NXMCD

Building A Basic Digital Twin



SIEMENS

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Overview Of Document

This document aims to provide a basic insight into building a basic digital twin. The project consists of a small conveyor belt transporting a cardboard box. When the simulation is started, the cardboard box will drop onto the conveyor belt.

Starting the conveyor belt will move the cardboard box along the conveyor belt until it reaches the proximity switch. Once the proximity switch detects the cardboard box the conveyor belt will stop.

Required Software Packages

This demonstration requires the following software packages to be installed.

1. TIA Portal V16
 - a. Including licence STEP 7 Professional V16
 - b. A trial licence can be activated for 21 days.
 - c. Download of this software available from <https://support.industry.siemens.com/cs/document/109772803/simatic-step-7-incl-safety-and-wincc-v16-trial-download?dti=0&lc=en-WW>
2. PLCSim Advanced V5.0
 - a. Including Licence PLCSim Advanced V5
 - b. A trial licence can be activated for 21 days.
 - c. Download of this software available from <https://support.industry.siemens.com/cs/document/109809300/simatic-s7-plcsim-advanced-v5-0-update-1-trial-download?dti=0&lc=en-WW>
3. NX-MCD
 - a. Including relevant licence

Virtual Machine

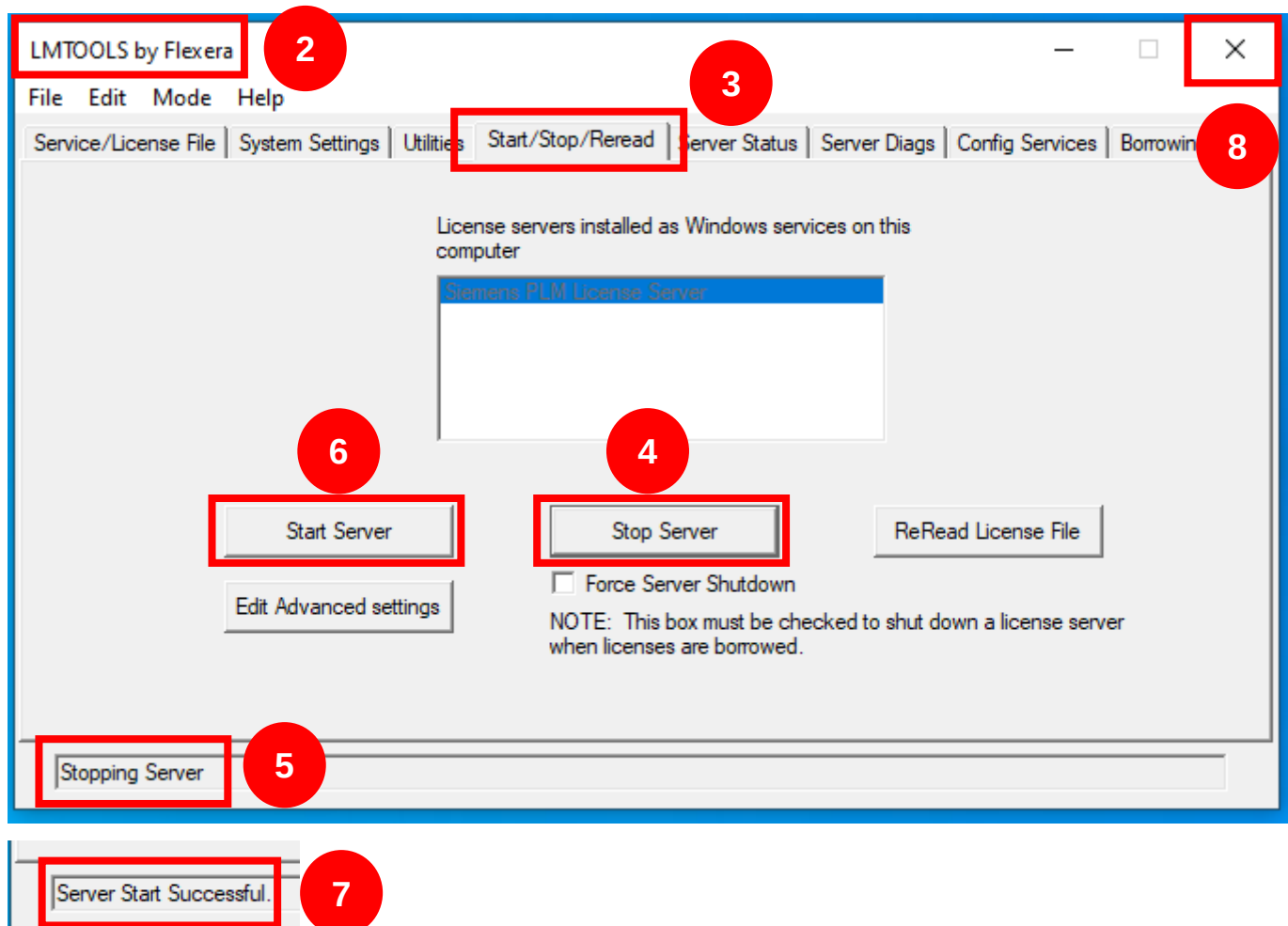
A virtual machine for training purposes is available upon request.

Refreshing The Licence Server

1. Open **LMTOOLS** by clicking on the icon in the Windows taskbar as shown below.



2. A new window will appear called **LMTOOLS BY FLEXERA** shown below.
3. Click on the **START/STOP** Tab.
4. Click on the **STOP SERVER** button.
5. After a period, the status bar at the bottom of the window will update to **STOPPING SERVER**.
6. Click the **START SERVER** button.
7. After a period, the status bar at the bottom of the window will update to **SERVER START SUCCESSFUL**.
8. **LMTOOLS** can now be closed by clicking the **X** in the top right corner of the window.

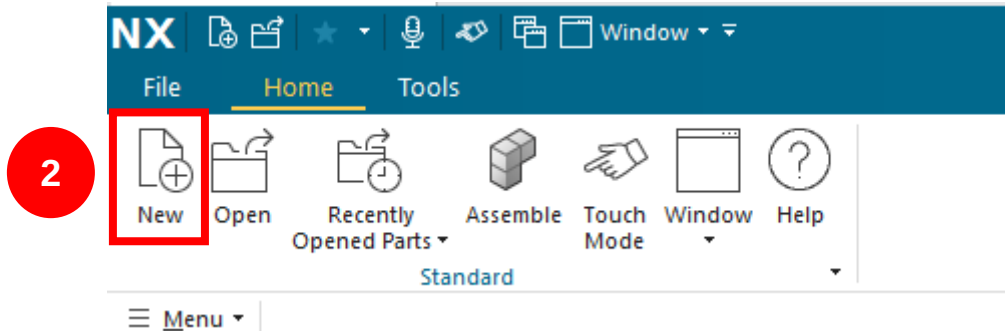


Opening Siemens NX and Starting an Assembly

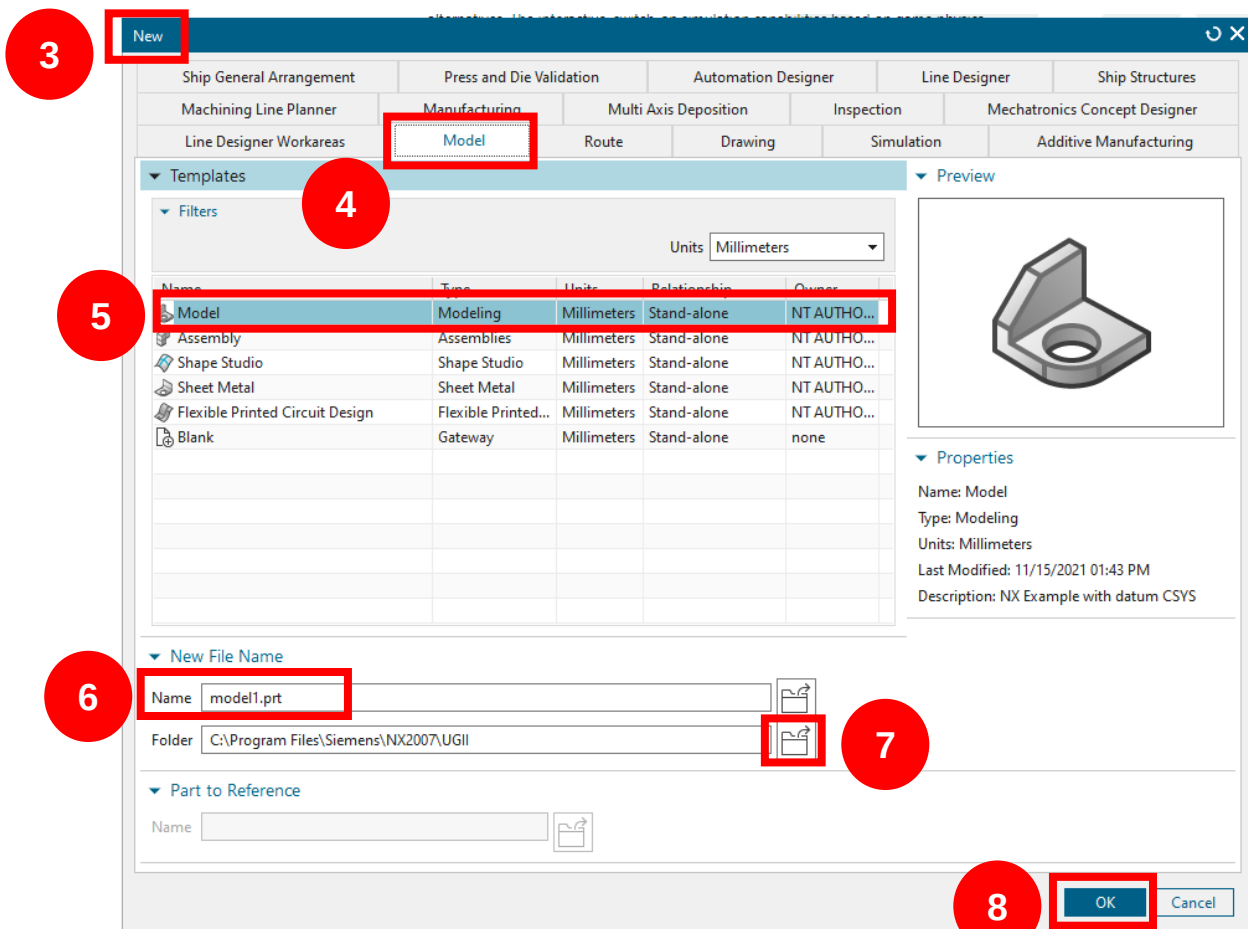
1. To open **SIEMENS NX**, click on the icon in the Windows taskbar as shown below.



2. Once **SIEMENS NX** has opened, click on the **NEW** button in the top left corner.

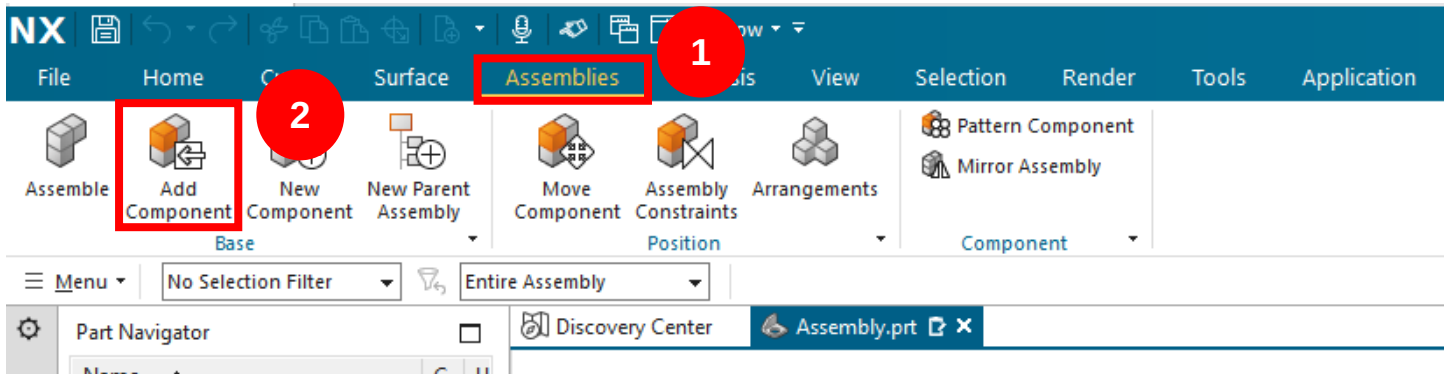


3. A new window will appear called **NEW**, shown below.
4. Click on the **MODEL** tab.
5. Click on **MODEL** in the Templates section.
6. Rename the file to **ASSEMBLY.PRT**.
7. Click the **BROWSE** button and navigate to the folder where the file will be saved.
8. Click **OK**.

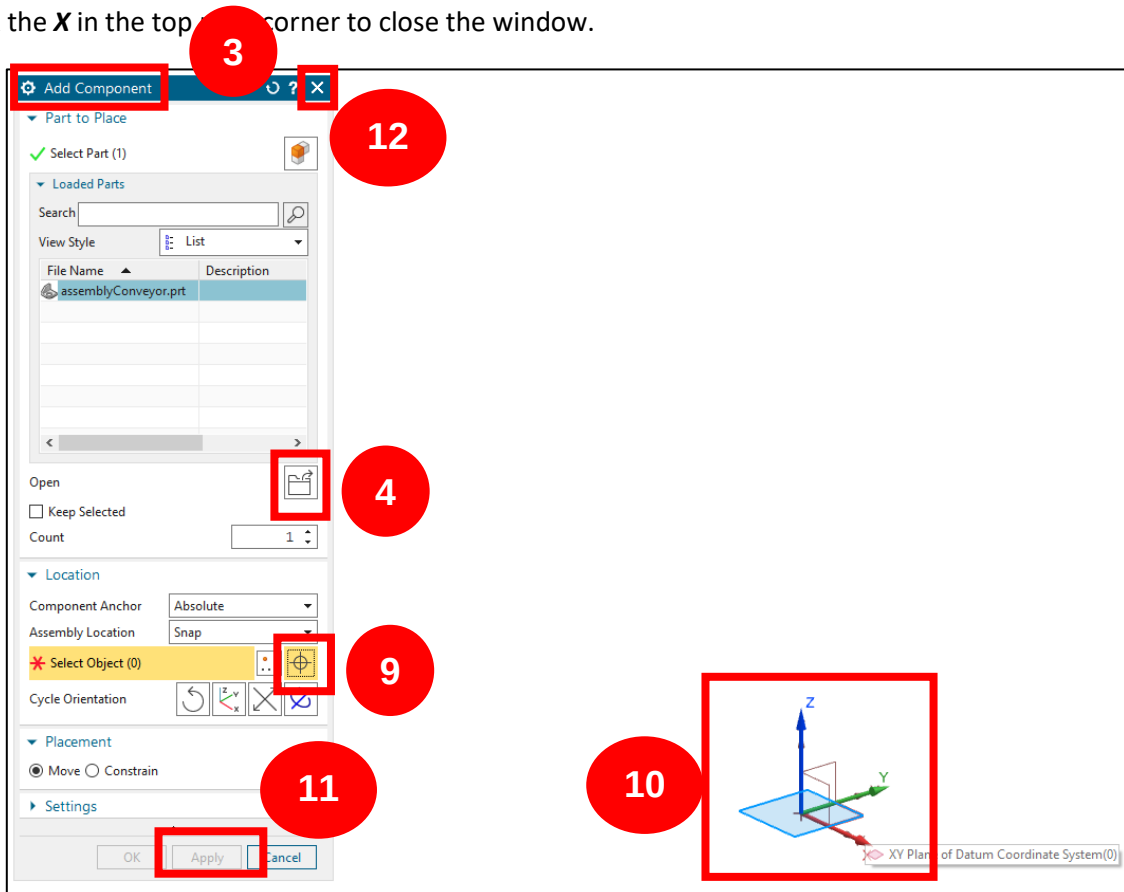


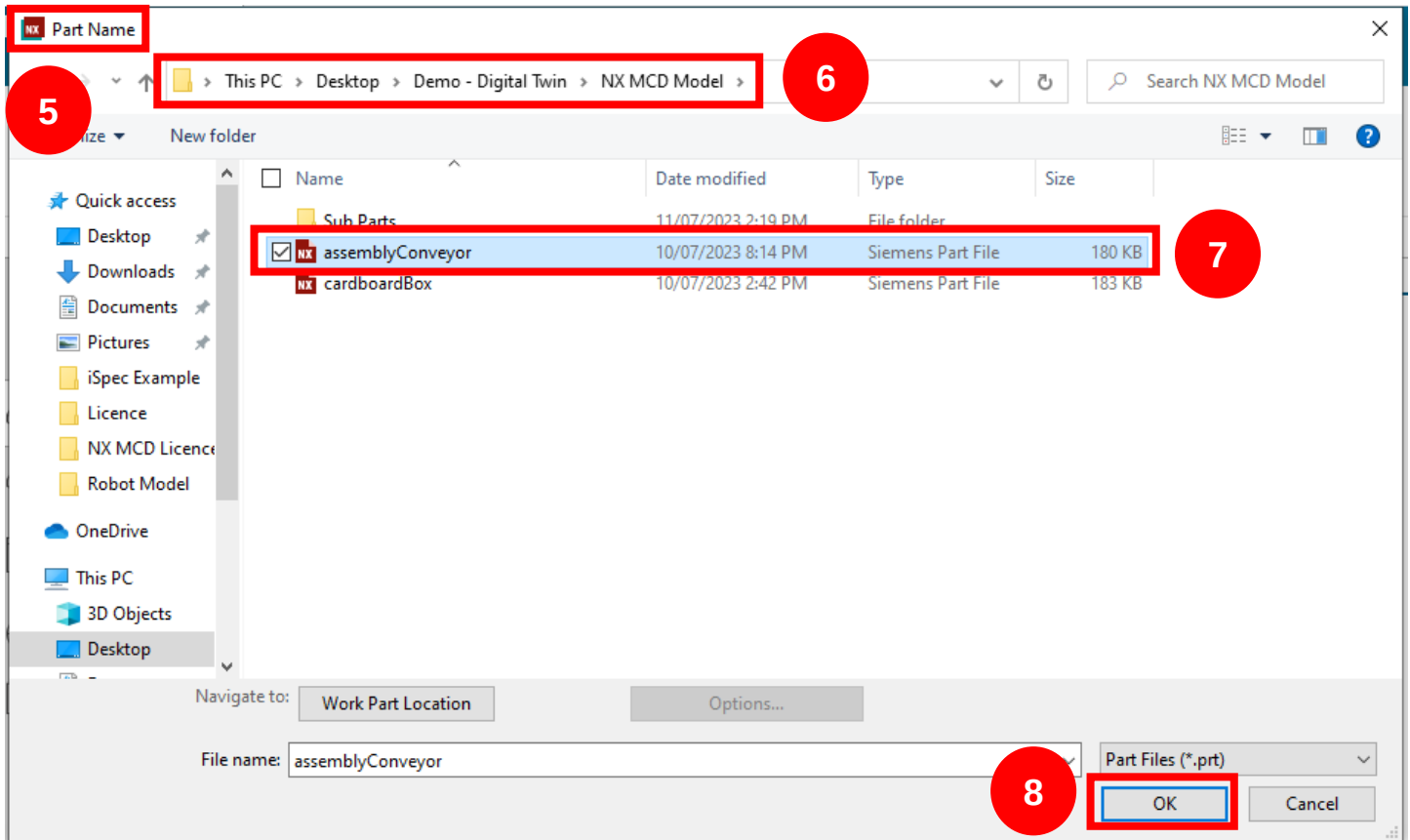
Inserting Conveyor Model within the Siemens NX Environment

1. Click on the **ASSEMBLIES** tab.
2. Click on the **ADD COMPONENT** button.



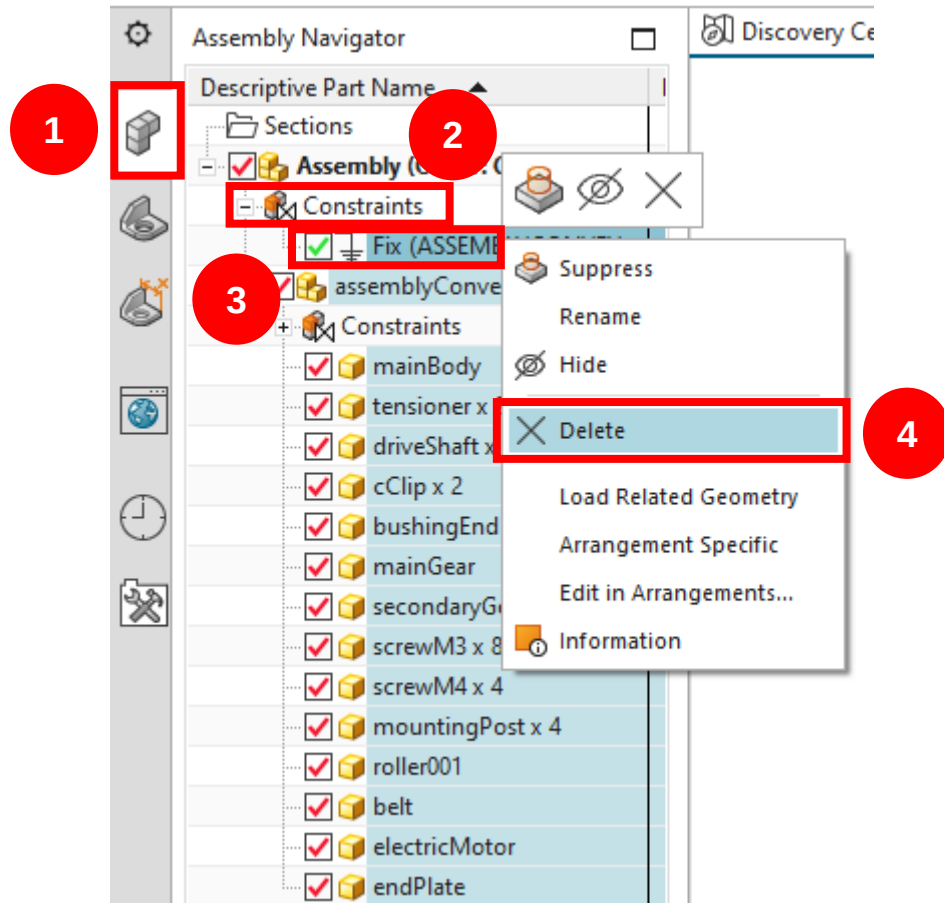
3. A new window will appear called **ADD COMPONENT**, shown below.
4. Click the **OPEN** button.
5. A new window will appear called **PART NAME**, shown below.
6. Browse to the **FOLDER** where the project files are stored, (i.e. **C:\Desktop\Demo - Digital Twin\NX MCD Model**).
7. Click on the file called **ASSEMBLYCONVEYOR.PRT**.
8. Click **OK**.
9. Click the **TARGET** icon in the **LOCATION** tab.
10. In the **MODEL** space, select the **XY DATUM COORDINATE**.
11. Click **APPLY**.
12. Click the **X** in the top corner to close the window.





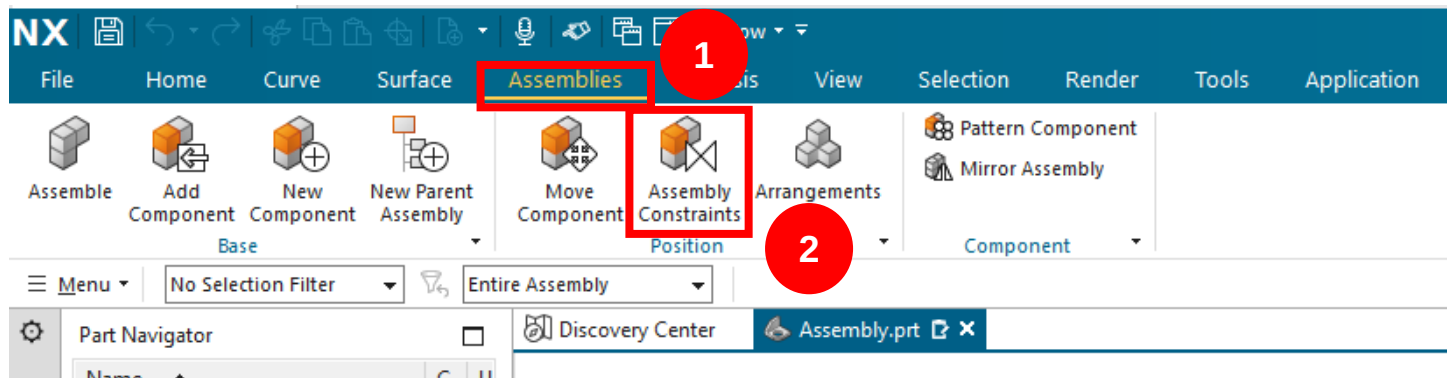
Delete the fixed constraint on the Conveyor Model

1. Click on the **ASSEMBLY NAVIGATOR** icon.
2. Expand the **CONSTRAINTS** tab under the **ASSEMBLY (ORDER: CHRONOLOGICAL)**.
3. Right Click on the **FIX** constraint.
4. Click on the **DELETE** option.

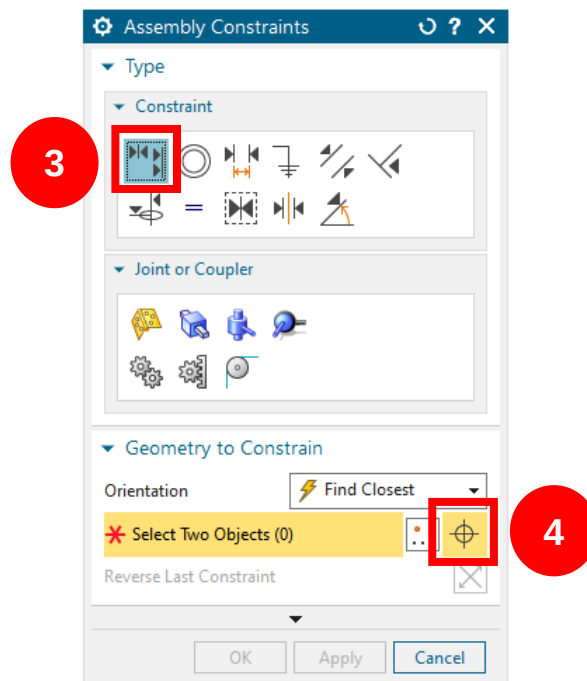


Constraining the Conveyor Model within the Siemens NX Environment

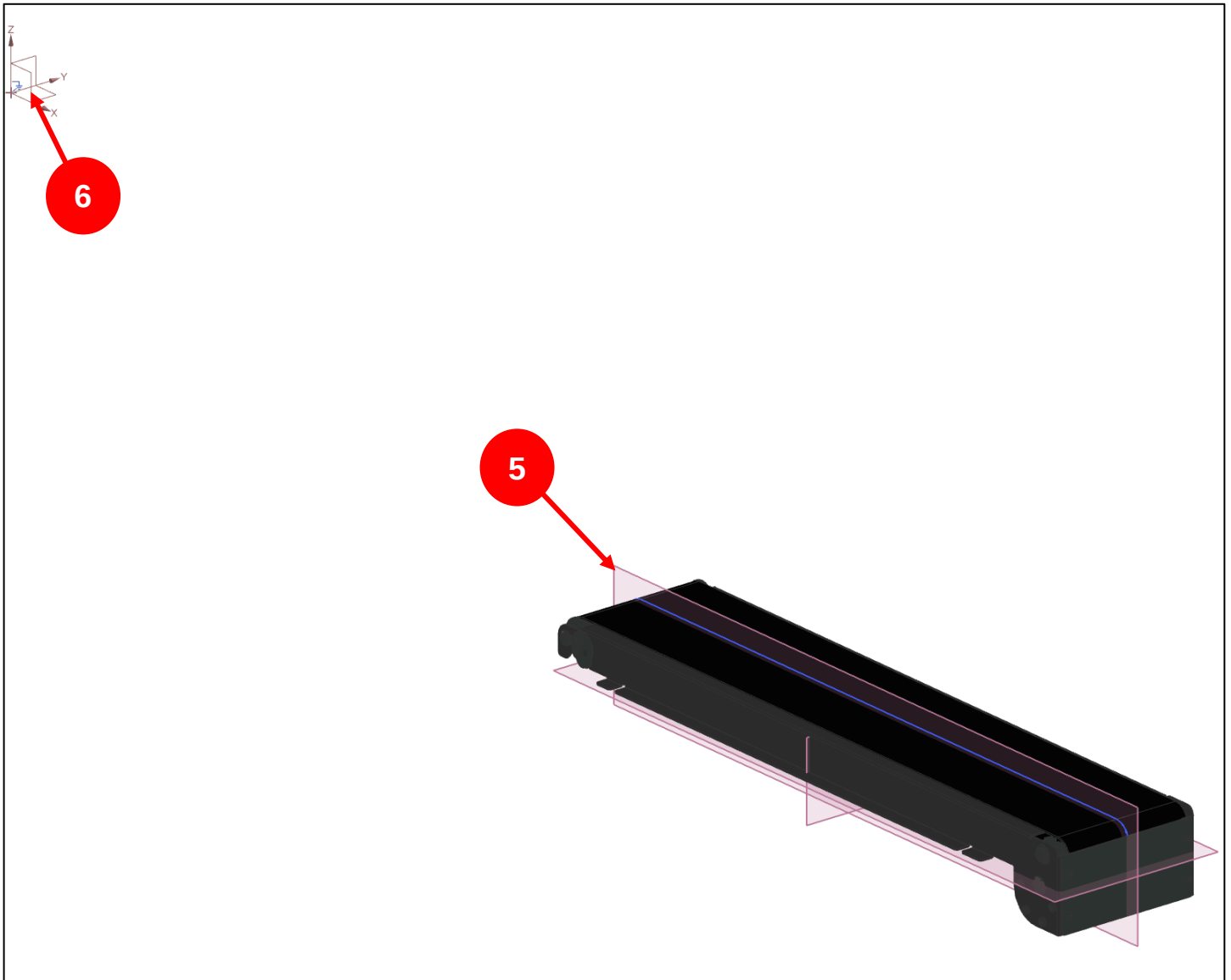
1. Click on the **ASSEMBLIES** tab.
2. Click on the **ASSEMBLY CONSTRAINTS** button.



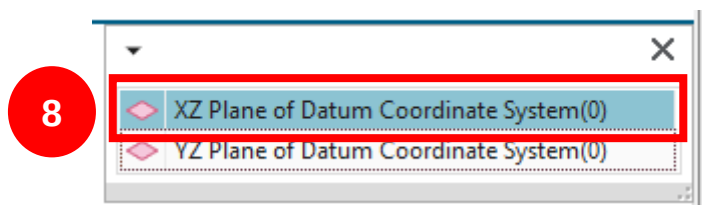
3. Click on the **TOUCH ALIGN** icon.
4. Click the **TARGET** icon in the **LOCATION** tab.



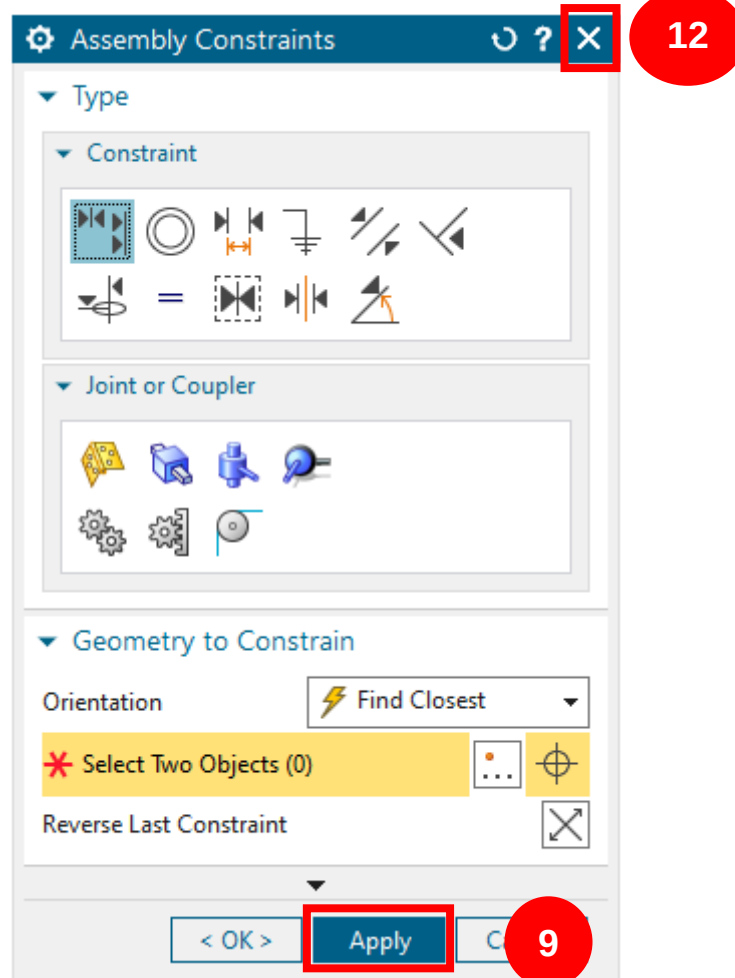
5. Click the ***XZ PLANE*** of the conveyor model.
6. Click on the ***XZ PLANE*** of the Siemens NX environment.



7. A new window will appear as shown below.
8. Select the ***XZ PLANE OF DATUM COORDINATE SYSTEM*** from the new window.



9. Click **APPLY**.



10. **REPEAT** steps 3 – 9 for the **YZ AND XY PLANE OF DATUM COORDINATE SYSTEM**.

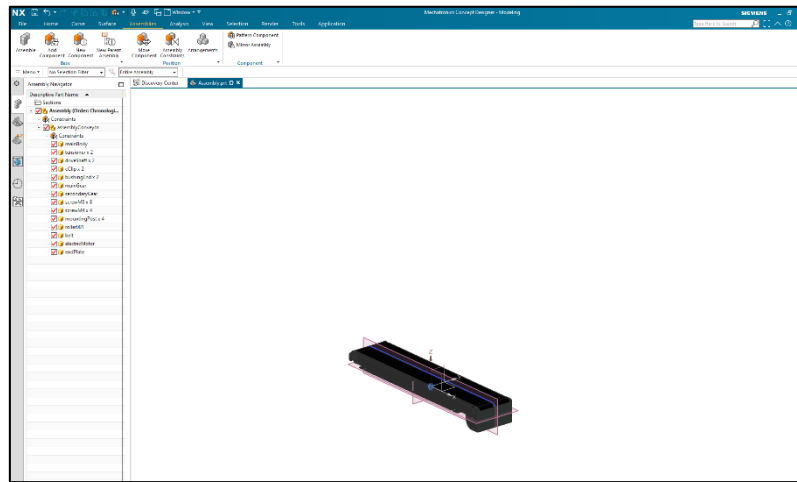
11. This will **CONSTRAIN** the model to the Siemens NX **WCS (WORLD COORDINATE SYSTEM)**.

12. When finished, close the **ASSEMBLY CONSTRAINTS** window with the **X** button, shown above.

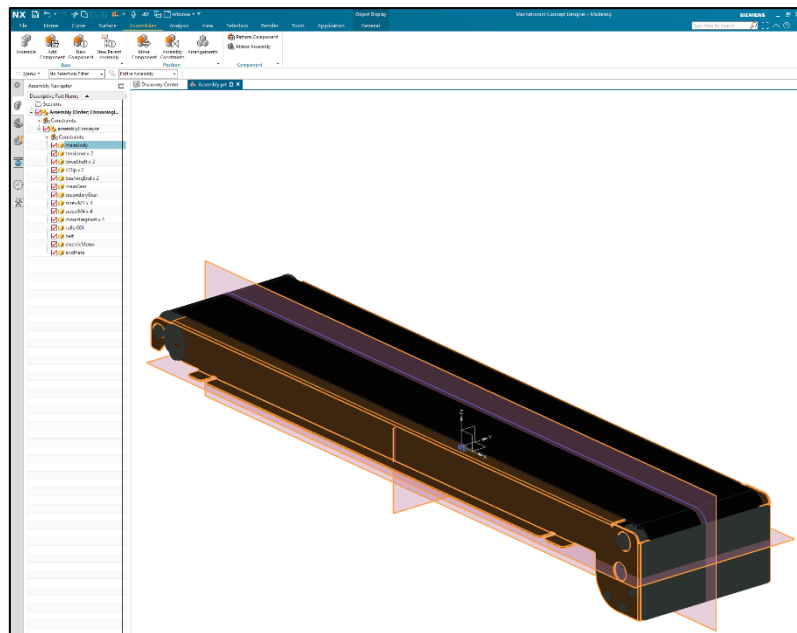
QUICK HINT – Fitting Model to Window

To fit the model to the WCS window, double click anywhere in the window.

BEFORE

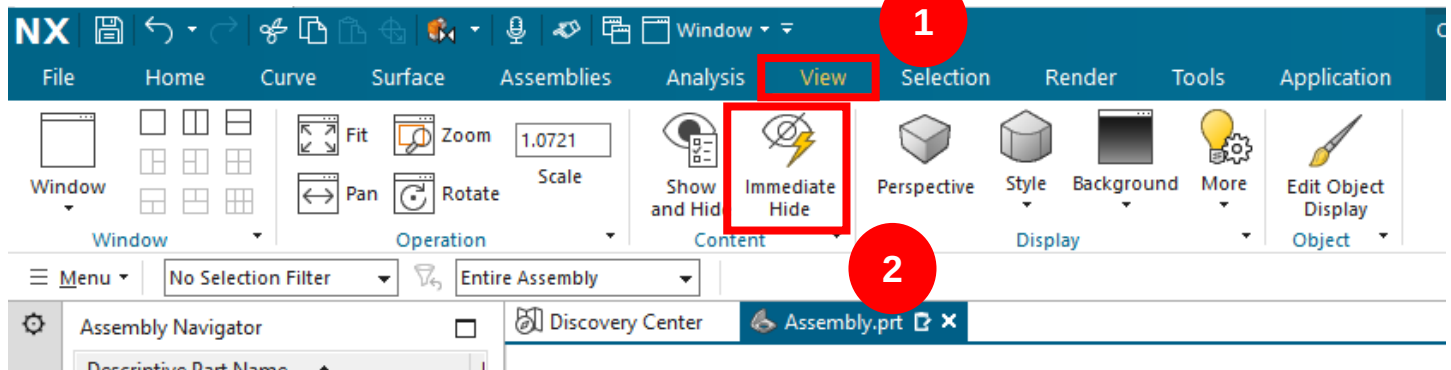


AFTER

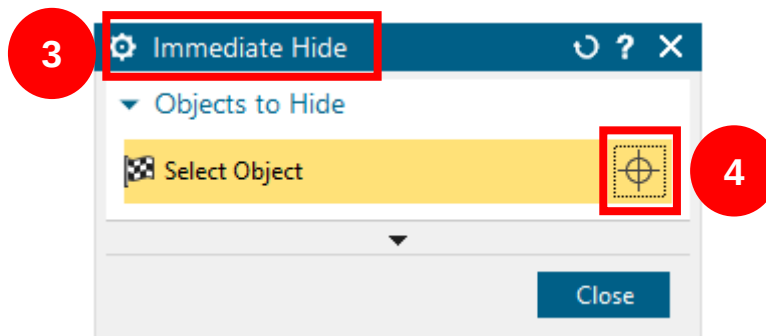


Hide Datum Planes and unused sketches

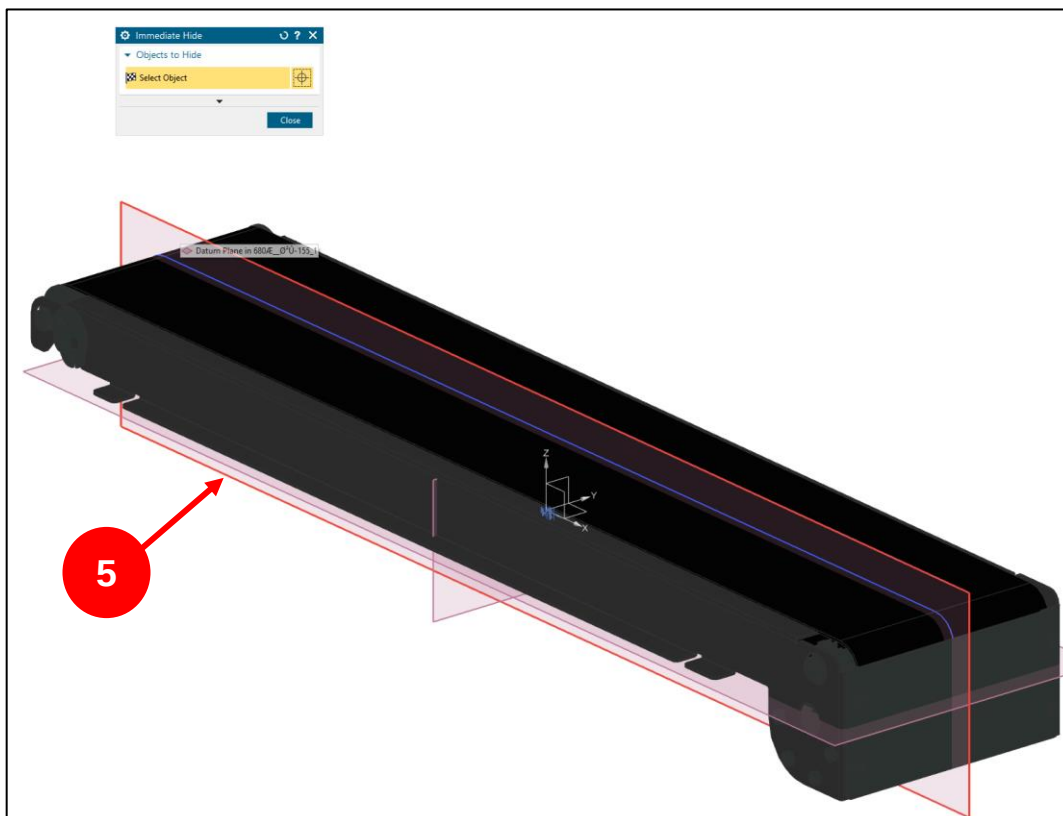
1. Click on the **VIEW** tab.
2. Click on the **IMMEDIATE HIDE** button.



3. A new window will appear called **IMMEDIATE HIDE**, shown below.
4. Click on the **SELECT OBJECT** target icon.

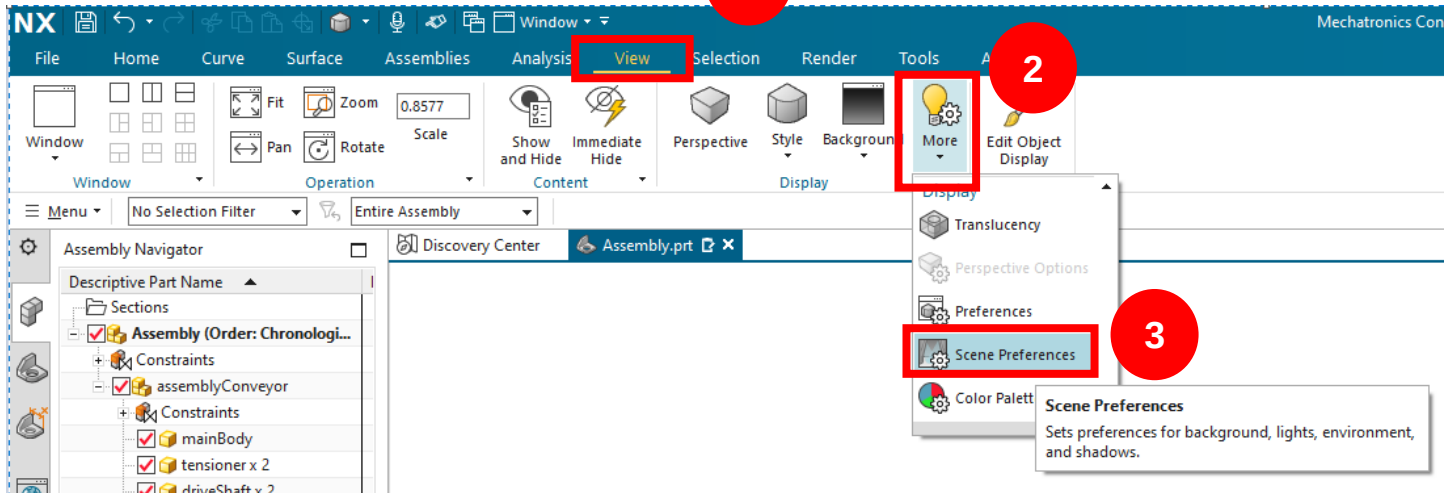


5. Select the **DATUM PLANES** and **UNUSED SKETCHES** to hide them.

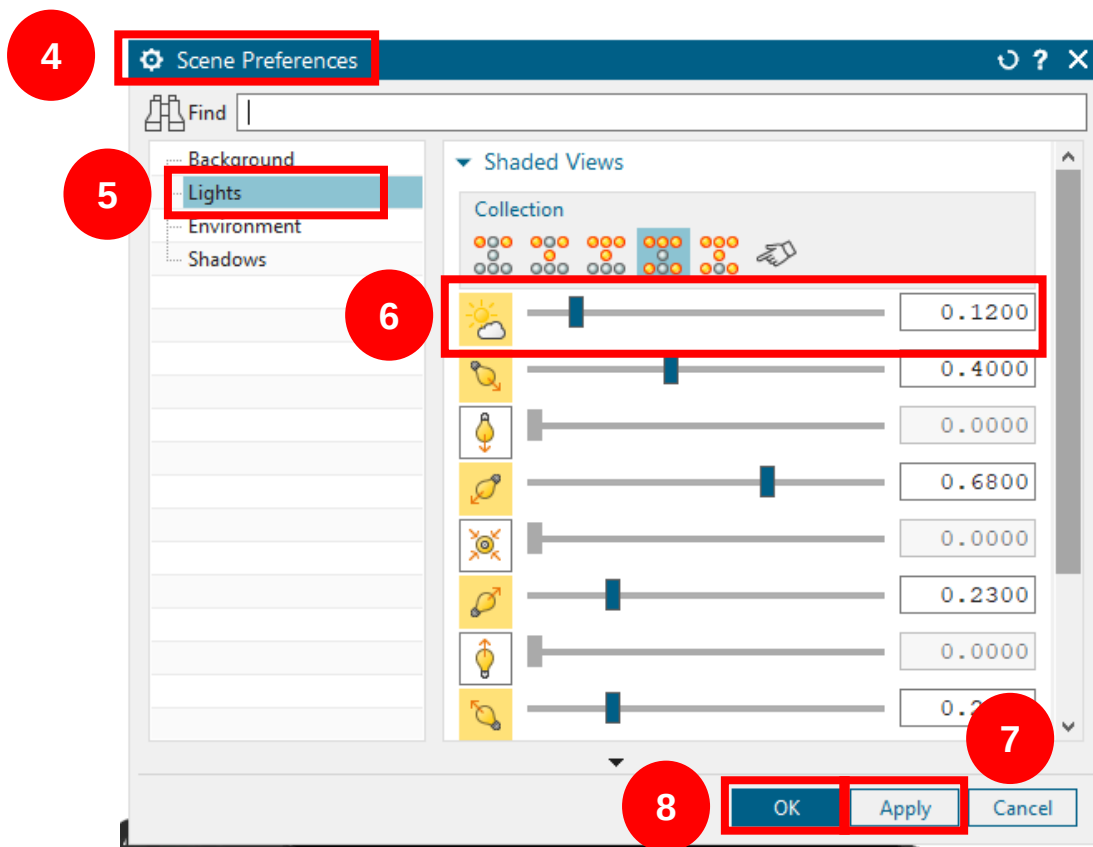


Adjust Lighting of WCS

1. Click on the **VIEW** tab.
2. Click on the **MORE** button in the **DISPLAY** area.
3. Click on the **SCENE PREFERENCES** button.

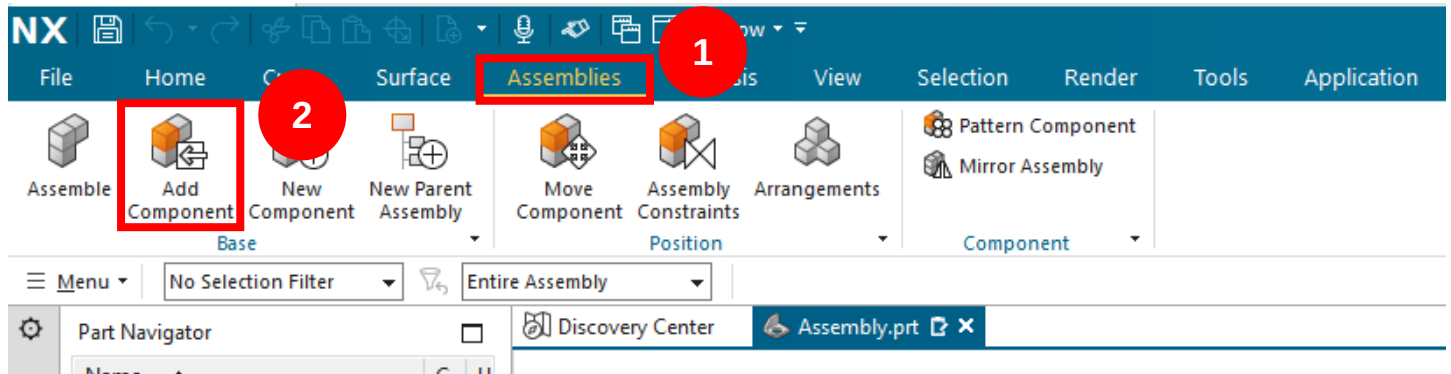


4. A new window will appear called **SCENE PREFERENCES**, shown below.
5. Click on the **LIGHTS** menu option.
6. Change the **SCENE AMBIENT** to **0.5**.
7. Click **APPLY**. This will make the model appear lighter and easier to view.
8. Click **OK** to close the window.

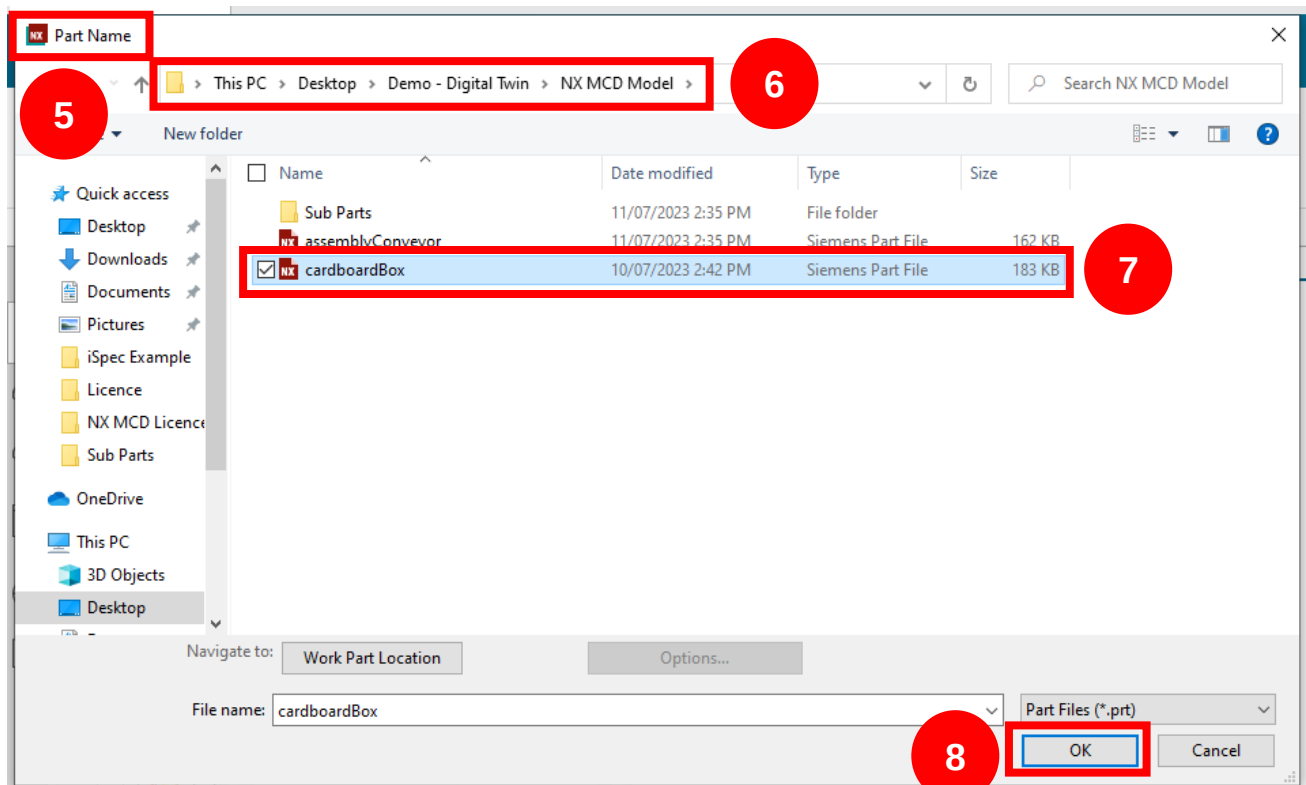
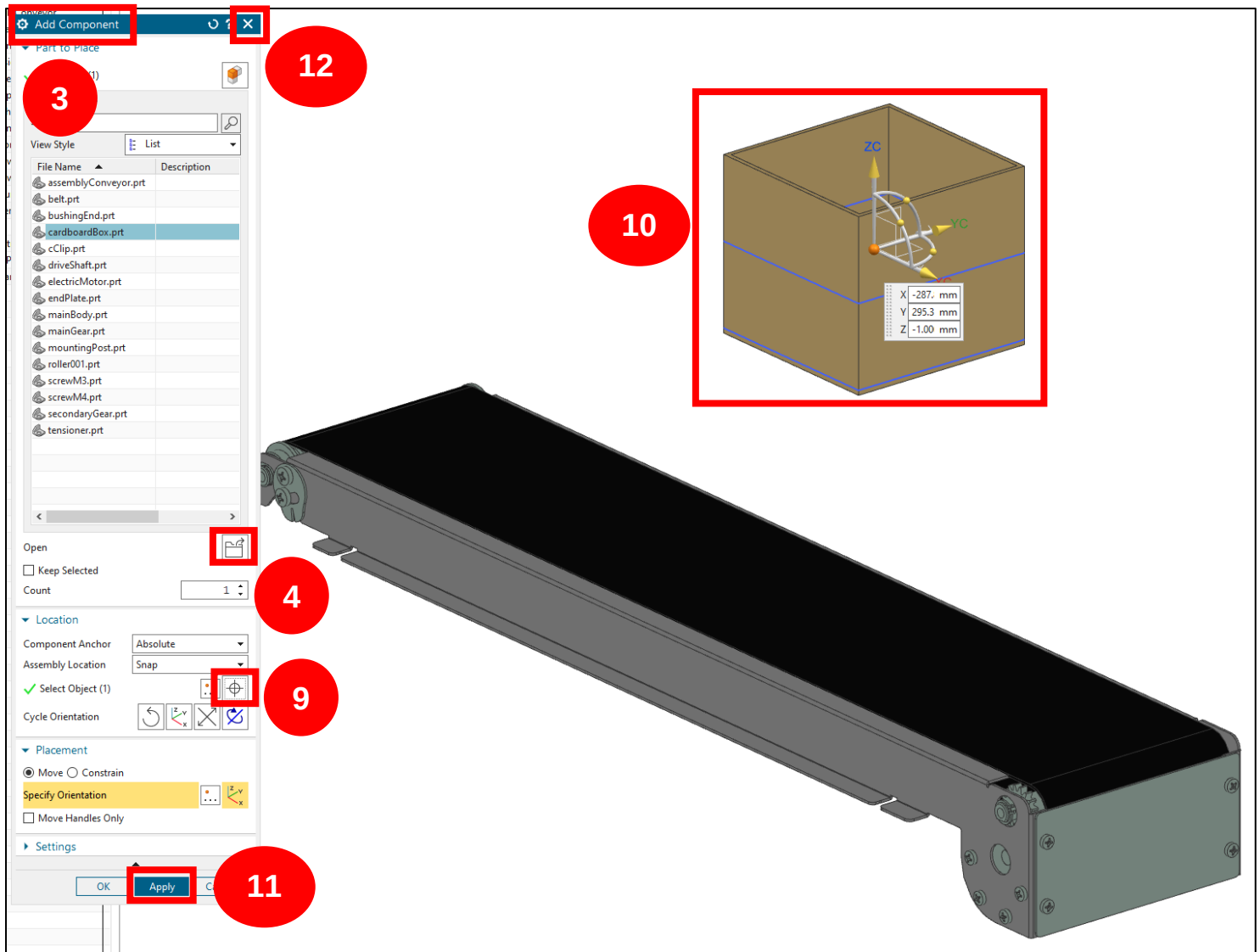


Inserting the Cardboard Box Model within the Siemens NX Environment

1. Click on the **ASSEMBLIES** tab.
2. Click on the **ADD COMPONENT** button.

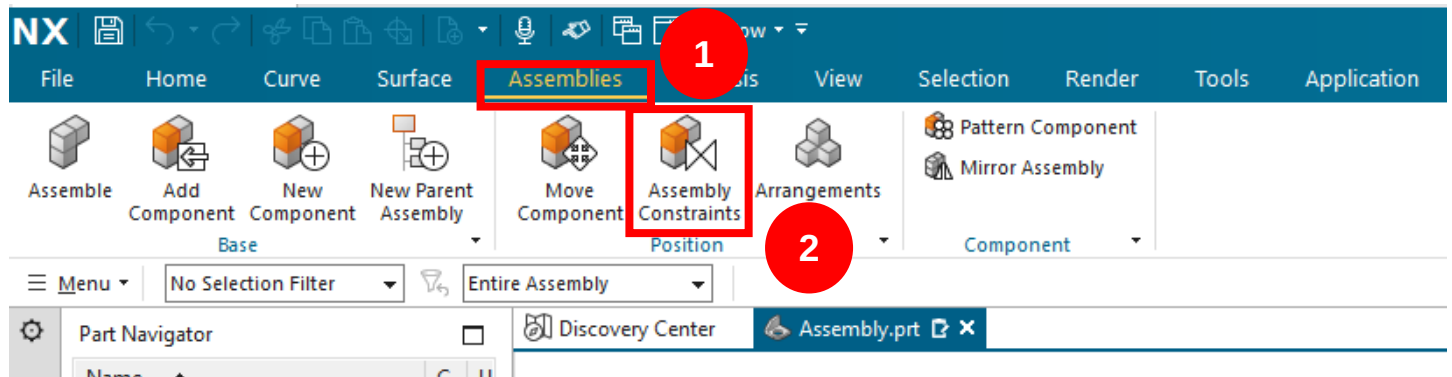


3. A new window will appear called **ADD COMPONENT**, shown below.
4. Click the **OPEN** button.
5. A new window will appear called **PART NAME**, shown below.
6. Browse to the **FOLDER** where the project files are stored,
(i.e. **C:\Desktop\Demo - Digital Twin\NX MCD Model**).
7. Click on the file called **CARDBOARDBOX.PRT**.
8. Click **OK**.
9. Click the **TARGET** icon in the **LOCATION** tab.
10. In the **MODEL** space, click anywhere to add the cardboard box.
11. Click **APPLY**.
12. Click the **X** in top right corner to close the window.

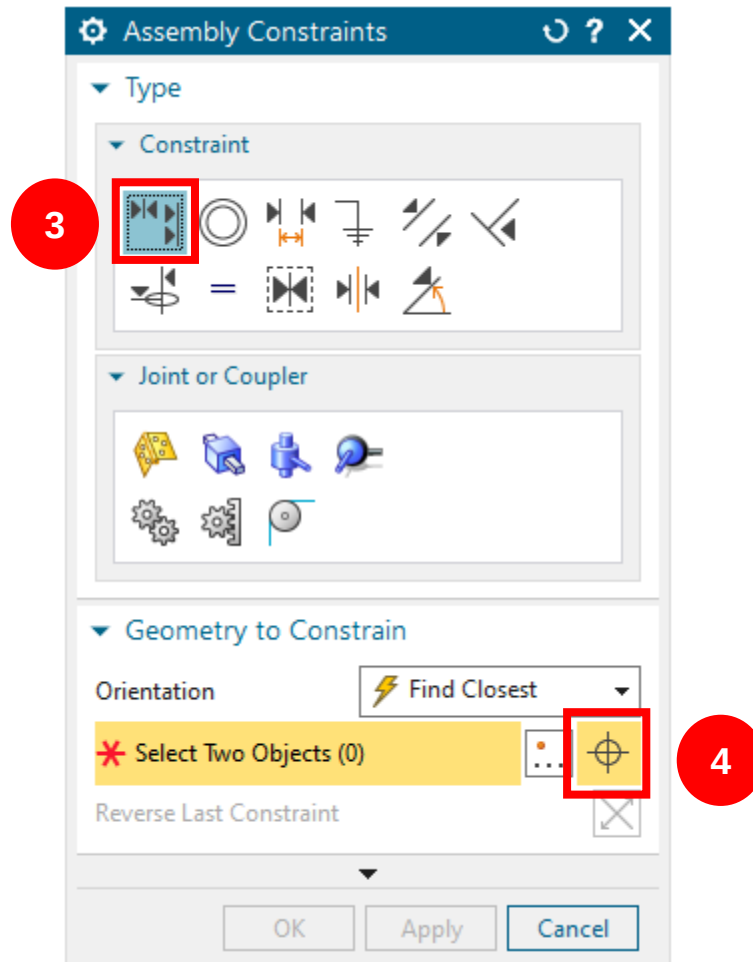


Constraining the Cardboard Box using Touch Align on the XZ Axis

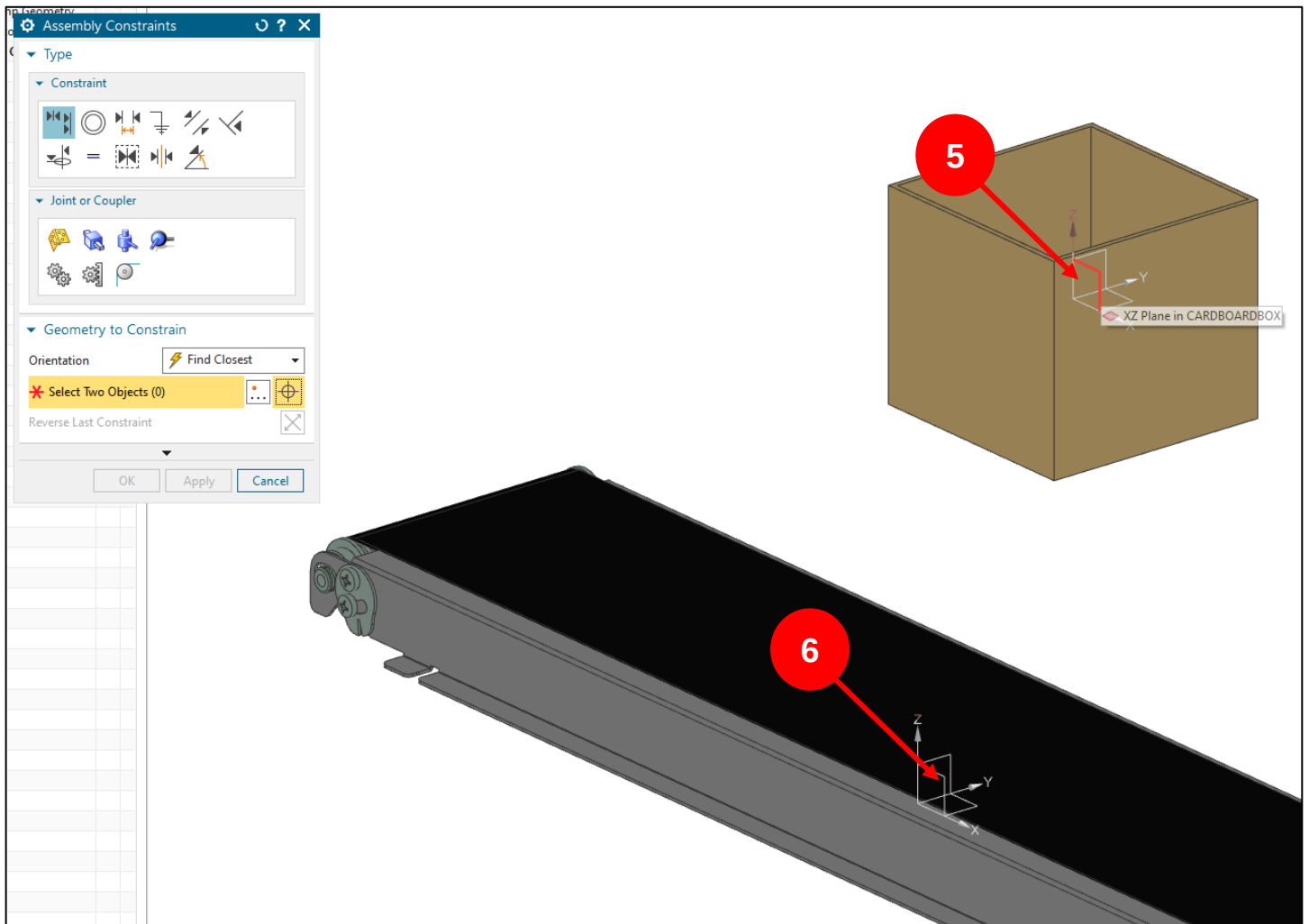
1. Click on the **ASSEMBLIES** tab.
2. Click on the **ASSEMBLY CONSTRAINTS** button.



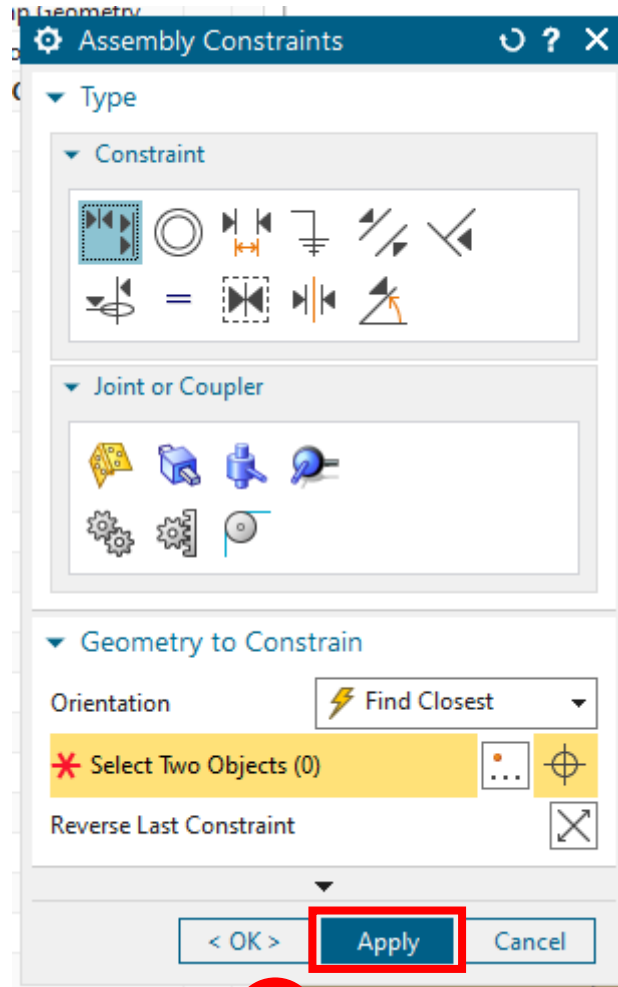
3. Click on the **TOUCH ALIGN** icon.
4. Click the **TARGET** icon in the **LOCATION** tab.



5. Click the **XZ PLANE** of the conveyor model.
6. Click on the **XZ PLANE** of the Siemens NX environment.



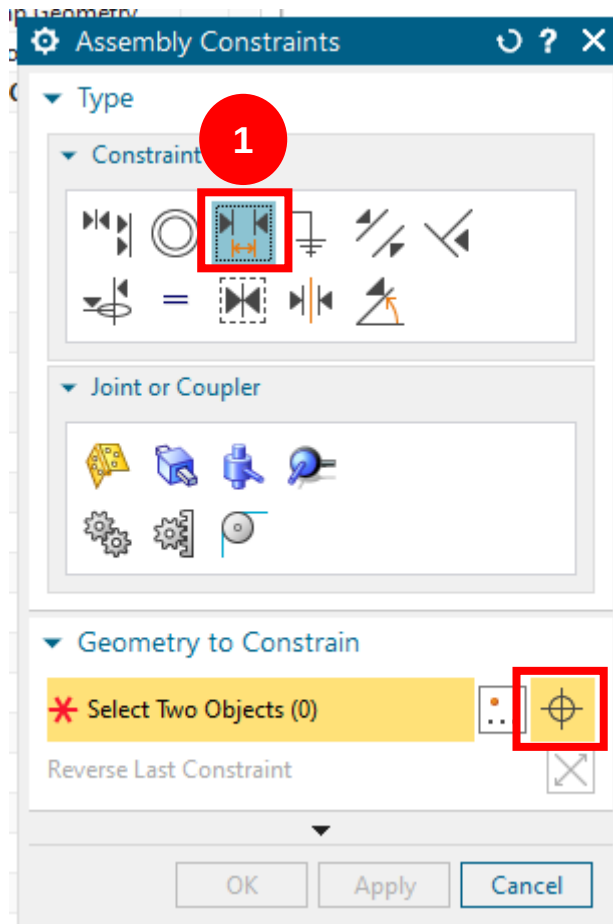
- Click **APPLY** to complete the constraint.



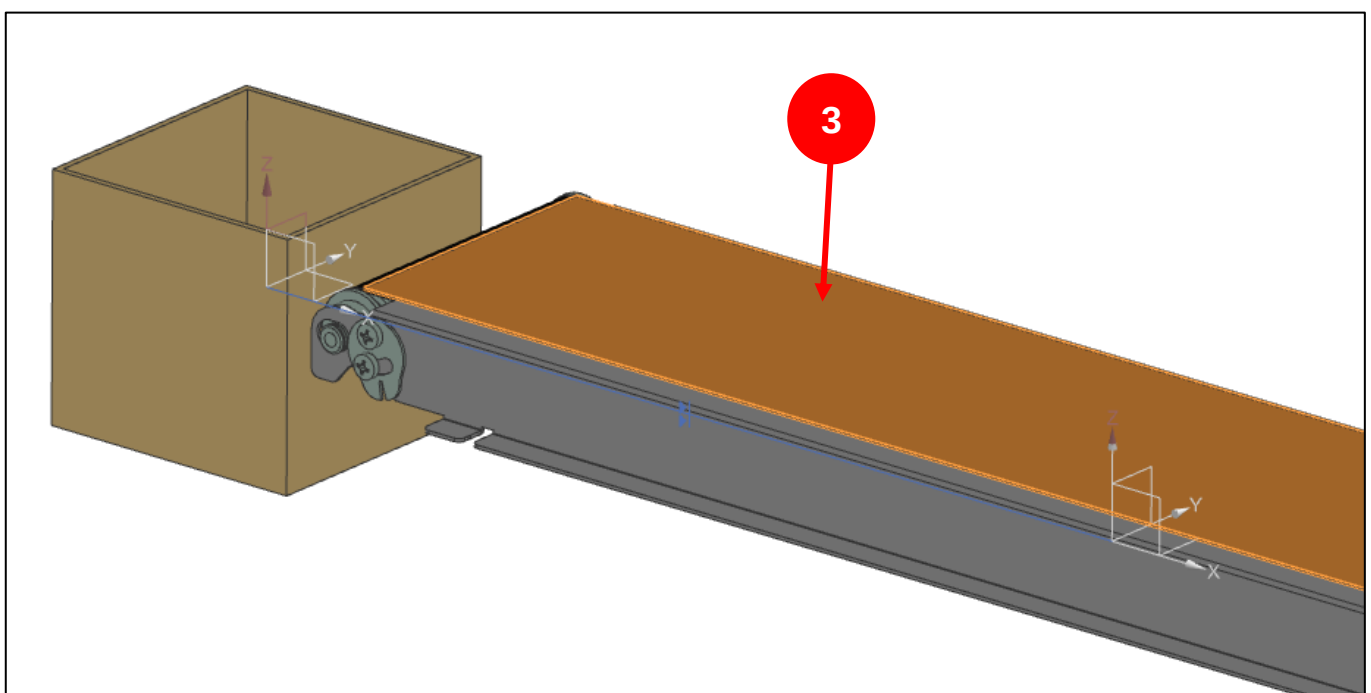
7

Constraining the Cardboard Box base using Distance Constraint from the conveyor

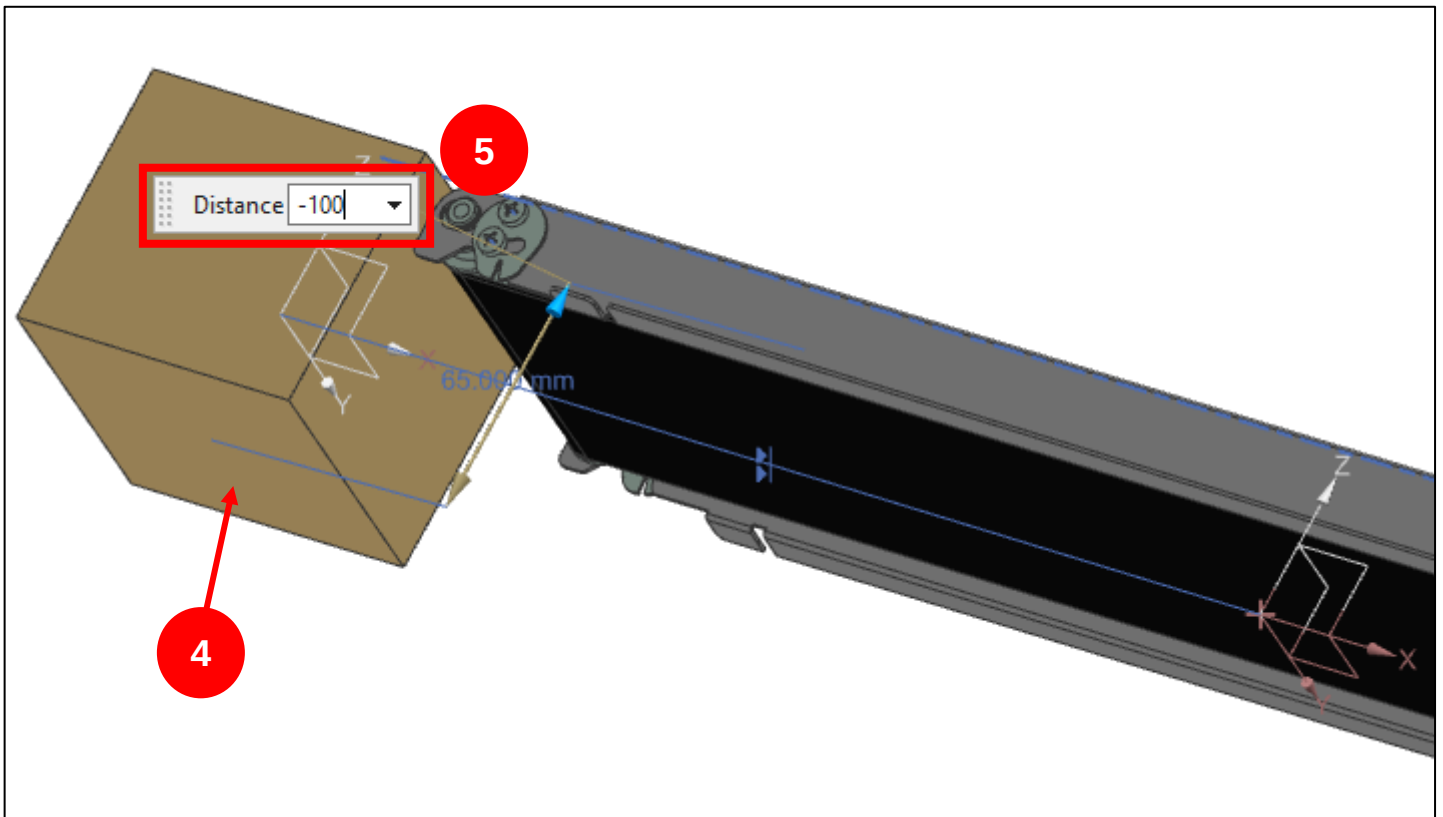
1. Click on the **DISTANCE CONSTRAINT** icon.
2. Click the **TARGET** icon in the **LOCATION** tab.



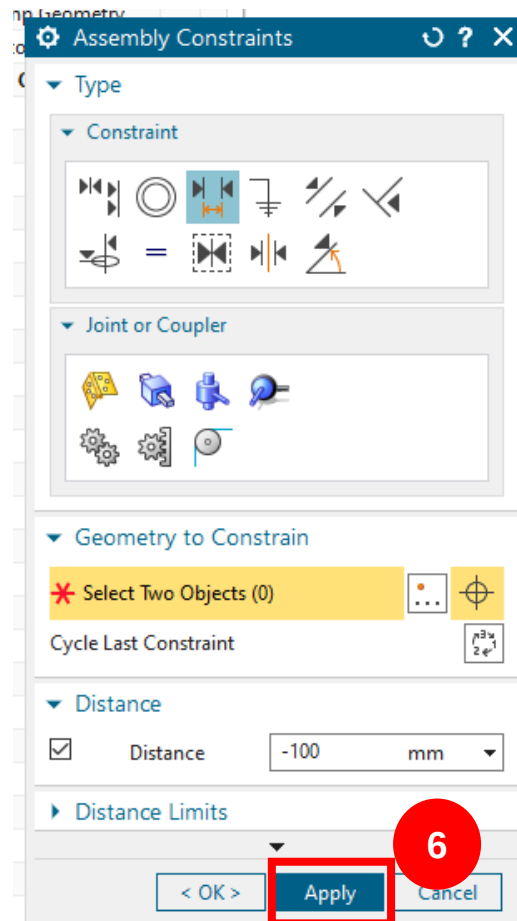
3. Click the **TOP OF CONVEYOR BELT**.



4. Click the **TOP OF CONVEYOR BELT**.
5. Enter a **DISTANCE** of **-100mm** (**MINUS ONE HUNDRED MILLIMETERS**)

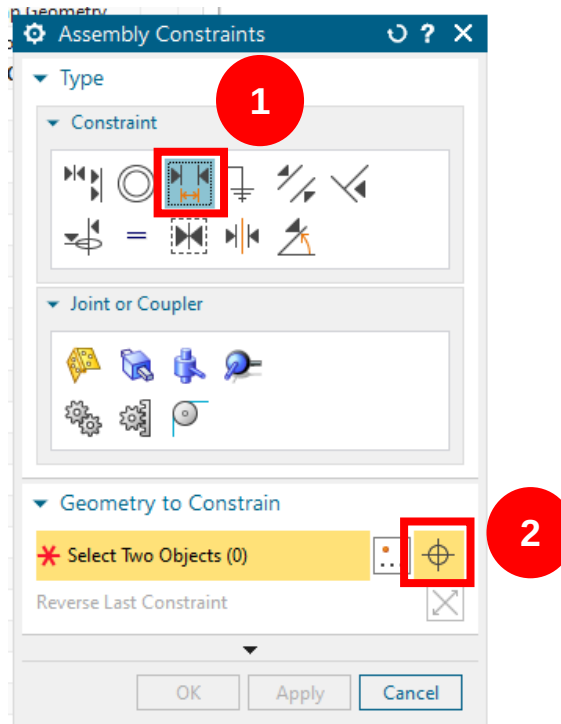


6. Click **APPLY**.

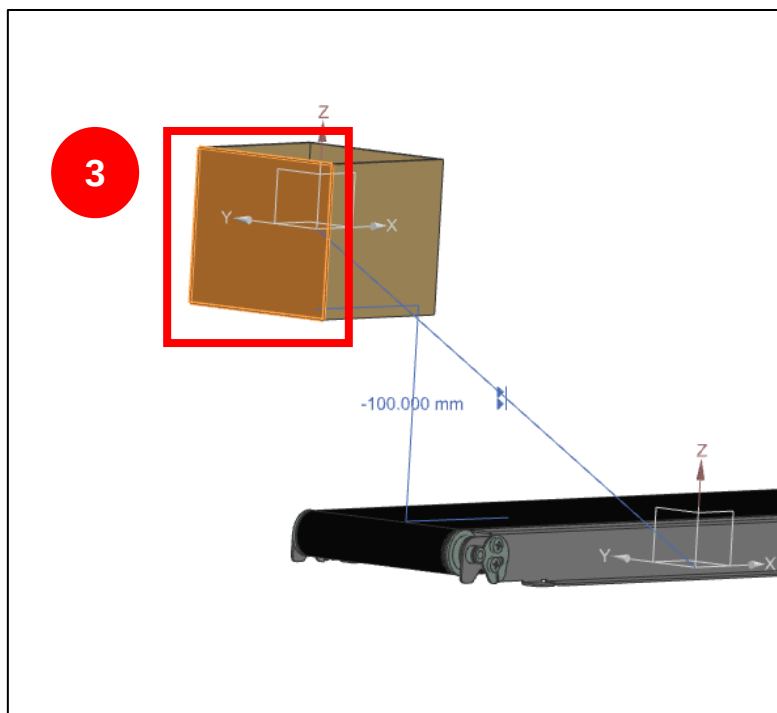


Constraining the Cardboard Box end using Distance Constraint from the conveyor

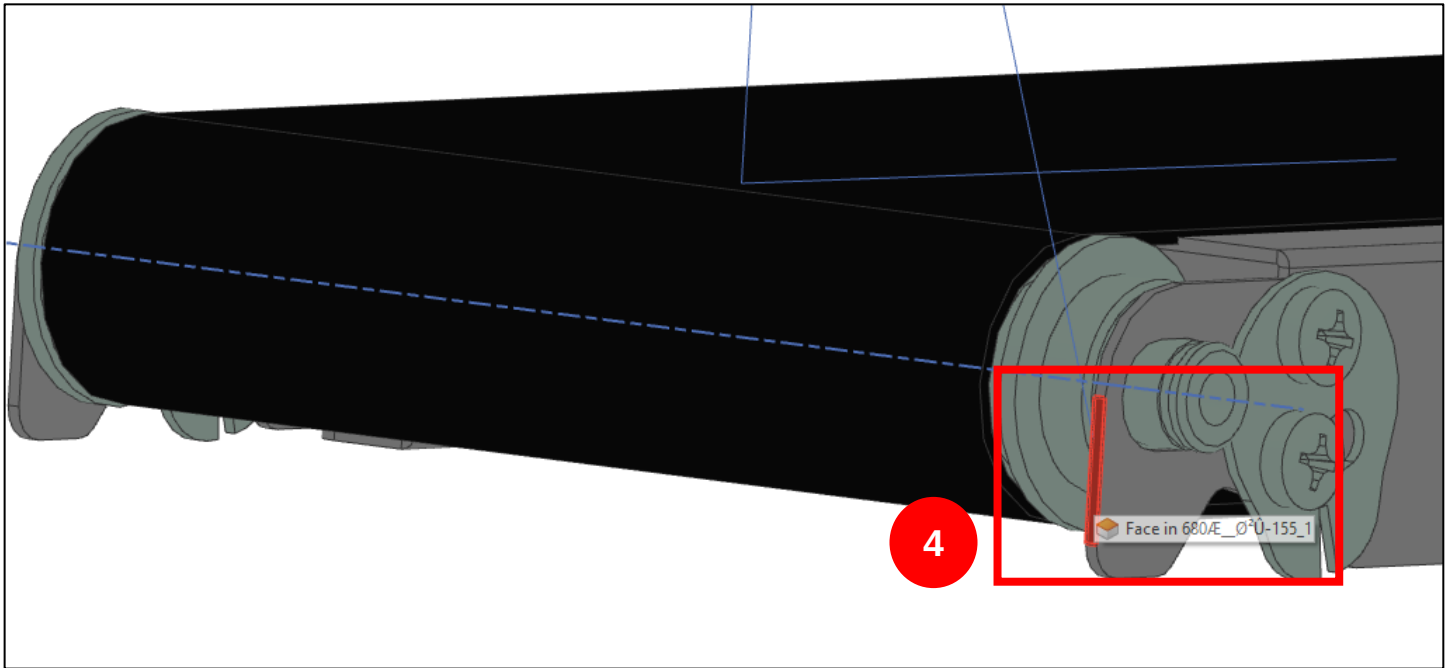
1. Click on the **DISTANCE CONSTRAINT** icon.
2. Click the **TARGET** icon in the **LOCATION** tab.



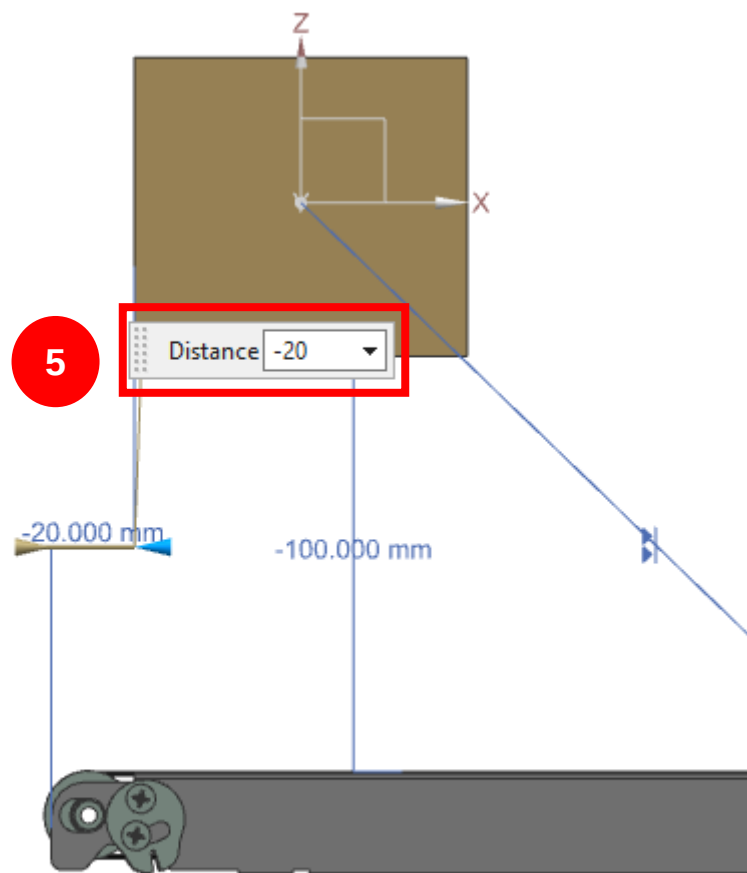
3. Click on the **LEFT-HAND SIDE** of the **CARDBOARD BOX**.



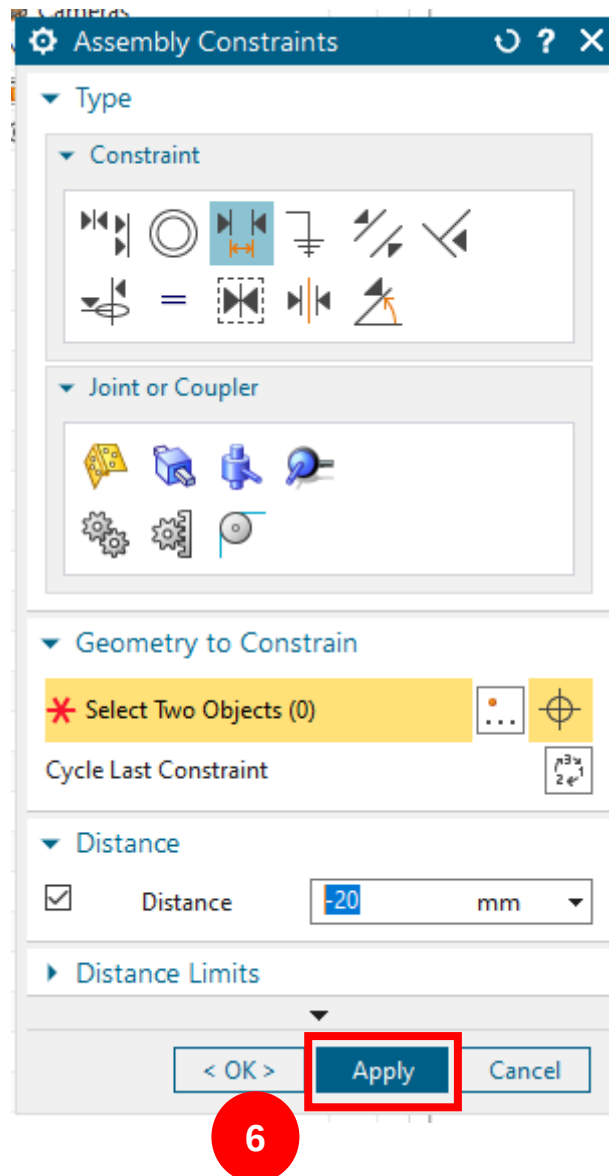
- Click on the **LEFT-HAND SIDE** of the **CONVEYOR FRAME**.



- Enter a **DISTANCE** of **-20mm (MINUS TWENTY MILLIMETERS)**

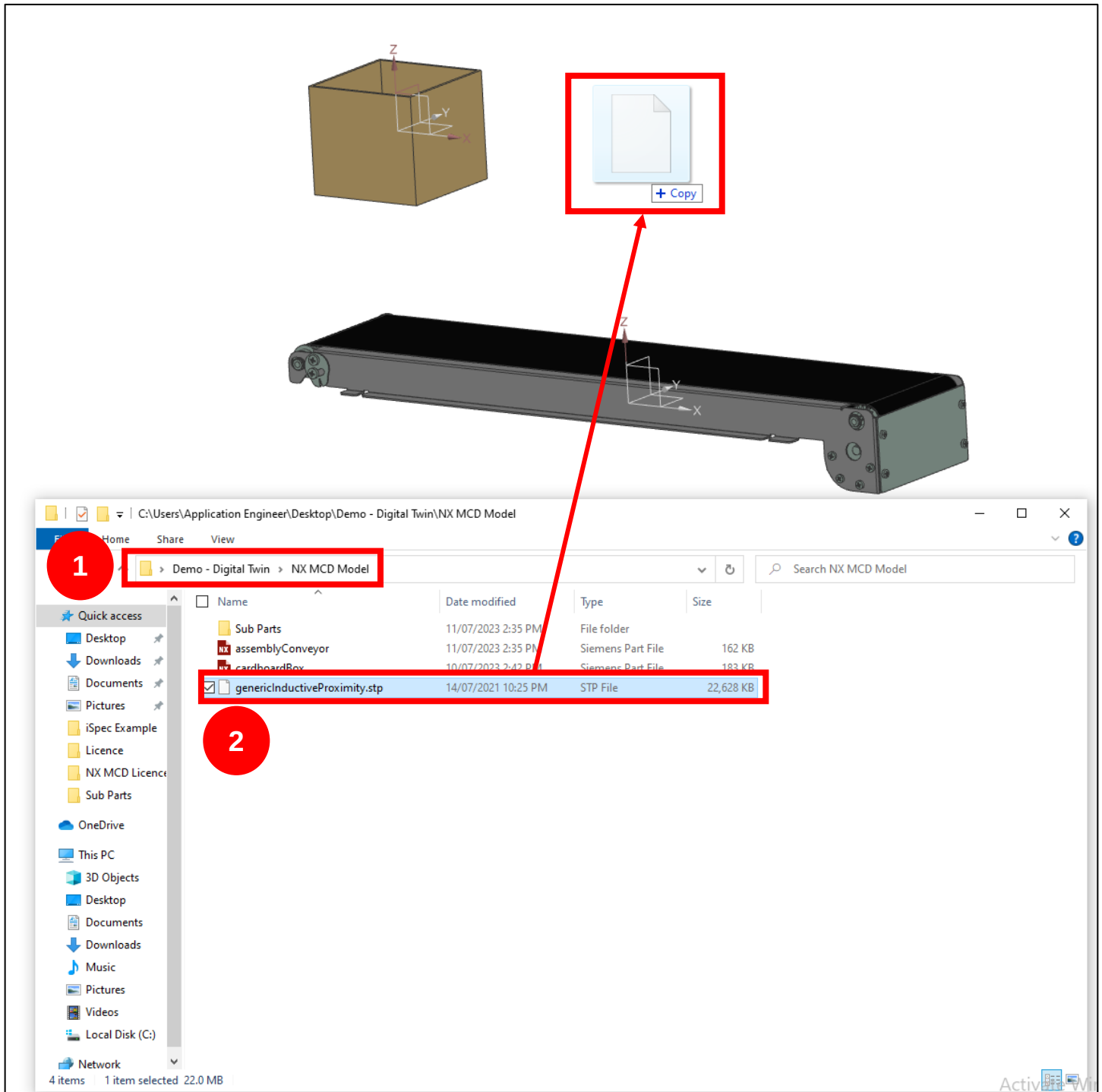


6. Click **APPLY**.
7. Click **X** in the top right corner to complete the parametrisation of the constraints.
8. Use the **IMMEDIATE HIDE** function as described earlier to clean up the model of dimensions.



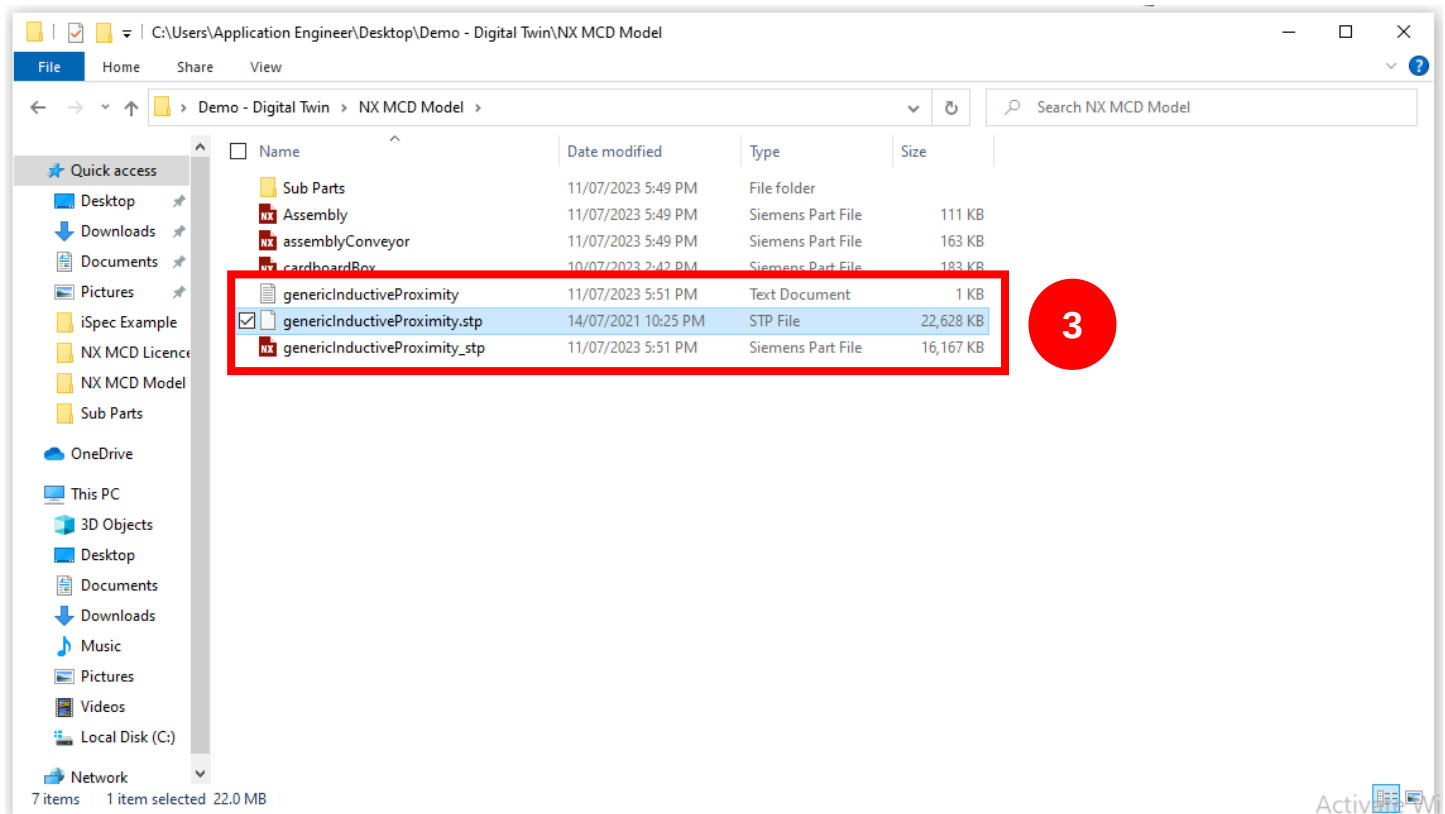
Insert Proximity Sensor as a STEP file using Drop and Drag Method

1. In Windows Explorer, navigate to the folder where the **GENERICINDUCTIVEPROXIMITY.STP** file is located. (i.e. **C:\Desktop\Demo - Digital Twin\NX MCD Model**).
2. **CLICK AND DRAG** the **STP** file into the work area of Siemens NX.



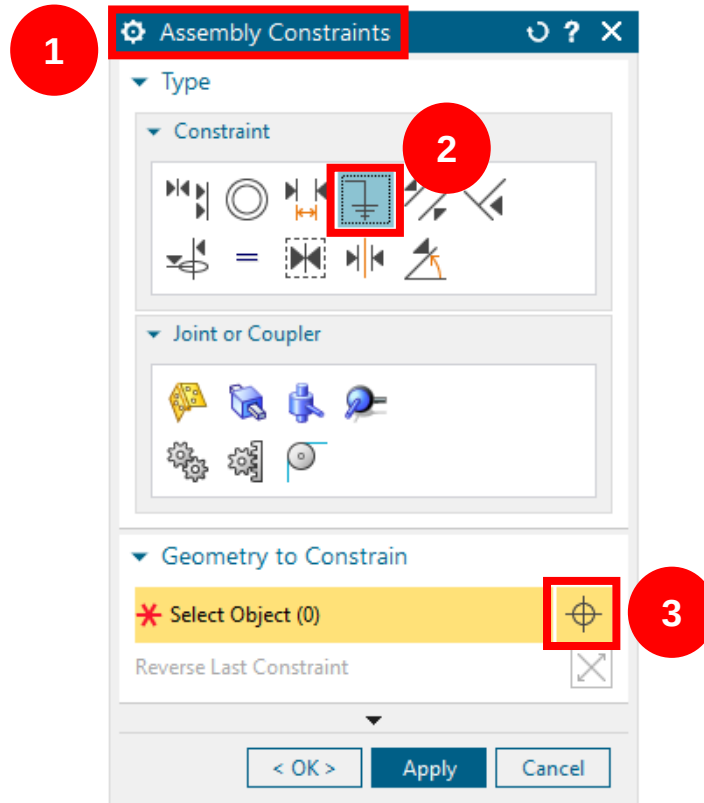
- Siemens NX will now start the process of importing the **STEP** file into a **PRT** file. This **PRT** file will be automatically added to the original folder where the STEP file was, shown below.

PLEASE BE PATIENT THIS MIGHT TAKE A LITTLE TIME.

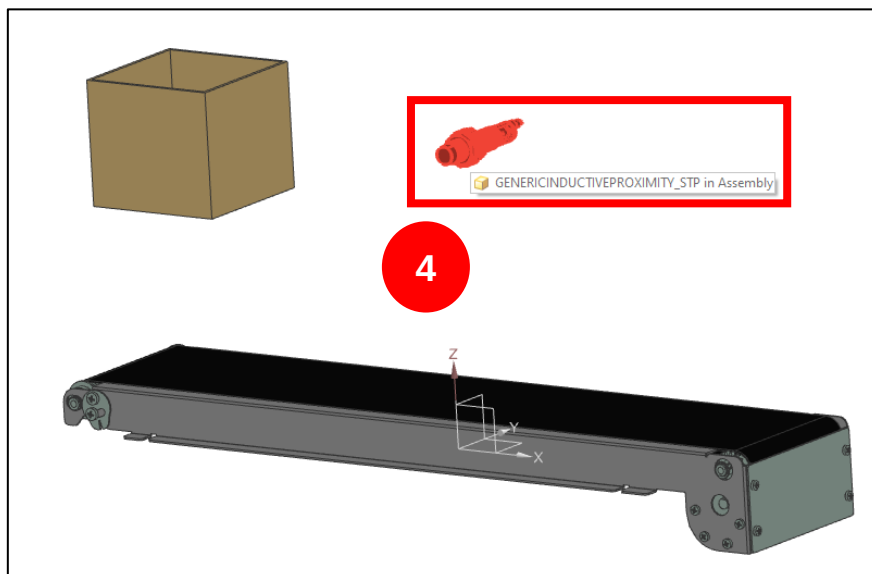


Using A Fixed Constraint on Proximity Sensor

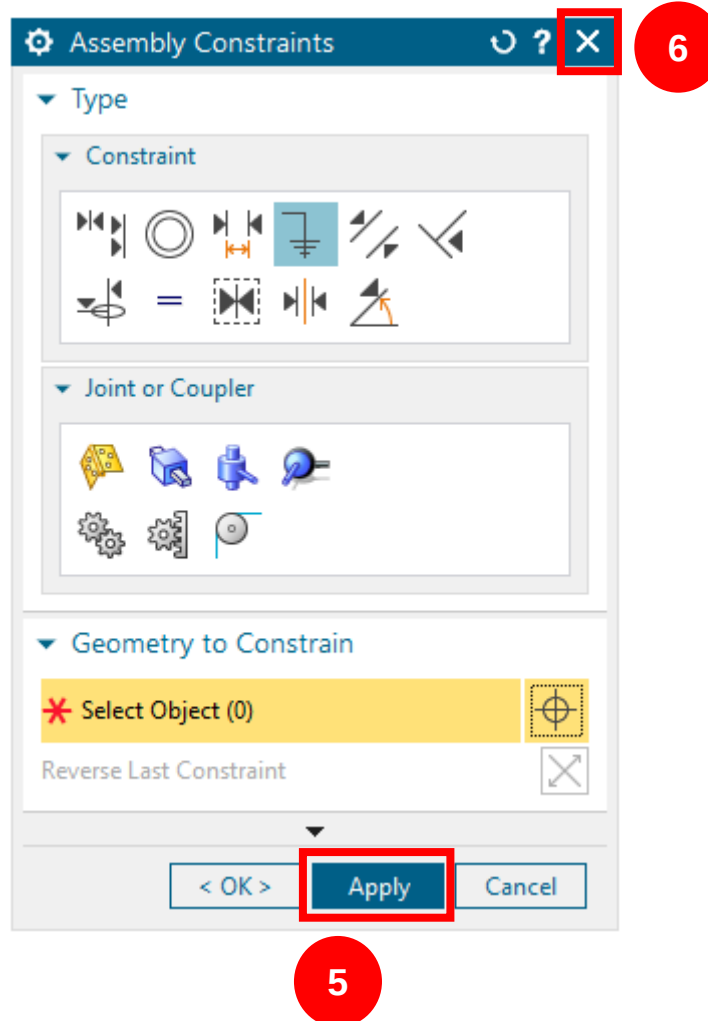
1. When the above **STEP** file import is complete, a new window will appear called **ASSEMBLY CONSTRAINTS**, shown below.
2. Click on the **FIXED** constraint icon.
3. Click the **TARGET** icon in the **LOCATION** tab.



4. Click on the **PROXIMITY SENSOR**.



5. Click **APPLY**.
6. Click **X** in the top corner to complete the operation.



QUICK HINT – Pan, Tilt and Zoom around the Siemens NX workspace.

It is possible to quickly pan, tilt and zoom around the Siemens NX workspace by just using the mouse and its associated buttons. The below image and table can be used to see these quick shortcuts.

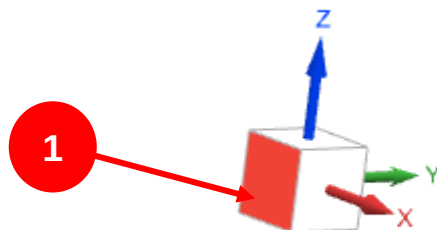
Shortcut Description	Action
Zoom Out	Scroll Down
Zoom In	Scroll Up
Orbit/Tilt	Hold Middle Button + Move Mouse
Pan	Hold Middle + Hold Right Button + Move Mouse



QUICK HINT – Using the WCS cube to change view

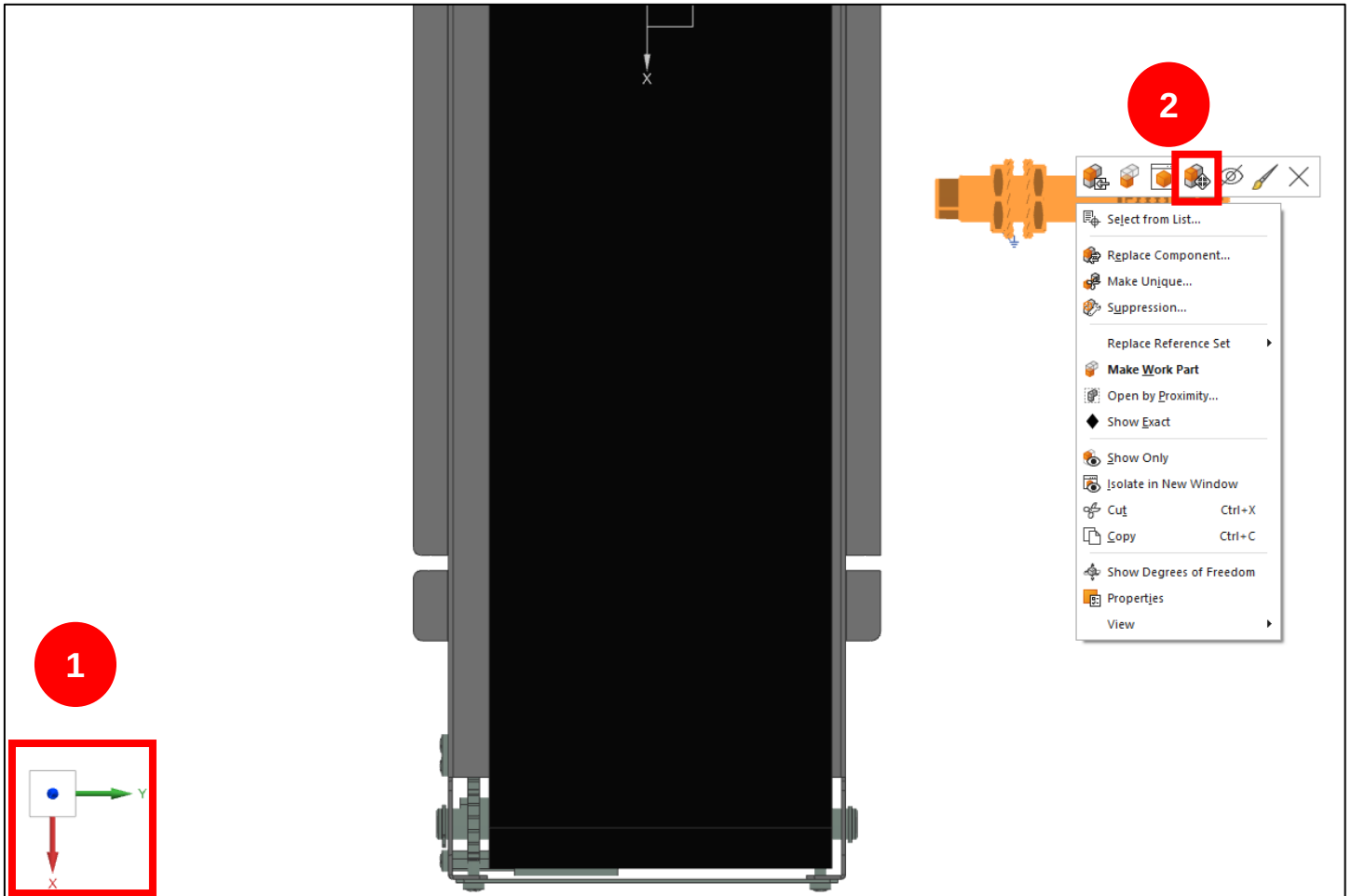
To change the view of the model, it is possible to use the axis cube located in the bottom left corner of the workspace, shown below. Clicking on the sides of the cube will change the view of the model to the relevant orientation.

1. Click on the side as shown below will change the model to the front view.

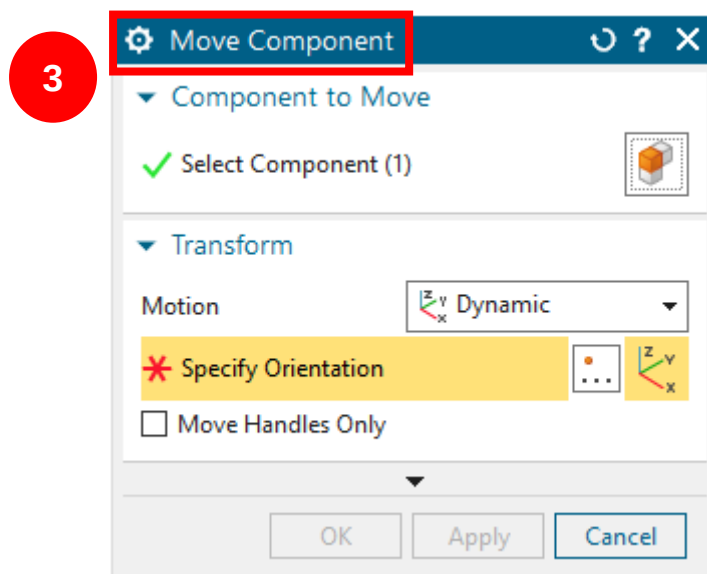


Move Proximity Sensor to rough sensing location from top view

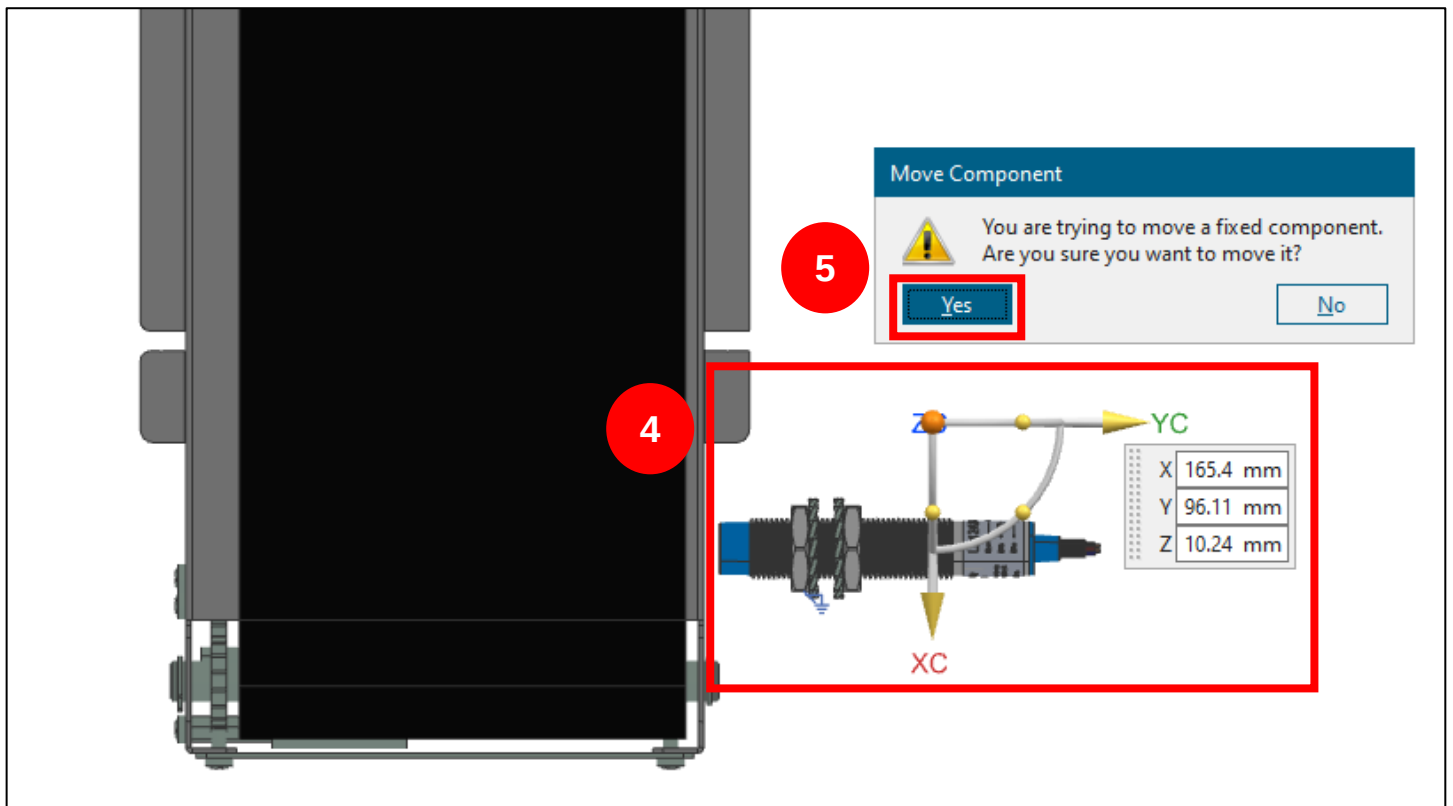
1. Using the **AXIS CUBE**, change the orientation of the model so that you are viewing the top of the conveyor. It should look like the image below.
2. Right Click on the proximity switch and select the **MOVE COMPONENT** icon.



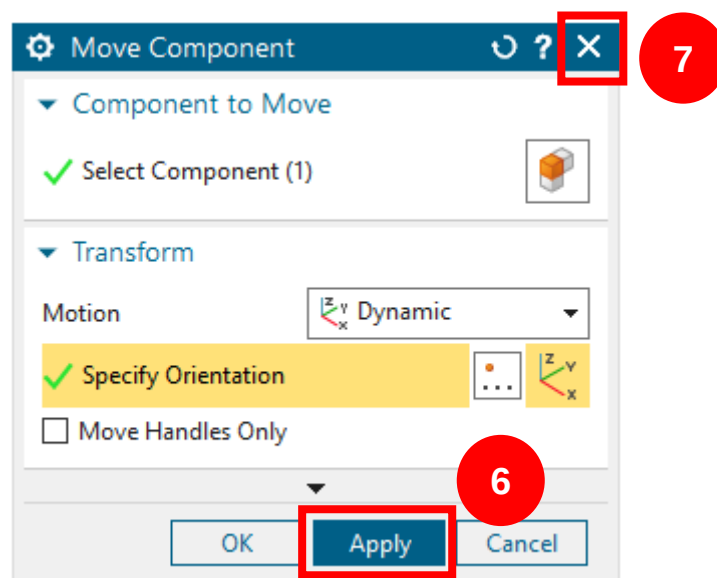
3. A new window will appear called **MOVE COMPONENT**, shown below.



4. The proximity switch will now have a **MOVE AXIS WIDGET** attached to it. Use the **GRAB POINTS** to move the proximity switch to a location just to the bottom right of the conveyor.
5. A **WARNING BOX** will appear stating that the proximity switch is a fixed component. Select **YES** to close this window.

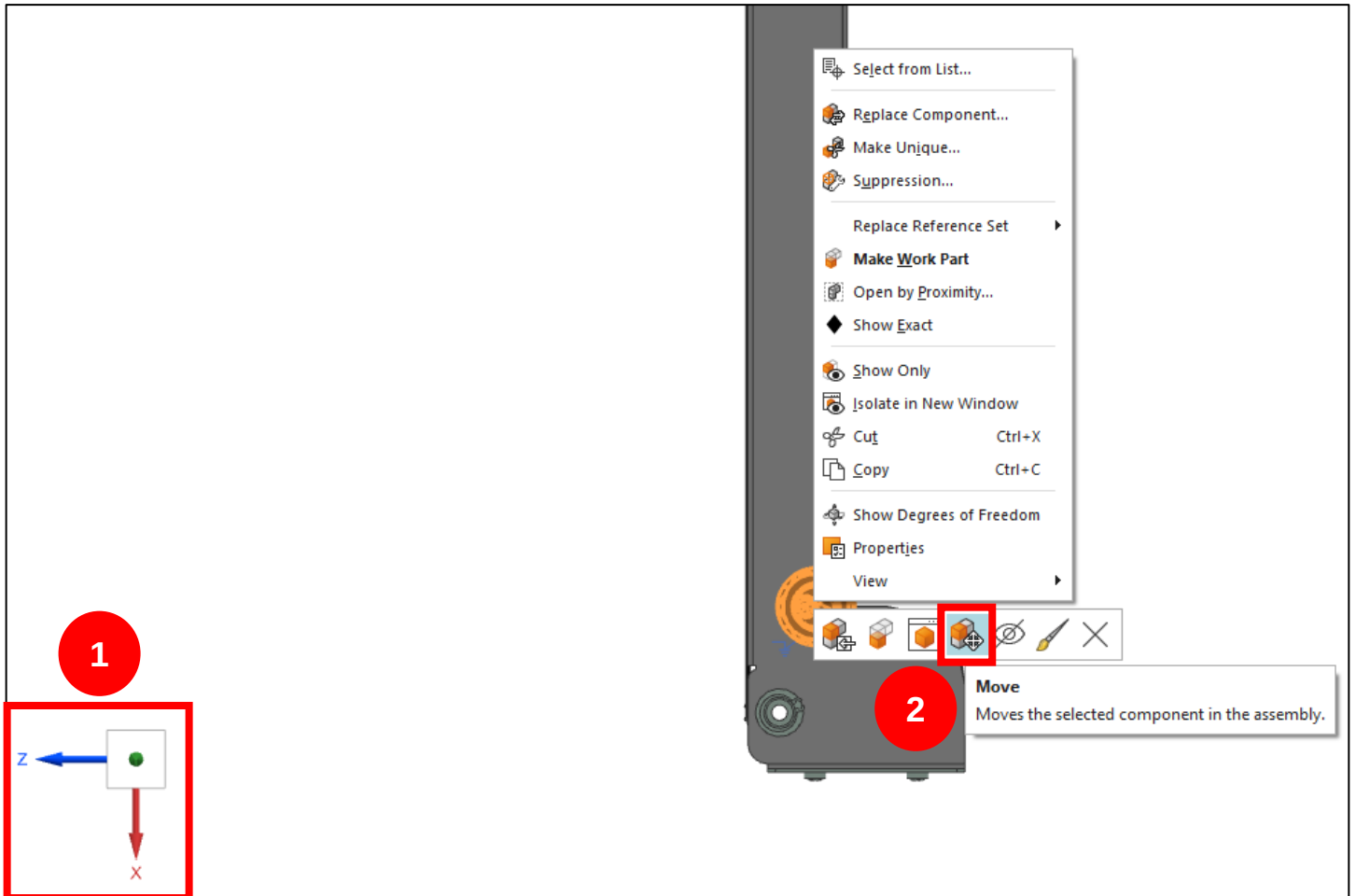


6. Once the proximity switch is moved to the correct location click **APPLY** on the **MOVE COMPONENT** window.
7. Click **X** in the top right corner to finish the operation.

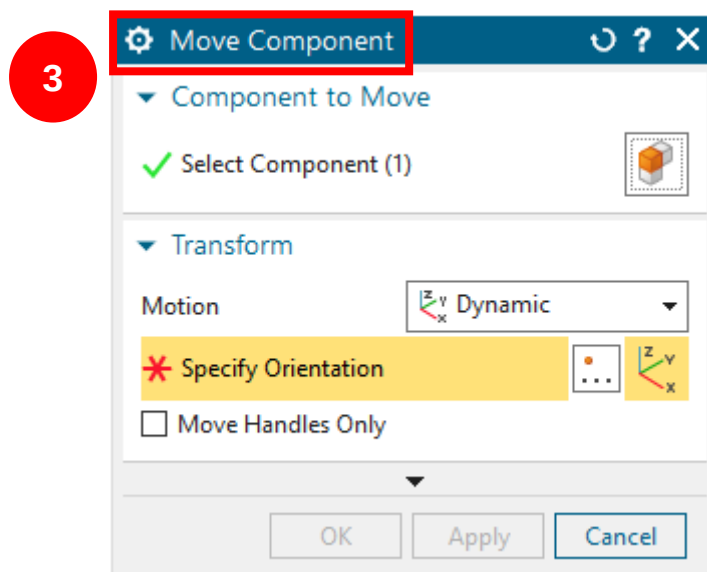


Move Proximity Sensor to rough sensing location from back view

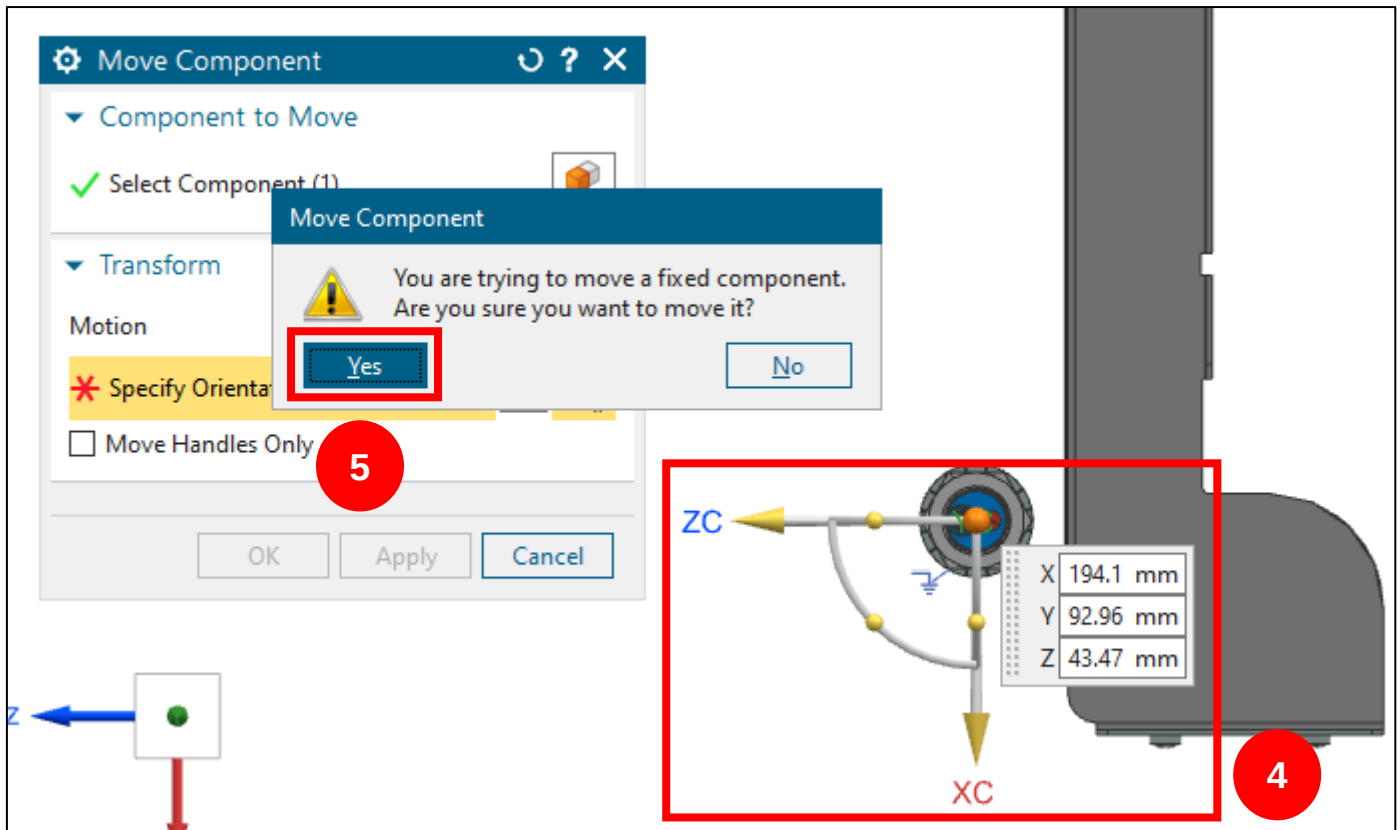
1. Using the **AXIS CUBE**, change the orientation of the model so that you are viewing the top of the conveyor. It should look like the image below.
2. Right Click on the proximity switch and select the **MOVE COMPONENT** icon.



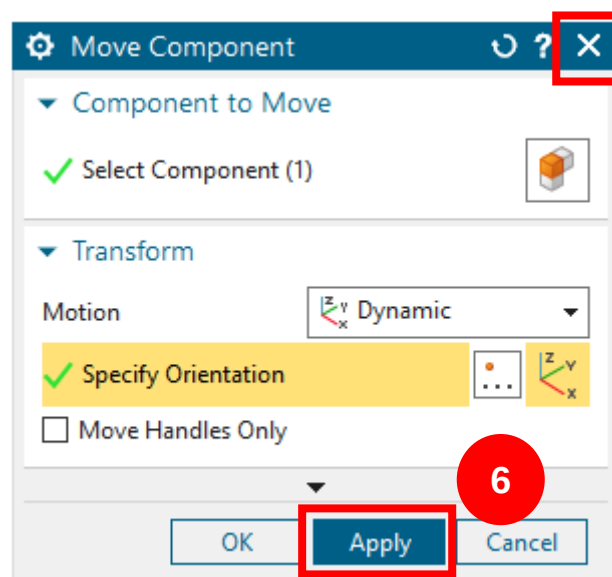
3. A new window will appear called **MOVE COMPONENT**, shown below.



4. The proximity switch will now have a **MOVE AXIS WIDGET** attached to it. Use the **GRAB POINTS** to move the proximity switch to a location just to the bottom right of the conveyor.
5. A **WARNING BOX** will appear stating that the proximity switch is a fixed component. Select **YES** to close this window.



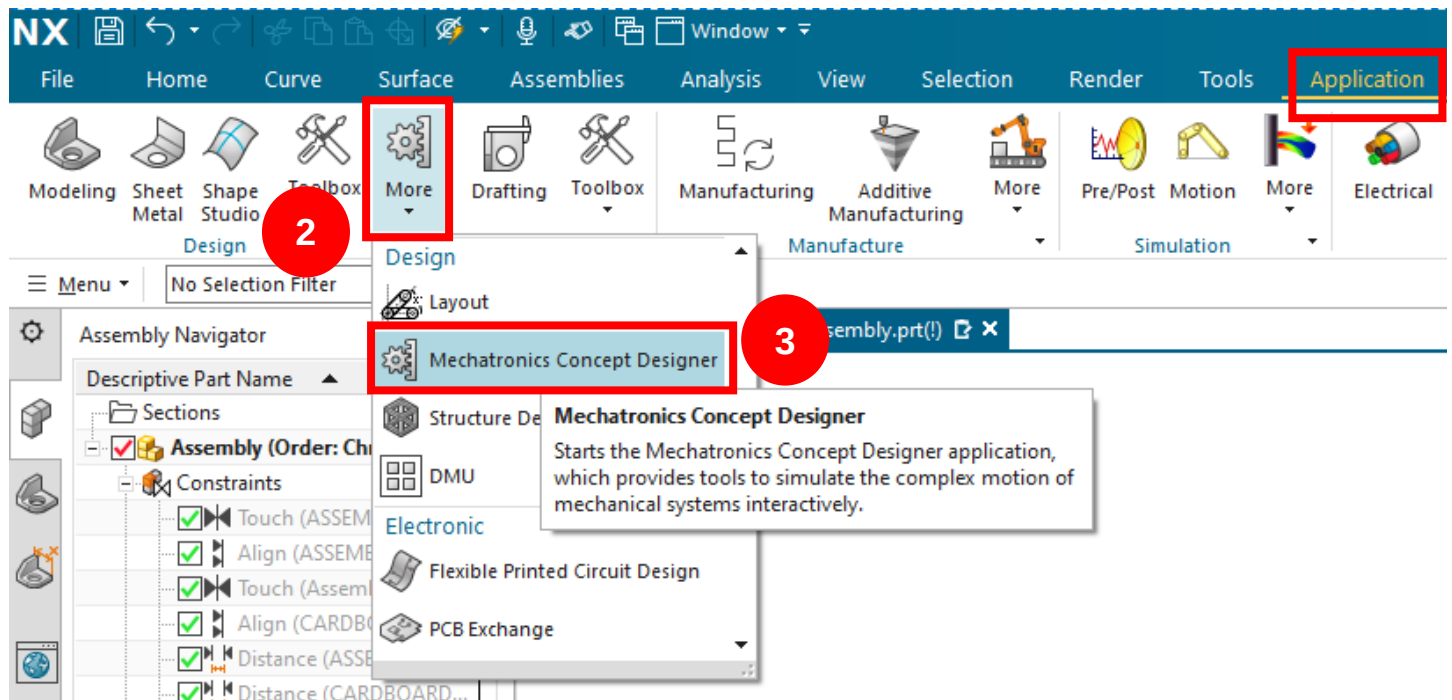
6. Once the proximity switch is moved to the correct location click **APPLY** on the **MOVE COMPONENT** window.
7. Click **X** in the top right corner to finish the operation.



Switching from Siemens NX (Modelling) to Siemens NX MCD environment

1. Click on the **APPLICATION** tab.
2. Click on the **MORE** button in the **DESIGN** area.
3. Click on the **MECHATRONICS CONCEPT DESIGNER** button.
4. This will change the **TOOLBAR** to have the relevant functions for NX MCD.

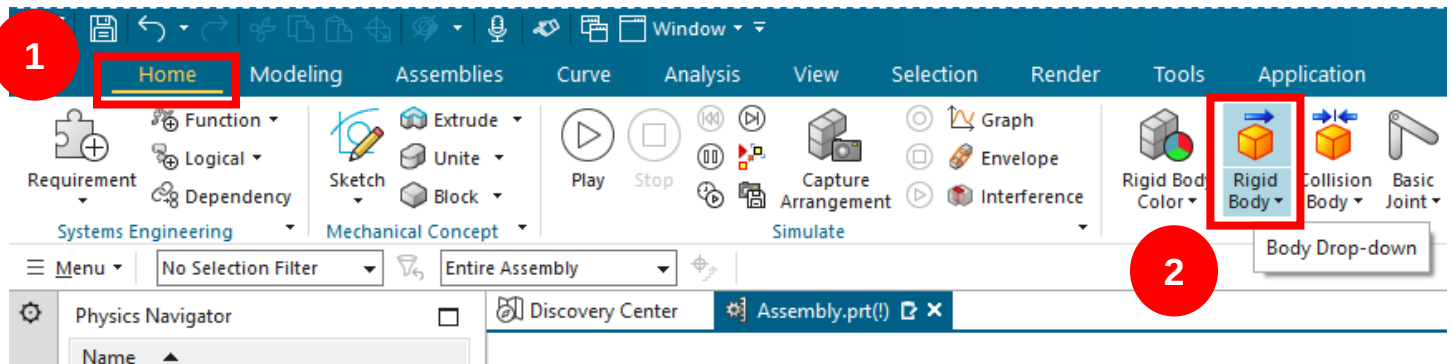
1



Defining the Cardboard Box as a Rigid Body

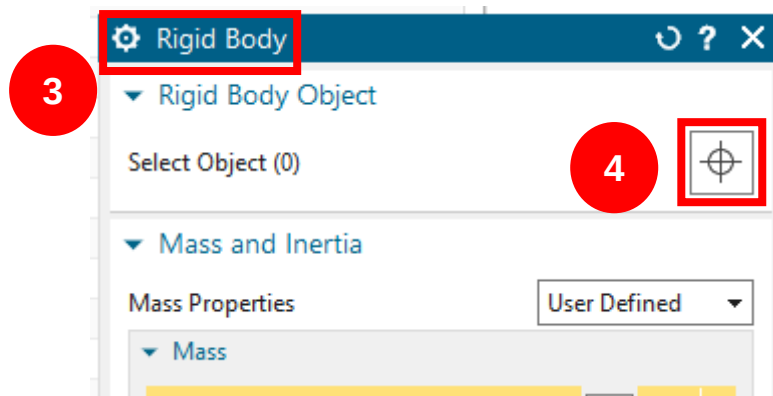
1. Click on the **HOME** tab.
2. Click on the **RIGID BODY** button in the **MECHANICAL** area.

1

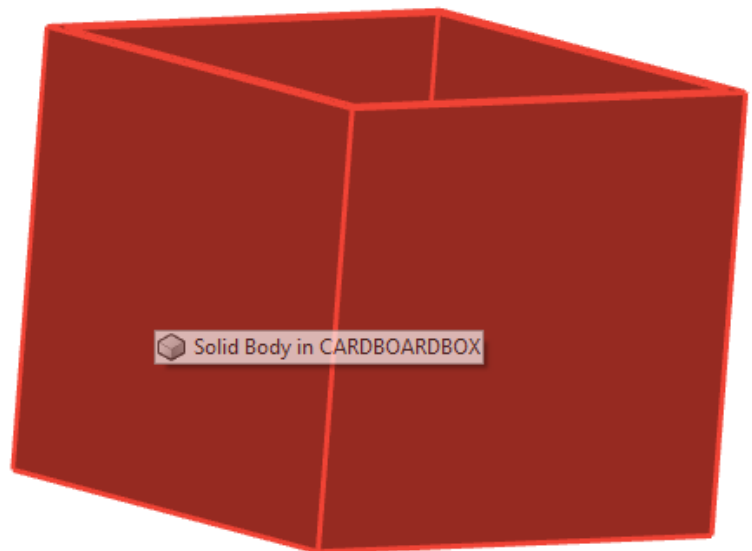
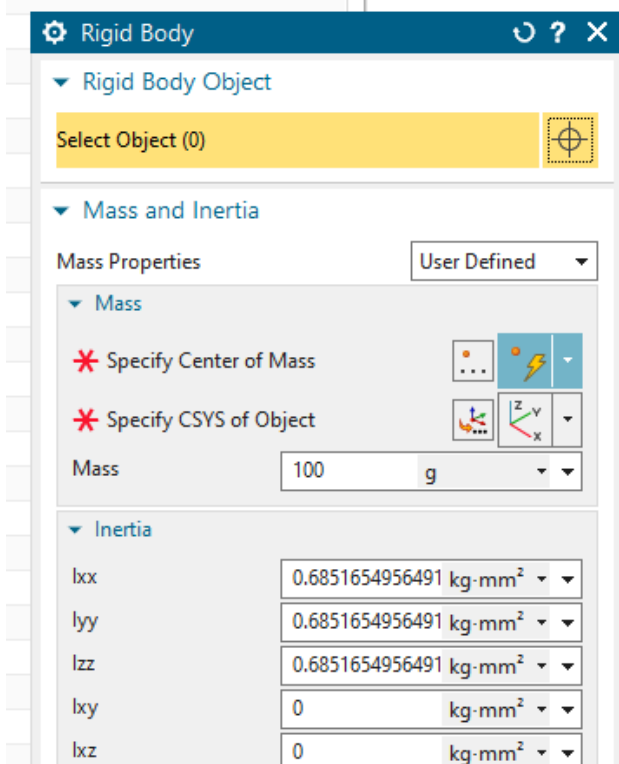


2

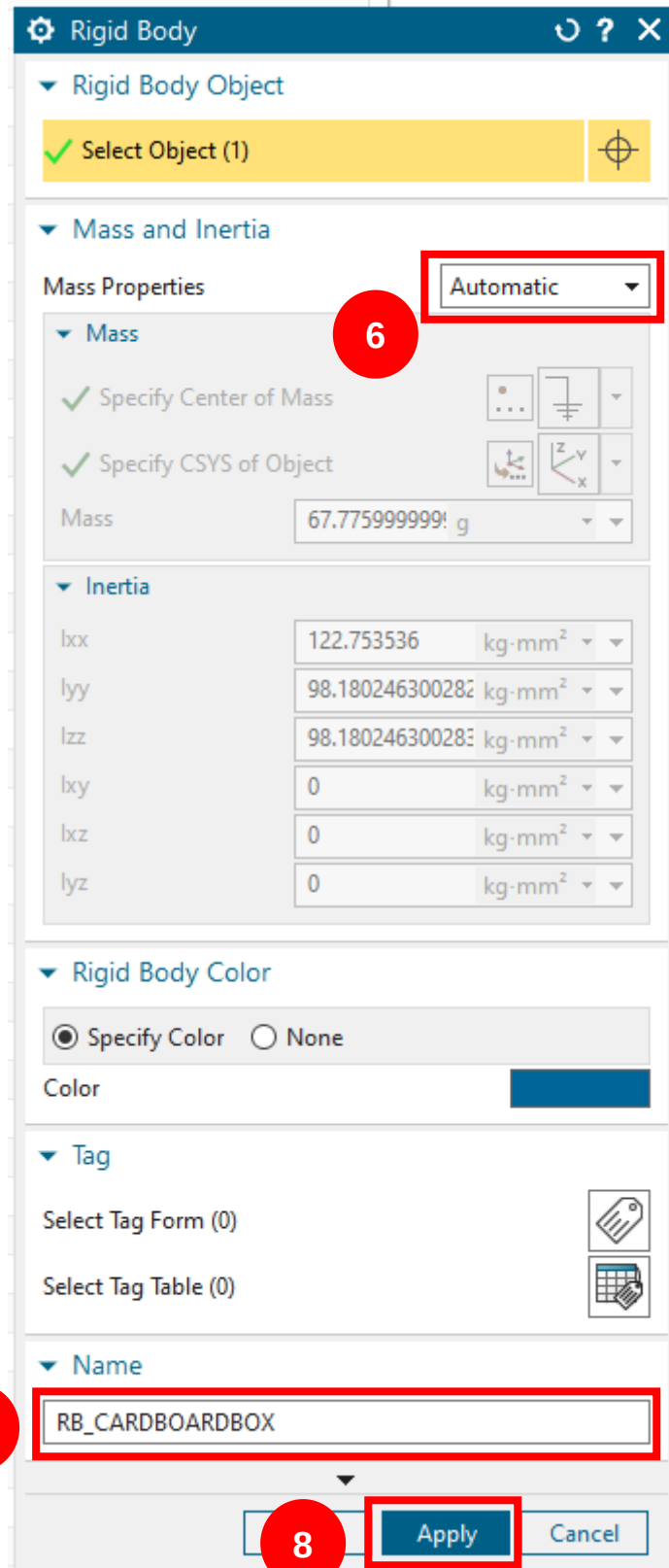
3. A new window will appear called **RIGID BODY**, shown below.
4. Click the **TARGET** icon in the **RIGID BODY OBJECT** tab.



5. Click on the **CARDBOARD BOX** to select it.

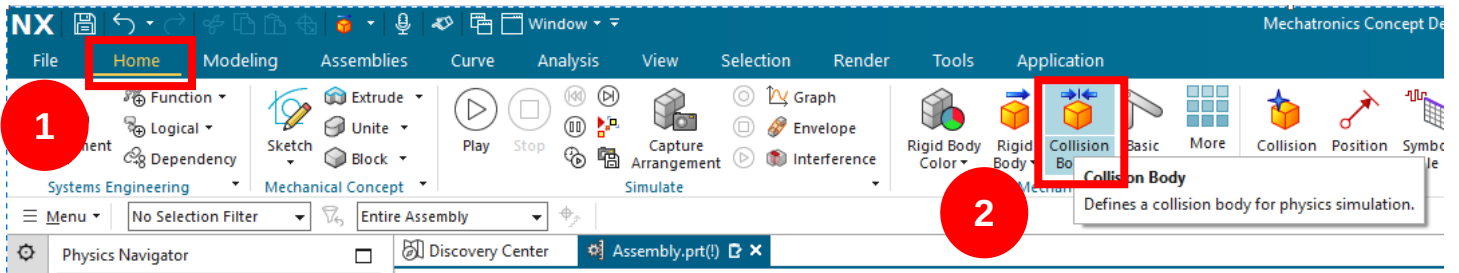


6. Ensure that **MASS PROPERTIES** are set to **AUTOMATIC**.
7. Enter a **NAME** for the Rigid Body in the **RIGID BODY** window, i.e. **RB_CARDBOARDBOX**.
8. Click **APPLY**.
9. Click **X** in the top right corner to finalise the operation.

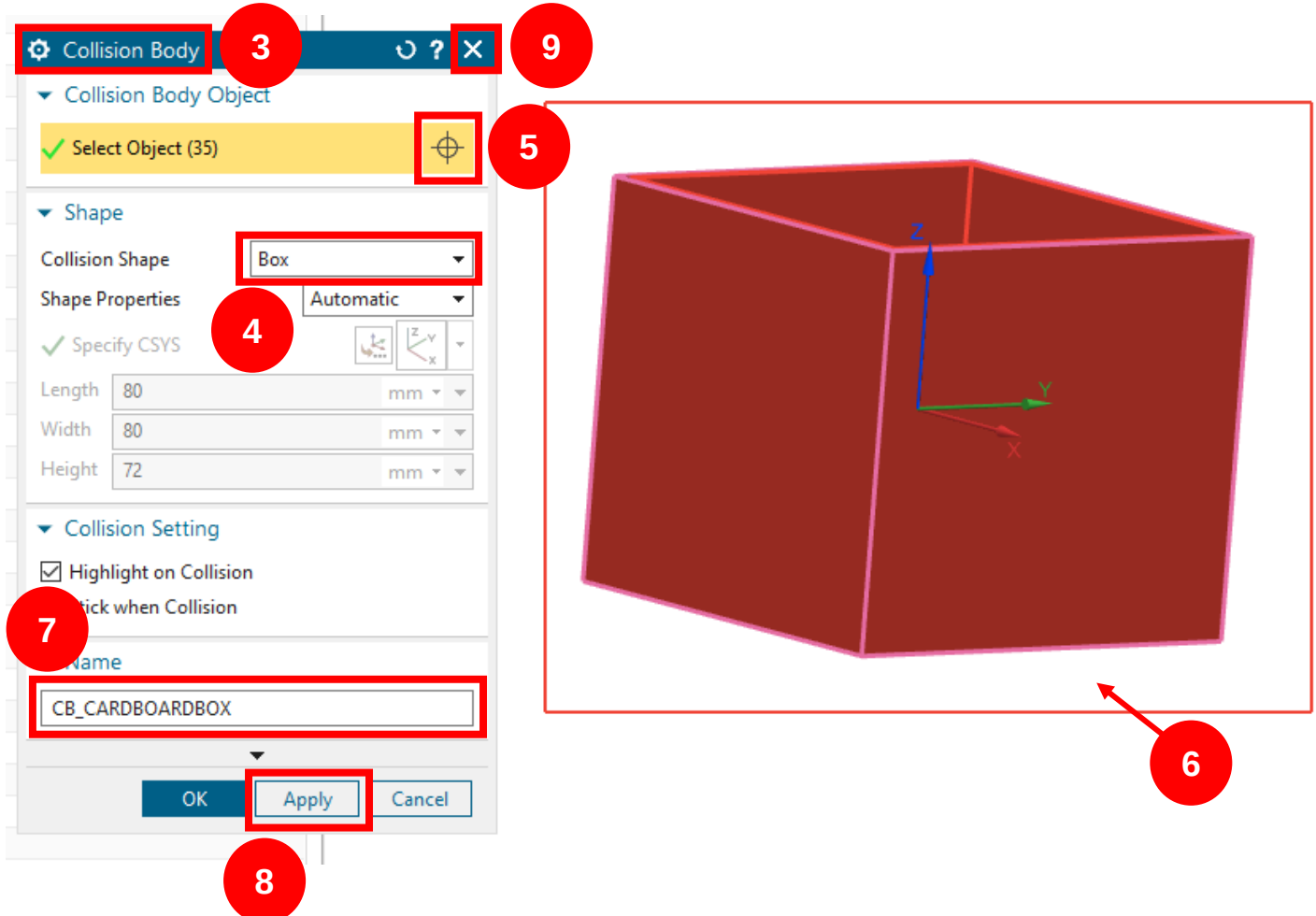


Defining the Cardboard Box as a Collision Body

1. Click on the **HOME** tab.
2. Click on the **COLLISION BODY** button in the **MECHANICAL** area.

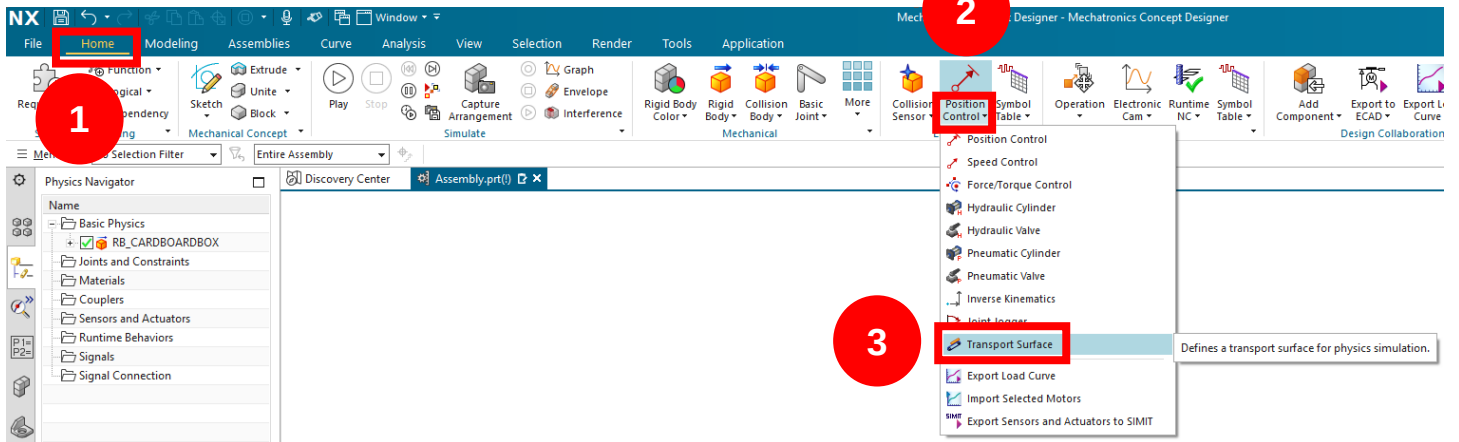


3. A new window will appear called **COLLISION BODY**, shown below.
4. Change the **COLLISION SHAPE** in the **SHAPE** tab to **BOX**.
5. Click the **TARGET** icon in the **COLLISION BODY OBJECT** tab.
6. Click and drag the **SELECTION** box so that it covers the **CARDBOARD BOX**, so that it is selected.
7. Enter a **NAME** for the Collision Body, i.e. **CB_CARDBOARDBOX**
8. Click **APPLY**.
9. Click **X** in top right corner to finalise the operation.

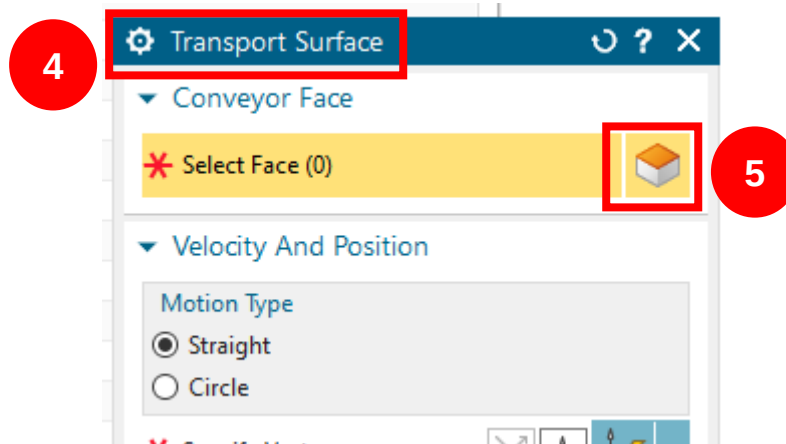


Defining the Conveyor Belt as a Transport Surface

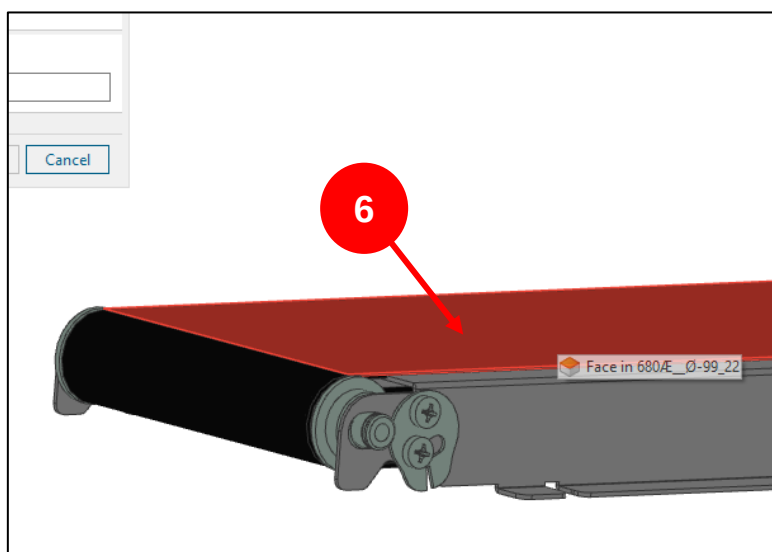
1. Click on the **HOME** tab.
2. Click on the **POSITION CONTROL EXPAND** button in the **ELECTRICAL** area.
3. Click on the **TRANSPORT SURFACE** dialog.



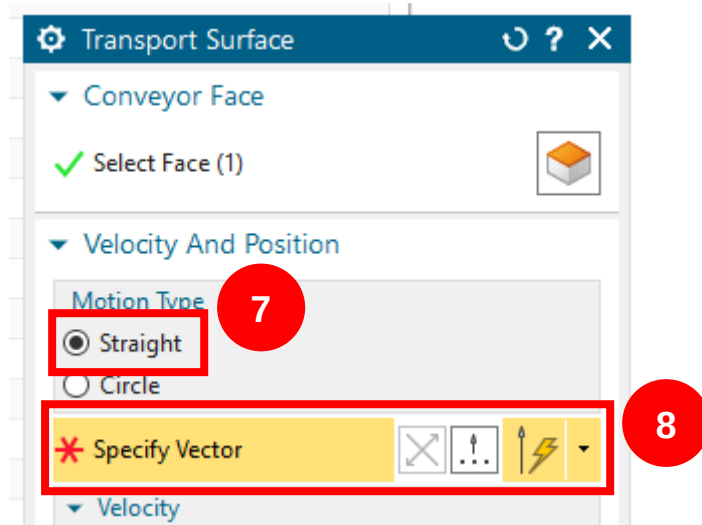
4. A new window will appear called **TRANSPORT SURFACE**, shown below.
5. Click the **BOX** icon in the **CONVEYOR FACE** tab.



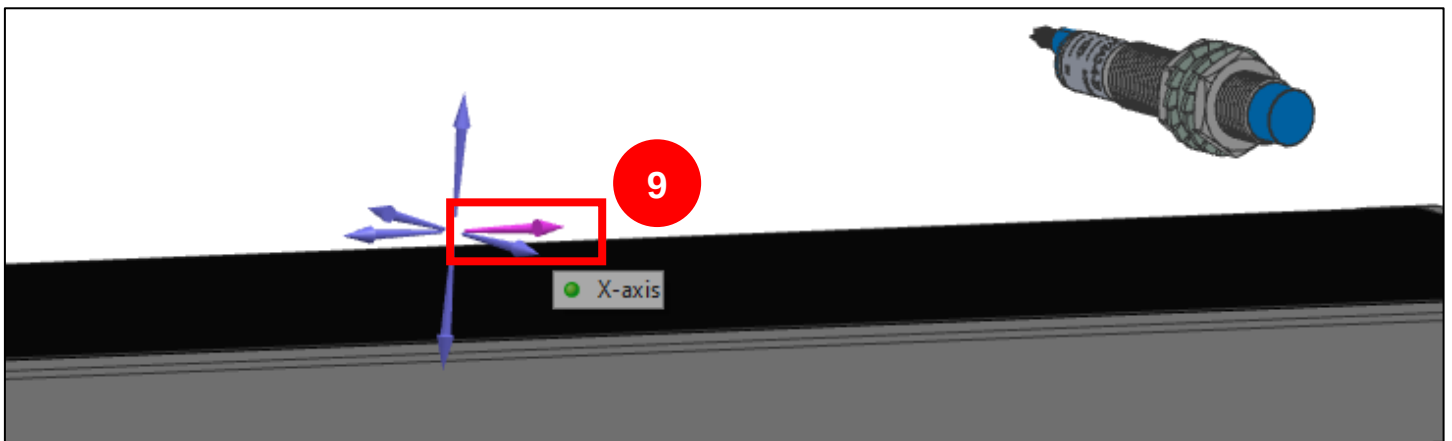
6. Click on the top of the **CONVEYOR BELT** to define this as the **TRANSPORT SURFACE**.



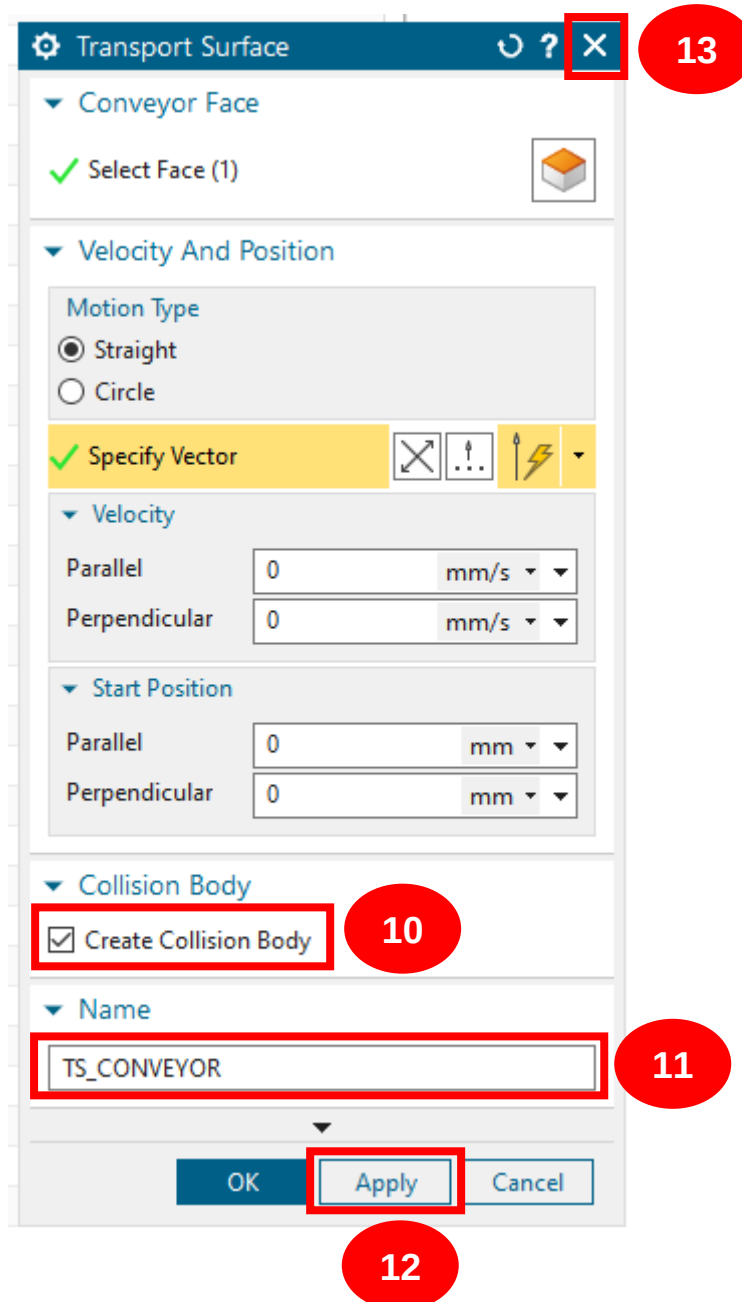
7. Click **STRAIGHT** for **MOTION TYPE**.
8. Click **SPECIFY VECTOR**, this will provide a **DIRECTIONAL ARROW** in the main workspace.



9. Select the **DIRECTION** that is toward the proximity sensor.

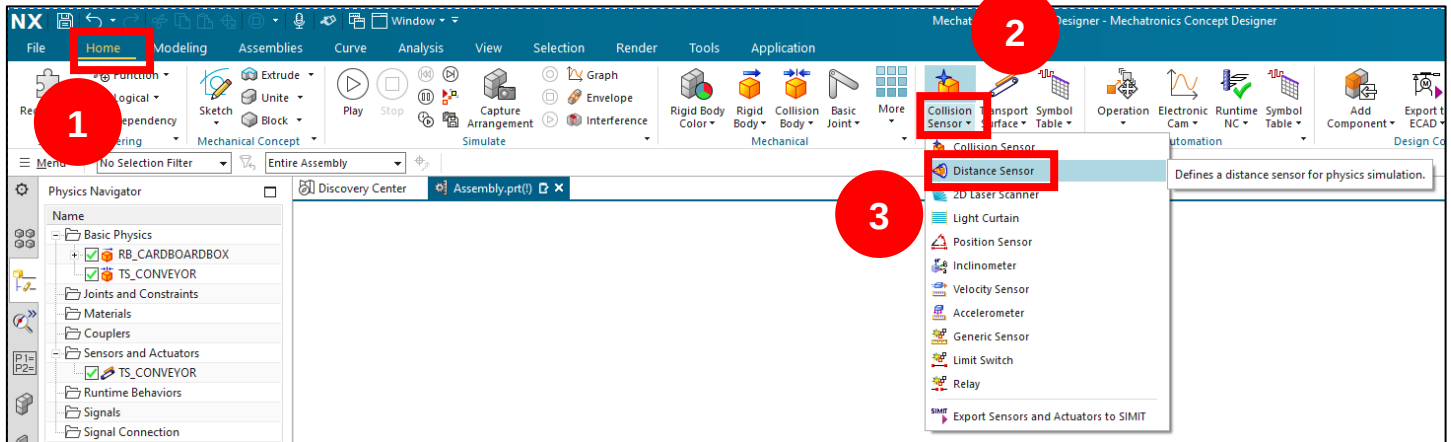


10. Ensure that the **CREATE COLLISION BODY** box is **TICKED**.
11. Enter a **NAME** for the **TRANSPORT SURFACE**, i.e. **TS_CONVEYOR**.
12. Click **APPLY**.
13. Click **X** in the top right corner to finalise the operation.

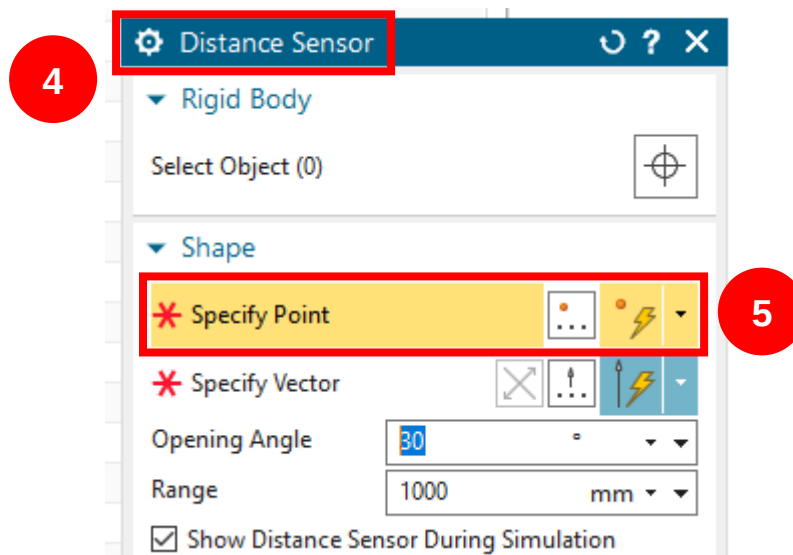


Defining the Proximity Sensor as a Distance Sensor

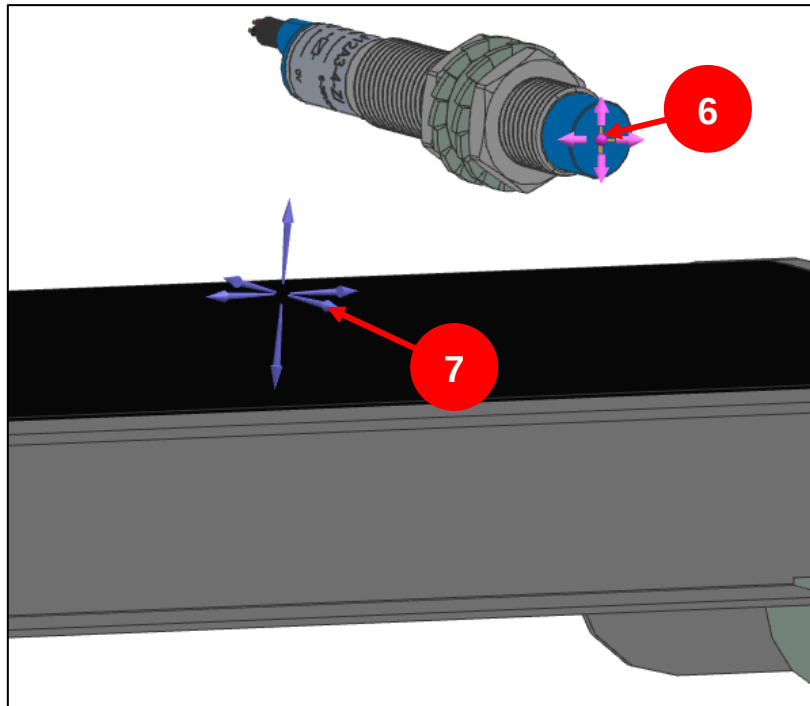
1. Click on the **HOME** tab.
2. Click on the **COLLISION SENSOR EXPANDED** button in the **ELECTRICAL** area.
3. Click on the **DISTANCE SENSOR** dialog.



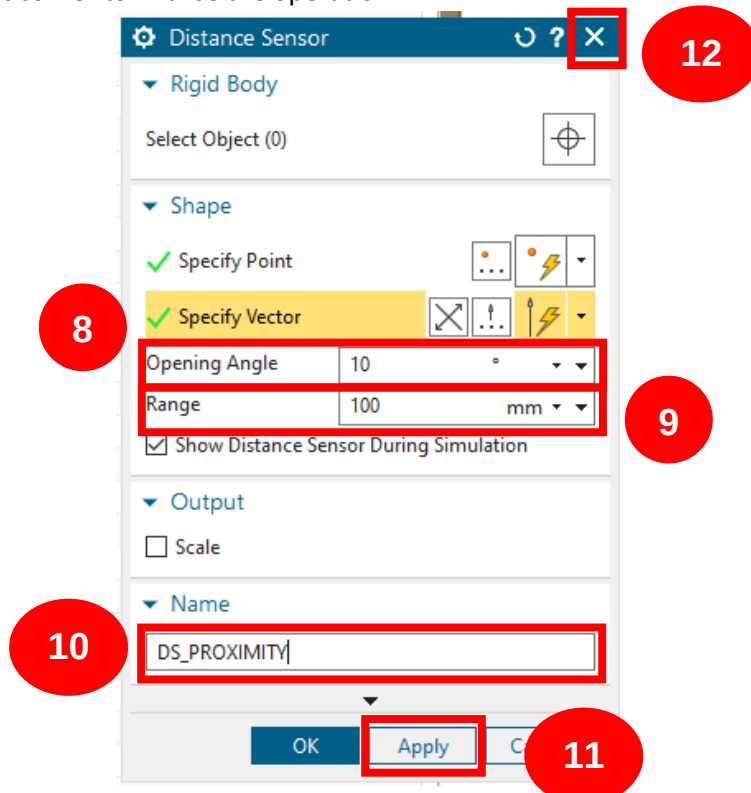
4. A new window will appear called **DISTANCE SENSOR**, shown below.
5. Click **SPECIFY POINT** to select where the sensor will originate from.



6. Click the **MIDDLE** of the proximity sensor face. A **HOVER** point will appear when the mouse pointer is over it.
7. A **DIRECTION ARROW** will appear, use this to select the arrow away from the proximity sensor.



8. Enter an **OPENING ANGLE**, this can usually be found on the data sheet of the device. For this application we will set this 10°.
9. Enter a **RANGE** in mm, this can also usually be found on the data sheet of the device. For this application we will set this to 100mm.
10. Enter a **NAME** for the **DISTANCE SENSOR**, i.e. **DS_PROXIMITY**.
11. Click **APPLY**.
12. Click **X** in the top right corner to finalise the operation.

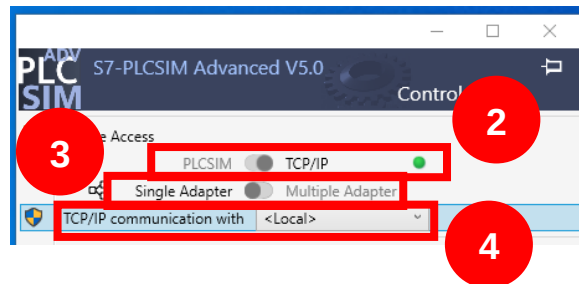


Start the simulated PLC via PLCSim Advanced V5.0

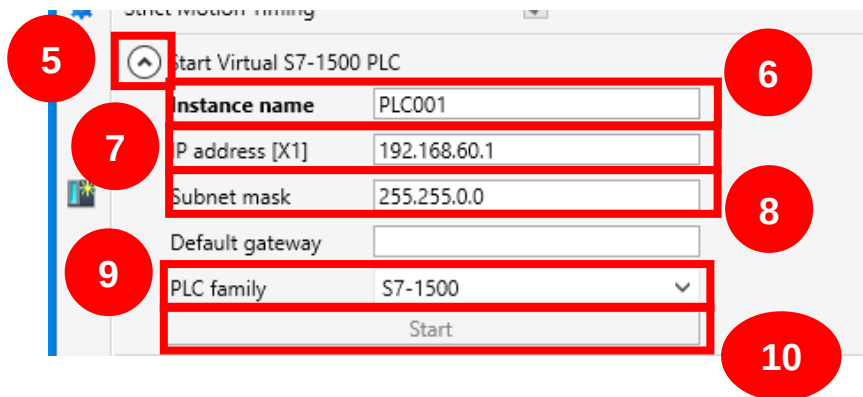
1. Open **PLCSIM ADVANCED V5.0** by clicking on the icon in the Windows taskbar as shown below.



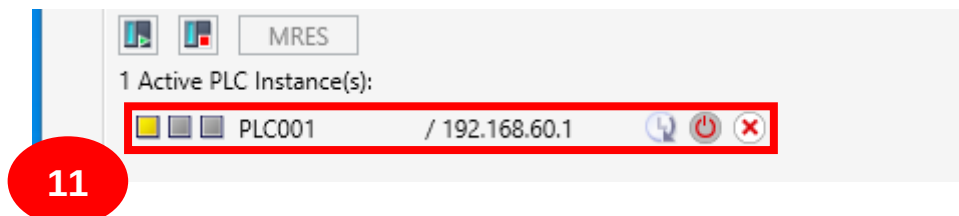
2. Set the **ONLINE ACCESS** to **TCP/IP**.
3. Set the **ADAPTER** to **SINGLE ADAPTER**.
4. Set the **TCP/IP COMMUNICATION** to **LOCAL**.



5. Expand the **START VIRTUAL S7-1500 PLC** section.
6. Enter a **PLC INSTANCE NAME**, for this application set this to **PLC001**.
7. Enter an **IP ADDRESS [X1]**, for this application set this to **192.168.60.1**
8. Enter a **SUBNET MASK**, for this application set this to **255.255.0.0**
9. Ensure that the **PLC FAMILY** is set to **S7-1500**.
10. Click **START**.

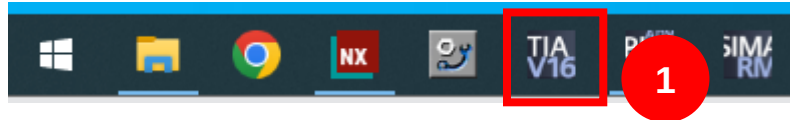


11. The **PLC001 INSTANCE** will start but in **STOP MODE**. This will be started later.

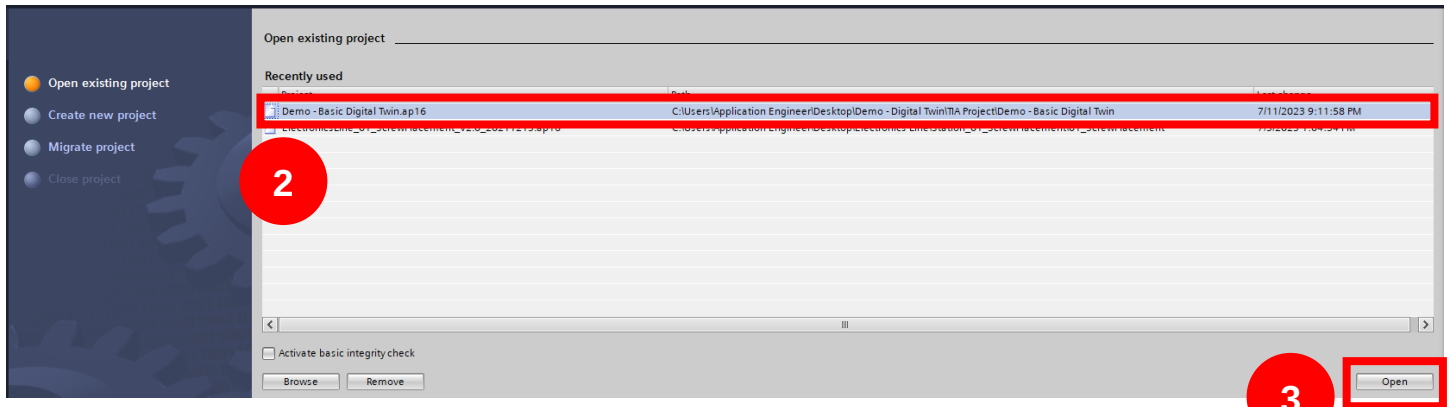


Start TIA Portal V16 and open the Pre-Engineered Program

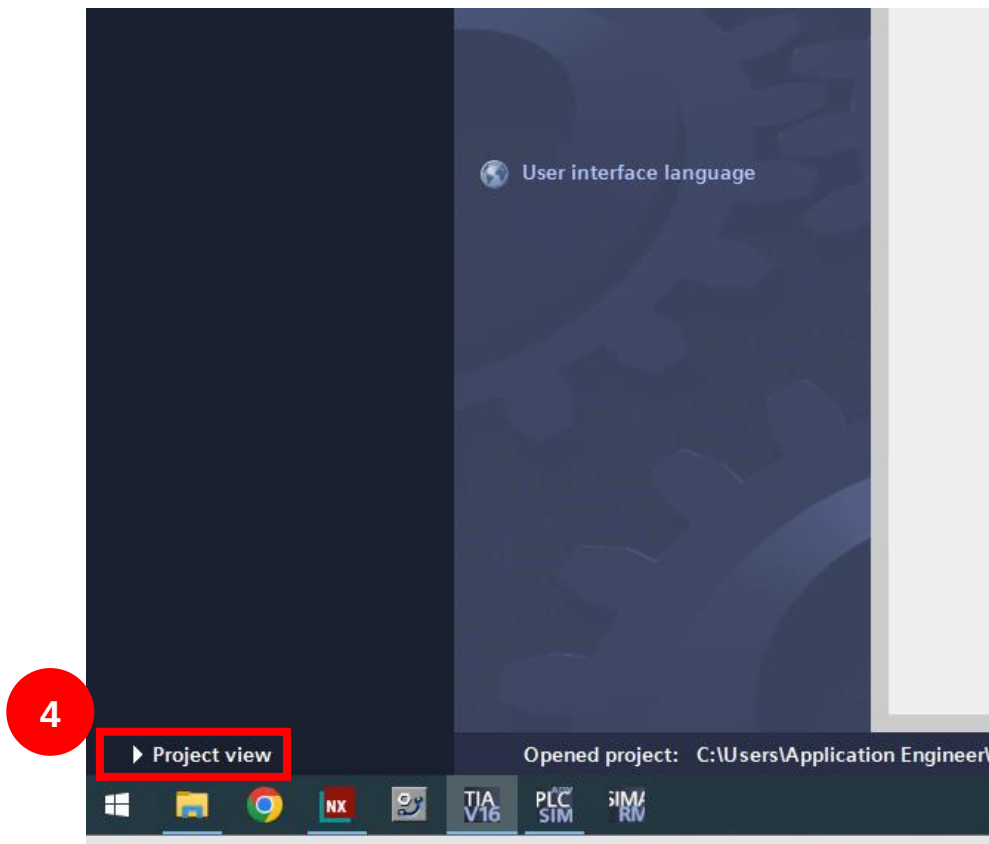
1. Open **TIA PORTAL V16.0** by clicking on the icon in the Windows taskbar as shown below.



2. Select the **DEMO – BASIC DIGITAL TWIN.AP16** project from the **RECENTLY USED**.
3. Click **OPEN**.

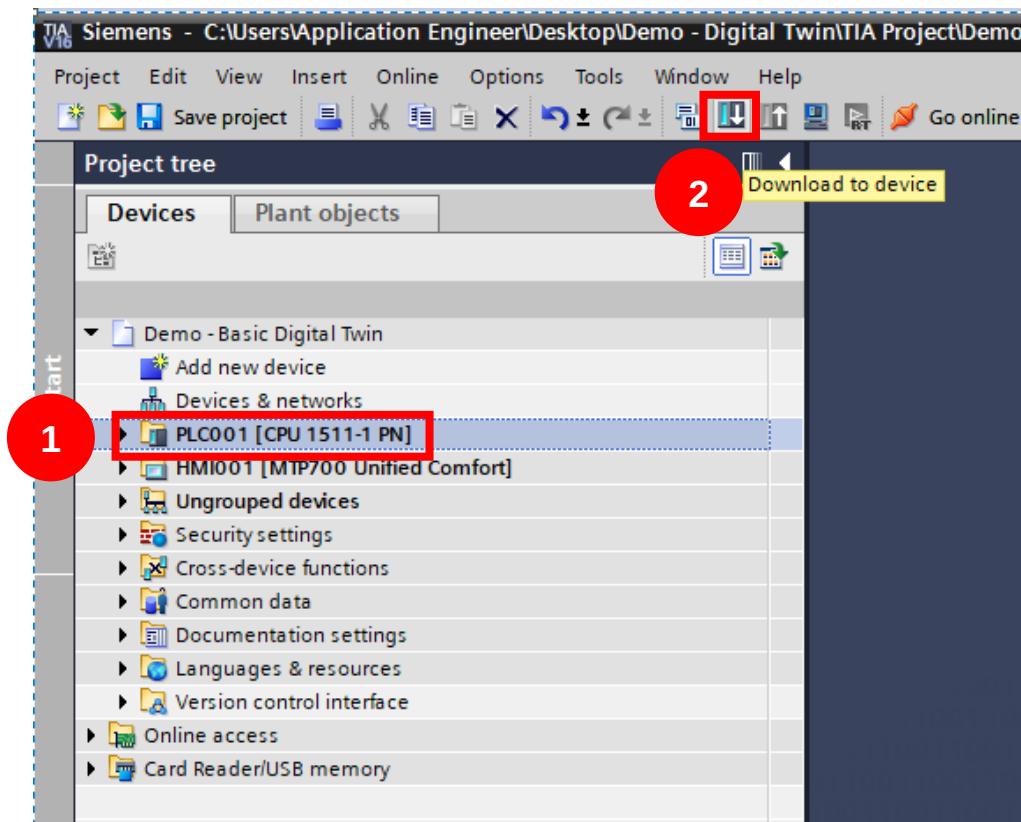


4. Click **PROJECT VIEW** in bottom left corner of window.

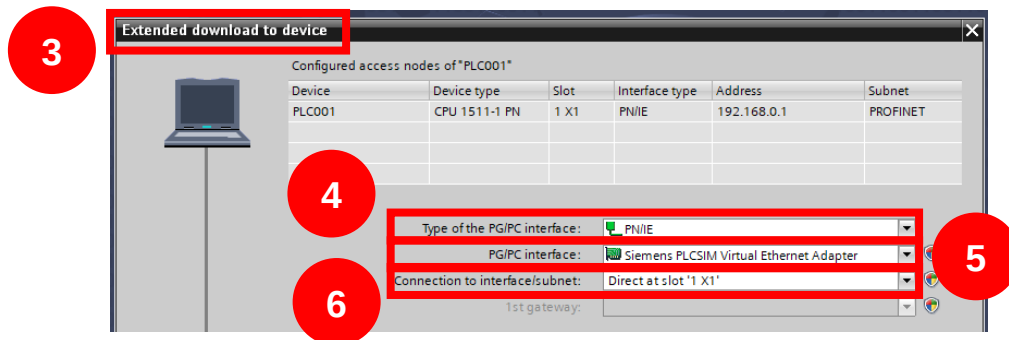


Download the Pre-Engineered PLC Program to PLCSim Advanced

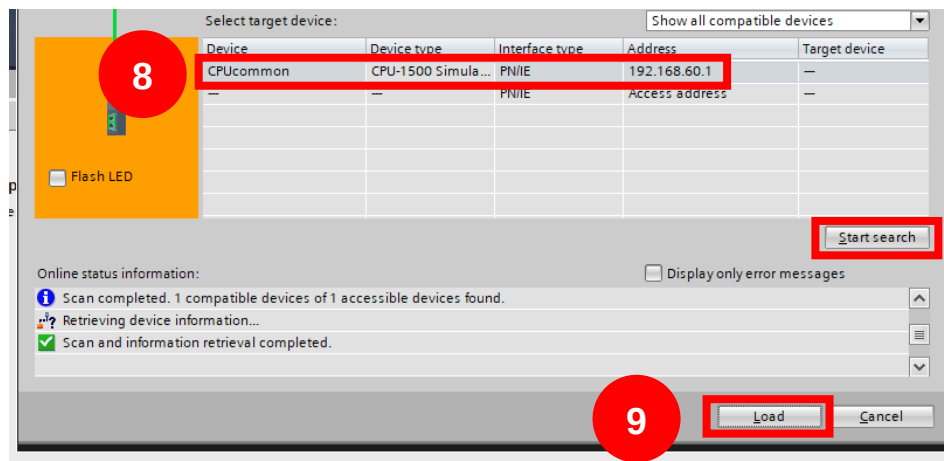
1. Highlight **PLC001** by clicking on the device in the TIA Portal project tree.
2. Click the **DOWNLOAD** icon in the **TOOLBAR**.



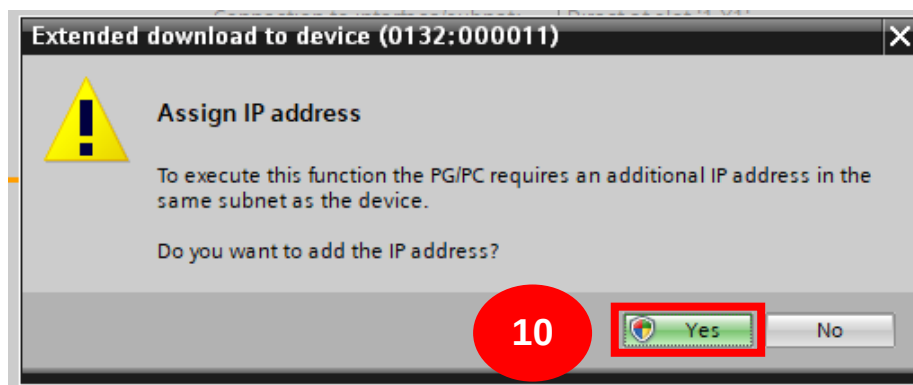
3. The **EXTENDED DOWNLOAD TO DEVICE** window will appear.
4. Select **PN/IE** in the **TYPE OF THE PG/PC INTERFACE** dropdown box.
5. Select **SIEMENS PLCSIM VIRTUAL ETHERNET ADAPTER** from the **PG/PC INTERFACE** dropdown box.
6. Select **DIRECT AT SLOT '1 X1'** from the **CONNECTION TO INTERFACE/SUBNET** dropdown box.



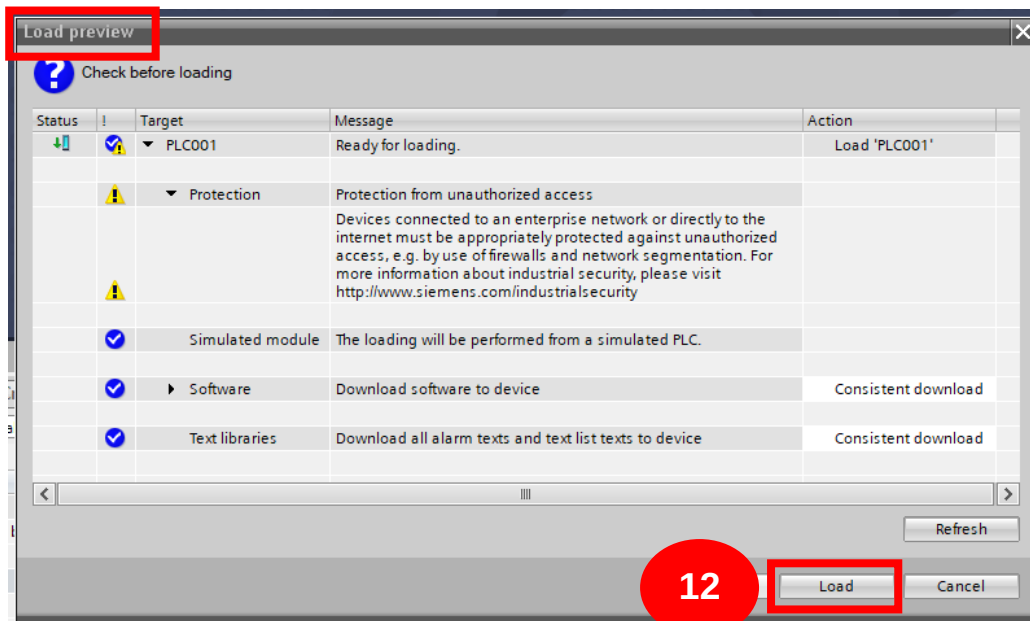
7. Select **START SEARCH** to interrogate the PLC via the set PG/PC interface.
8. Highlight the device named **CPUCommon** should appear with a matching **IP ADDRESS 192.168.60.1**
9. Click **LOAD**.



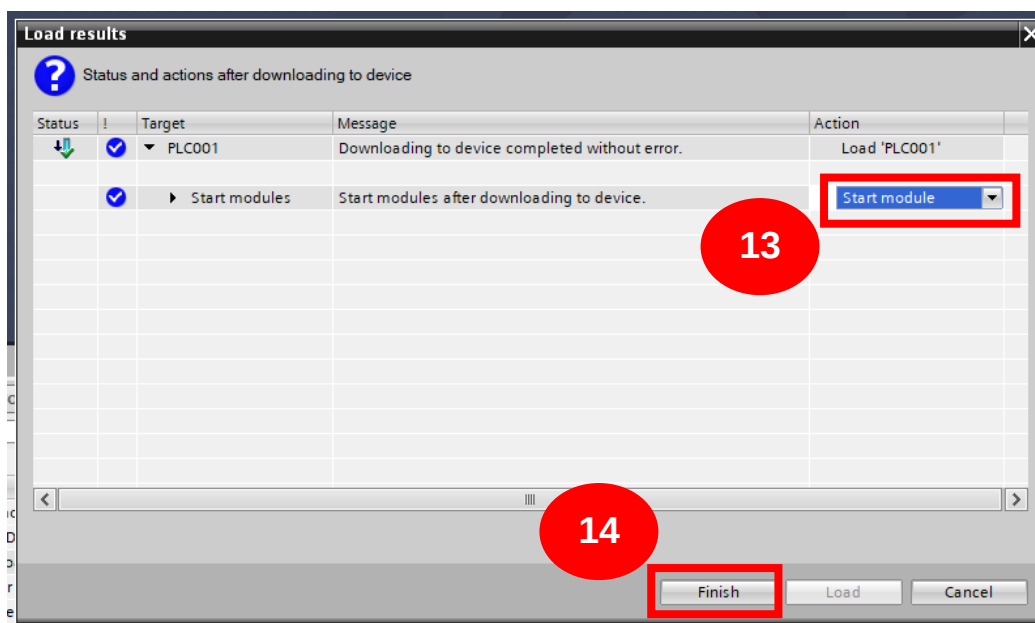
10. If a **WARNING** dialog appears asking to **ASSIGN IP ADDRESS**, click **YES**.



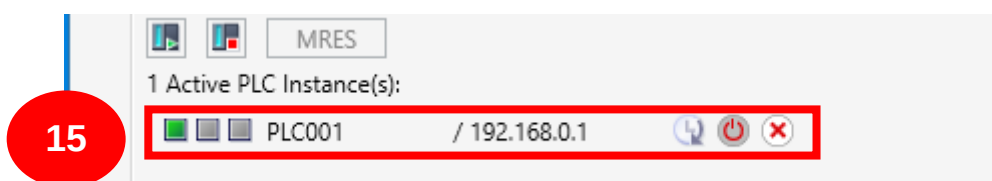
11. A dialog window will appear with a LOAD PREVIEW.
12. Click LOAD to download the program to the PLC.



13. After a small period, the download will finish. Change the **START MODULES** dropdown box to **START MODULE**.
14. Click **FINISH**.

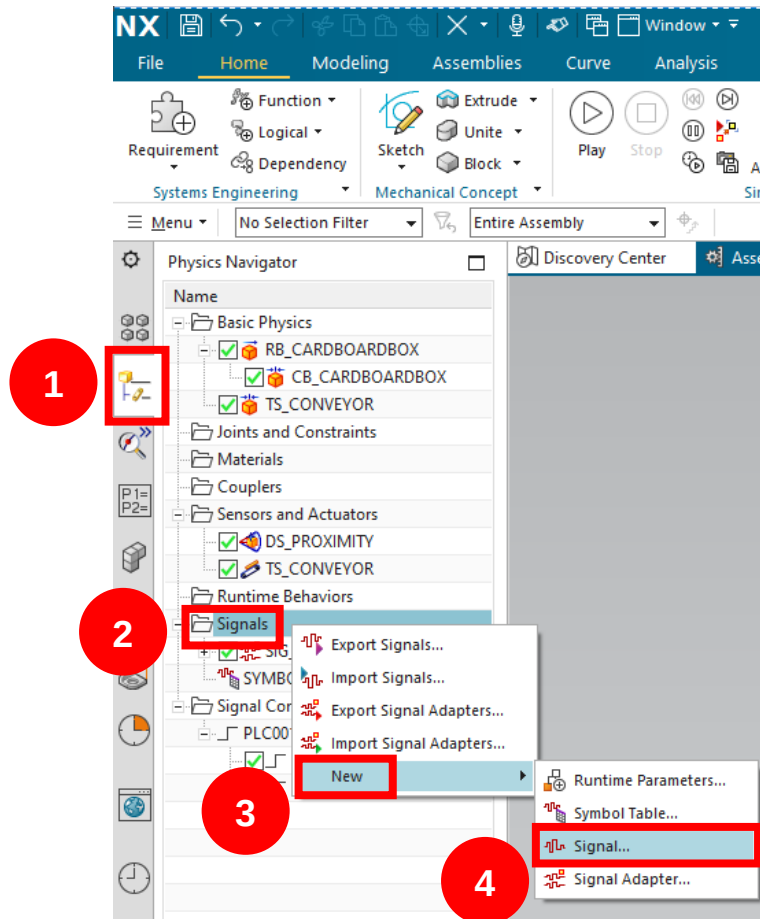


15. **PLC001** instance in PLCSim Advanced V5.0 should now be running.

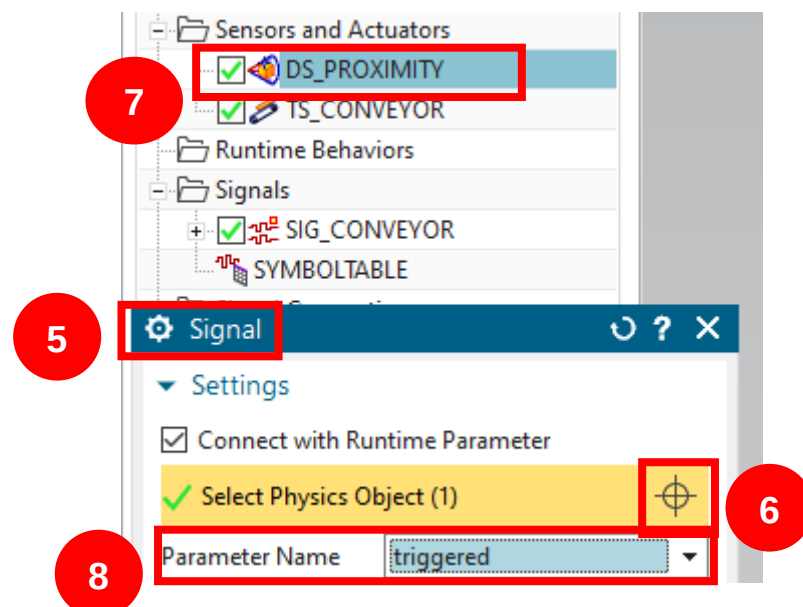


Add New Signal Adapter to DS_PROXIMITY

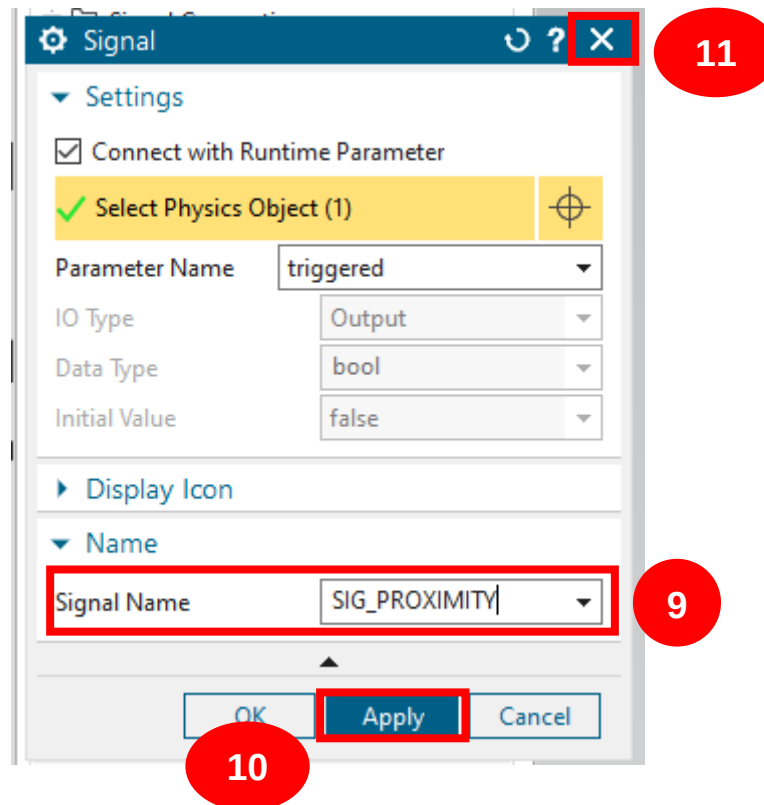
1. In Siemens NX-MCD, navigate to the **PHYSICS NAVIGATOR**.
2. Right Click on the **SIGNALS** folder.
3. Click **NEW**.
4. Click **SIGNAL**.



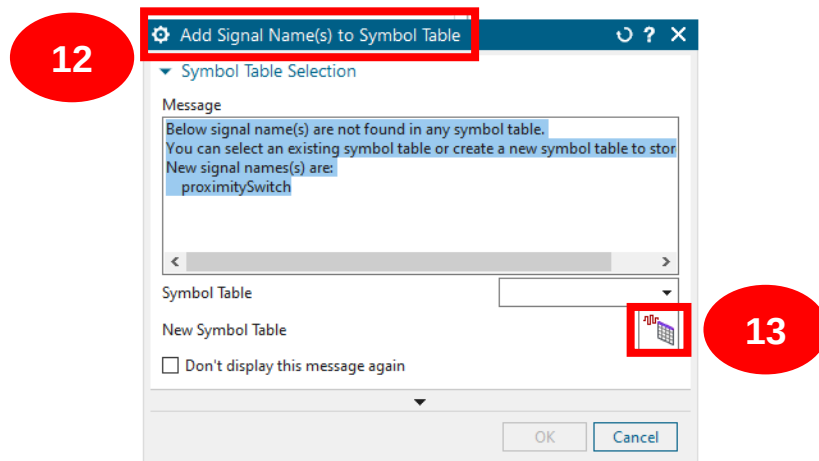
5. A new window will appear called **SIGNAL**, shown below.
6. Click the **TARGET** icon in the **PARAMETERS** tab.
7. Click **DS_PROXIMITY** in the **SENSORS AND ACTUATORS** section of the **PHYSICS NAVIGATOR**.
8. Change the **PARAMETER NAME** to **TRIGGERED**.



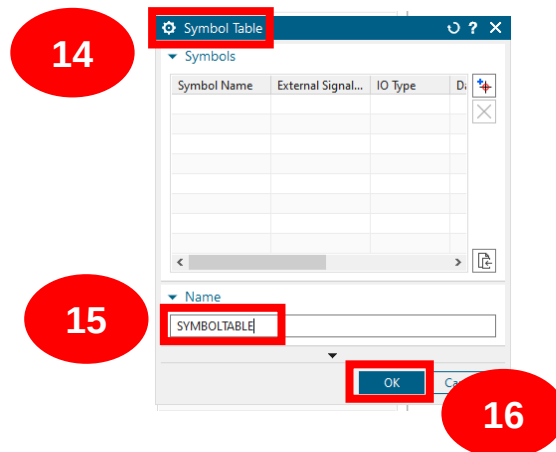
9. Enter a **NAME** for the **SIGNAL**. For this application, enter the **NAME SIG_PROXIMITY**.
10. Click **APPLY**.
11. Click **X** to finalise the operation.



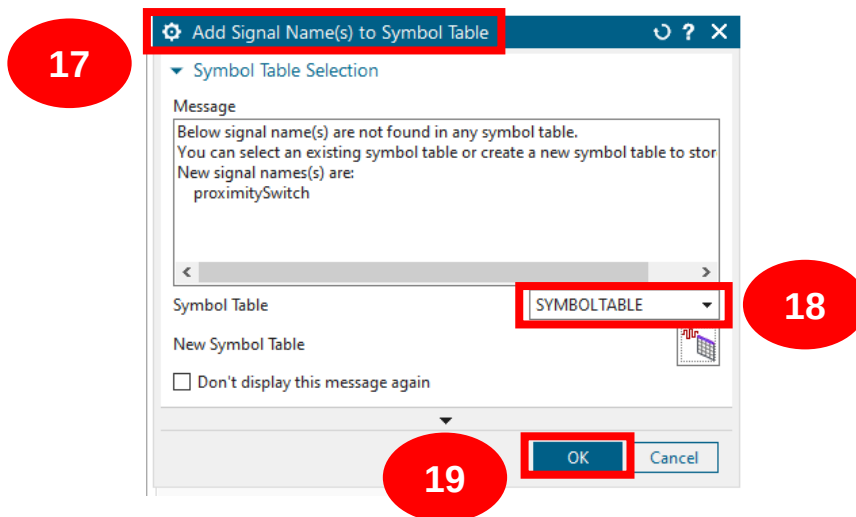
12. A new window will appear called **ADD SIGNAL NAME(S) TO SYMBOL TABLE**.
13. Click **NEW SYMBOL TABLE** icon.



14. A new window will appear called **SYMBOL TABLE**.
15. Enter a symbol table **NAME** called **SYMBOLTABLE**.
16. Click **OK**.

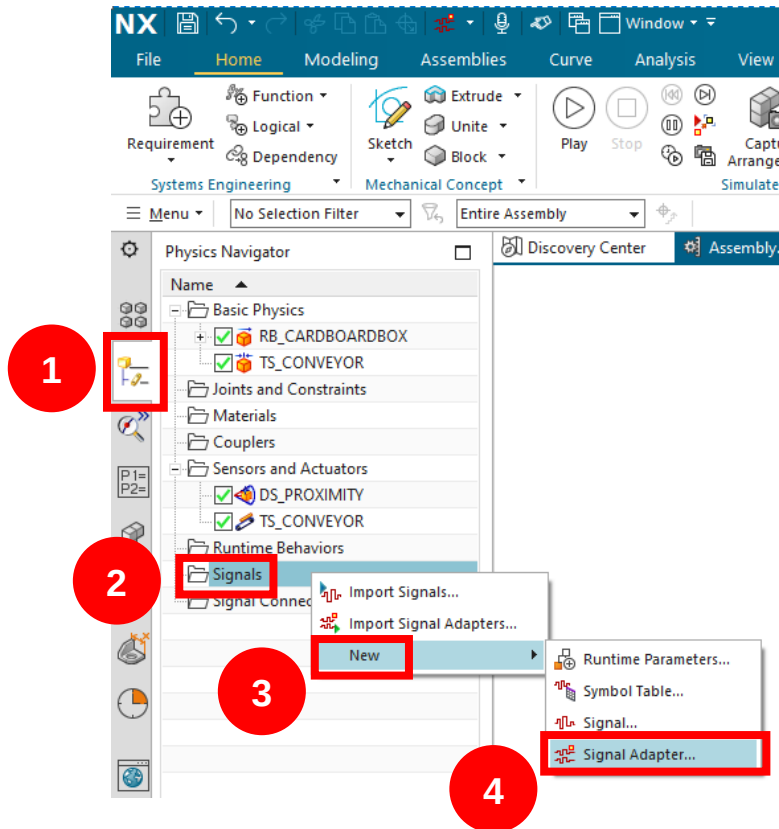


17. The **ADD SIGNAL NAME(S) TO SYMBOL TABLE** window will reappear.
18. The **SYMBOL TABLE** dropdown box will be prefilled with **SYMBOLTABLE**.
19. Click **OK**.

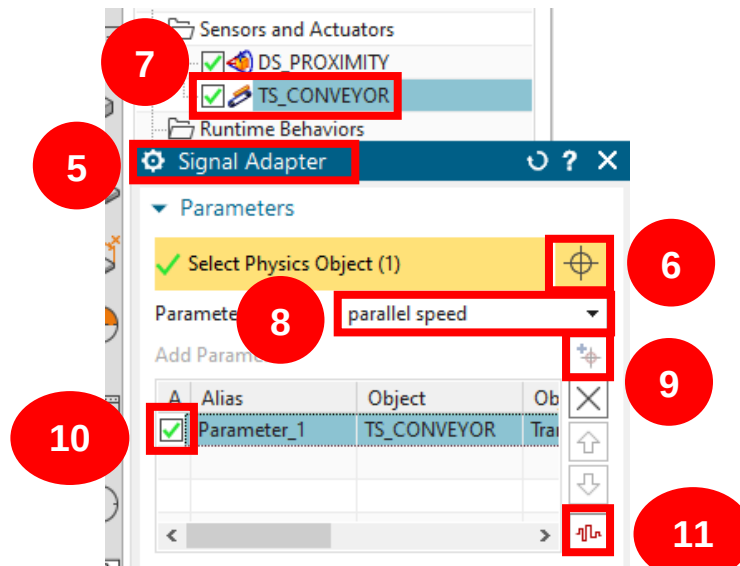


Add New Signal Adapter to TS_CONVEYOR

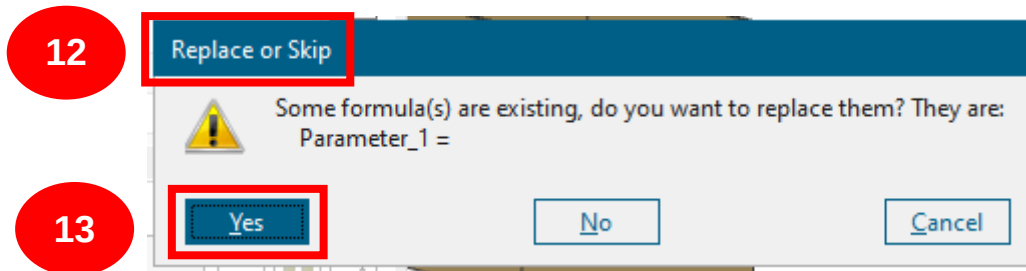
1. In Siemens NX-MCD, navigate to the **PHYSICS NAVIGATOR**.
2. Right Click on the **SIGNALS** folder.
3. Click **NEW**.
4. Click **SIGNAL ADAPTER**.



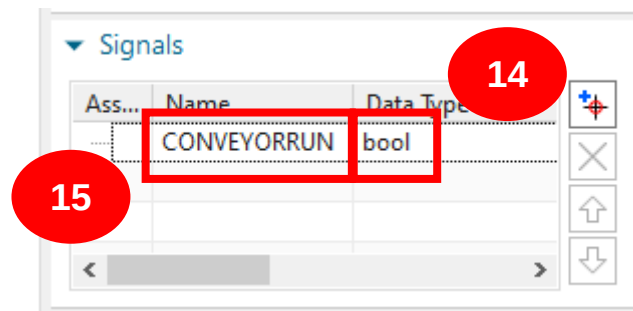
5. A new window will appear called **SIGNAL ADAPTER**, shown below.
6. Click the **TARGET** icon in the **PARAMETERS** tab.
7. Click **TS_CONVEYOR** in the **SENSORS AND ACTUATORS** section of the **PHYSICS NAVIGATOR**.
8. Change the **PARAMETER NAME** to **PARALLEL SPEED**.
9. Click the **ADD PARAMETER** icon to add **PARAMETER_1**.
10. Click the **TICK BOX** to assign the parameter.
11. Click the **GENERATE SIGNALS AND FORMULAS** icon.



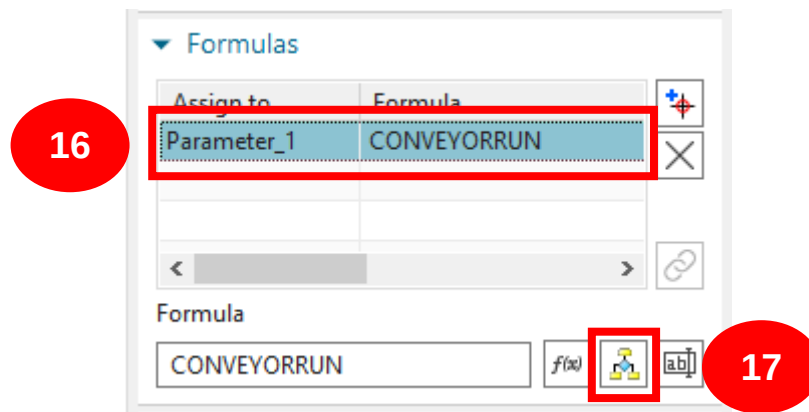
12. A warning box will appear called **REPLACE OR SKIP**.
13. Click **YES** to continue.



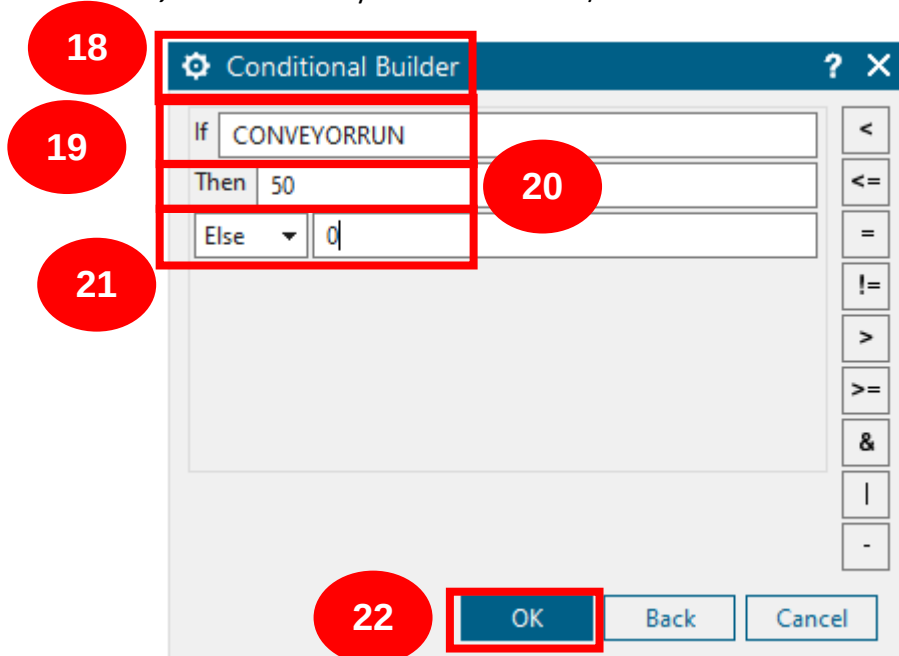
14. In the **SIGNALS** section, click on the **DATA TYPE** to change to **BOOL** using the dropdown box.
15. Click on the **NAME** and change to **CONVEYORRUN**.



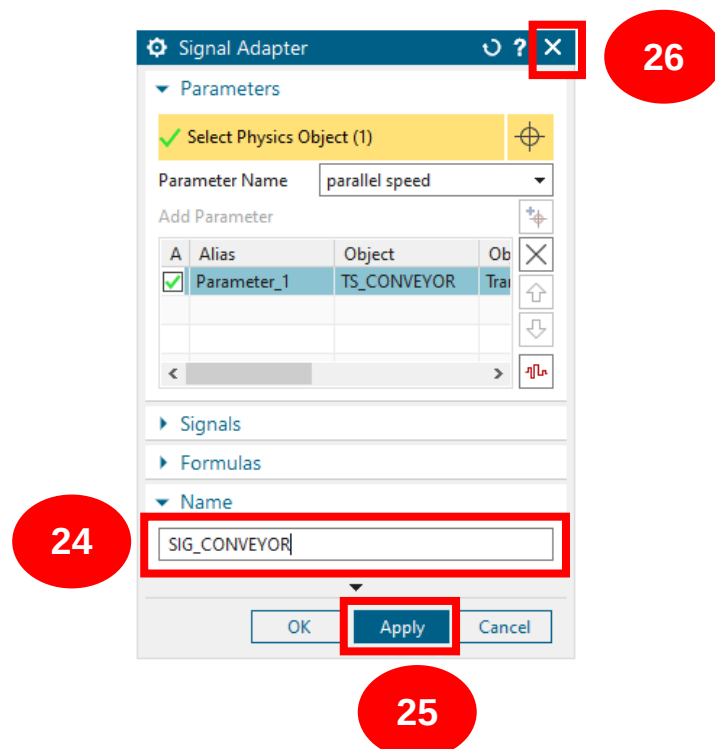
16. In the **FORMULAS** section, click on the generated formula to **HIGHLIGHT**.
17. Click on the **ADD CONDITIONAL** icon to define the **FORMULA**.



18. A new window will appear called **CONDITIONAL BUILDER**.
19. In the **IF** dialog box, enter the parameter **CONVEYORRUN**.
20. In the **THEN** dialog box, enter a value of **50**.
21. In the **ELSE** dialog box, enter a value of **0**.
22. Click **OK** to complete the operation.
23. This formula essentially looks at the parameter **CONVEYORRUN** and if it's **TRUE** then the conveyor will run at 50mm/s. If its **FALSE**, then the conveyor will run at 0mm/s or **STOP**.

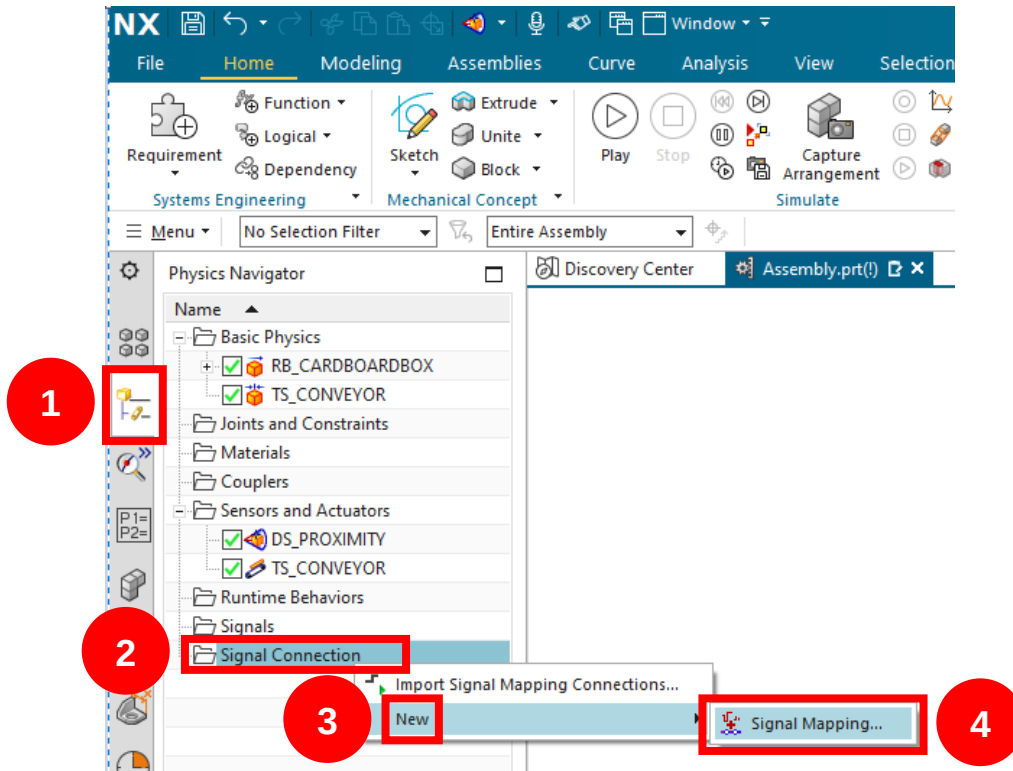


24. Enter a **NAME** for the signal adapter, for this application name it **SIG_CONVEYOR**.
25. Click **APPLY**.
26. Click **X** in the top right corner to complete the operation.

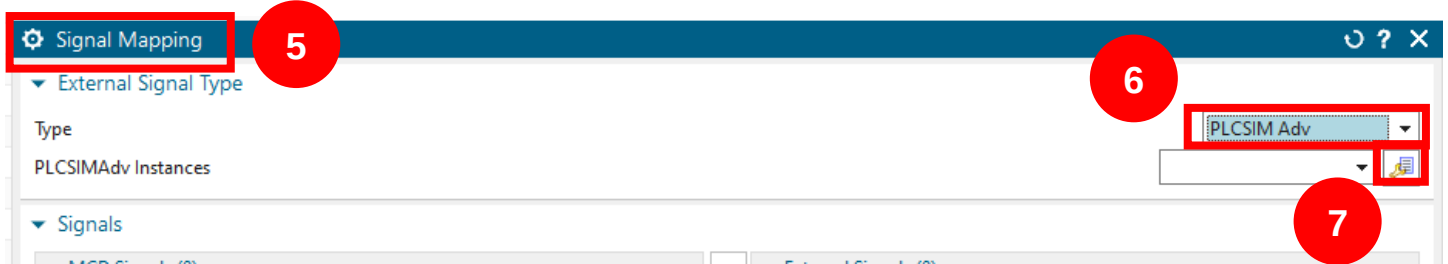


Add Signal Connection to Signal Adapter

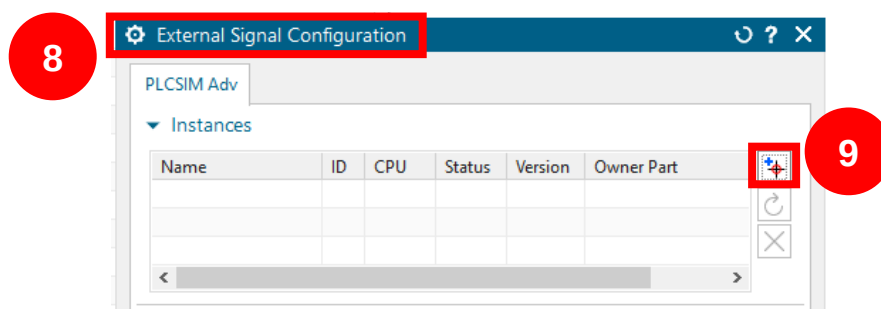
1. In Siemens NX-MCD, navigate to the **PHYSICS NAVIGATOR**.
2. Right Click on the **SIGNAL CONNECTION** folder.
3. Click **NEW**.
4. Click **SIGNAL MAPPING**.



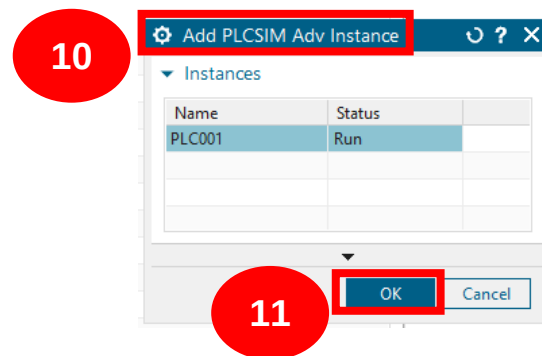
5. A new window will appear called **SIGNAL MAPPING**.
6. Select **PLCSIM ADV** from the **EXTERNAL SIGNAL TYPE** dropdown box.
7. Click the **CONNECTION SETTING** icon.



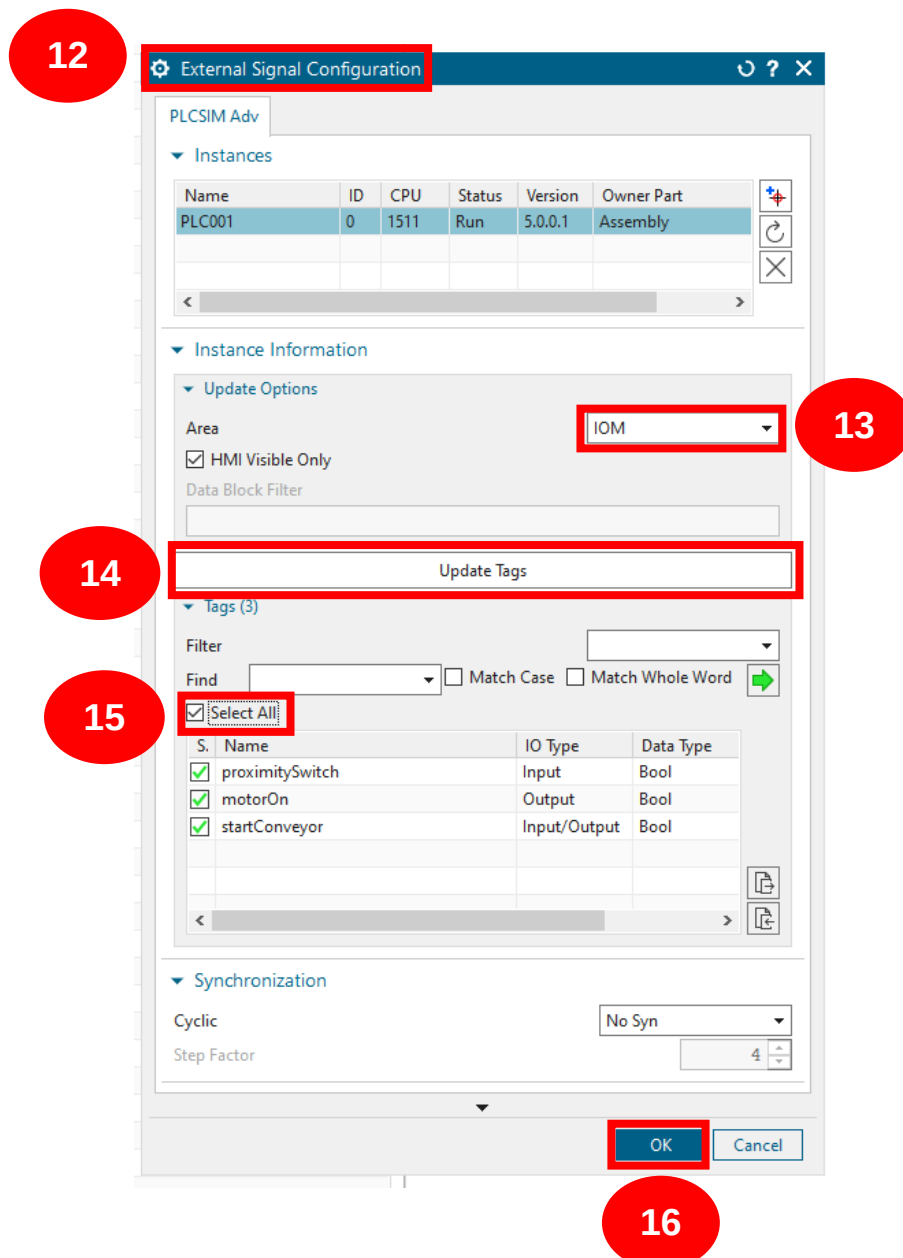
8. A new window will appear called **EXTERNAL SIGNAL CONFIGURATION**.
9. Click the **ADD INSTANCE** icon.



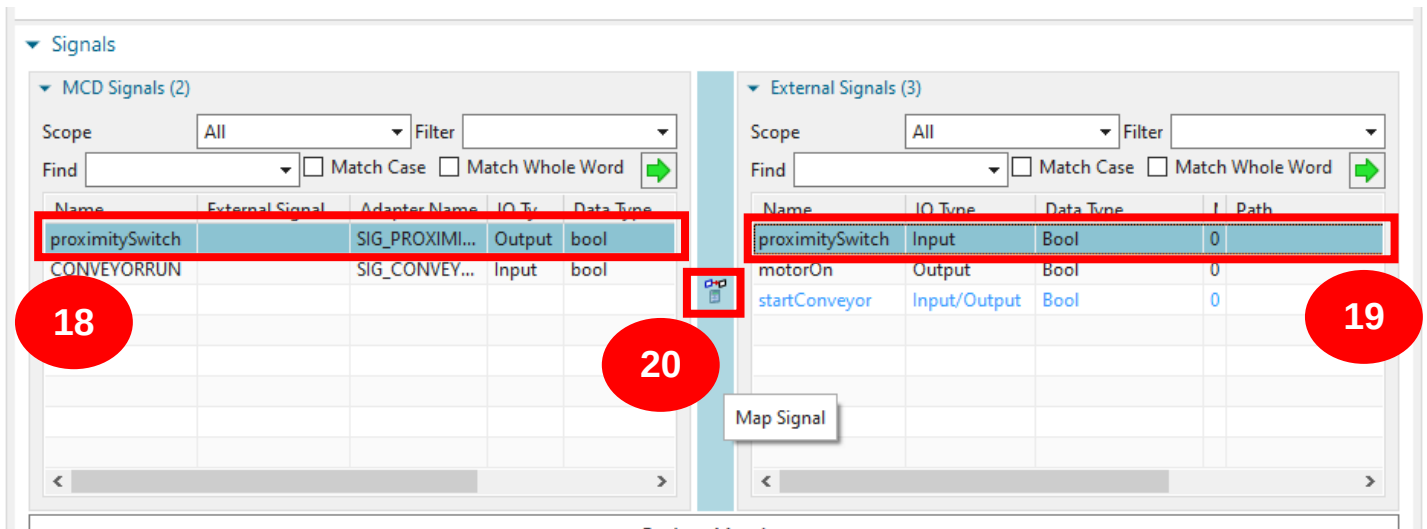
10. A new window will appear called **ADD PLCSIM ADV INSTANCE**.
11. Click **OK**.



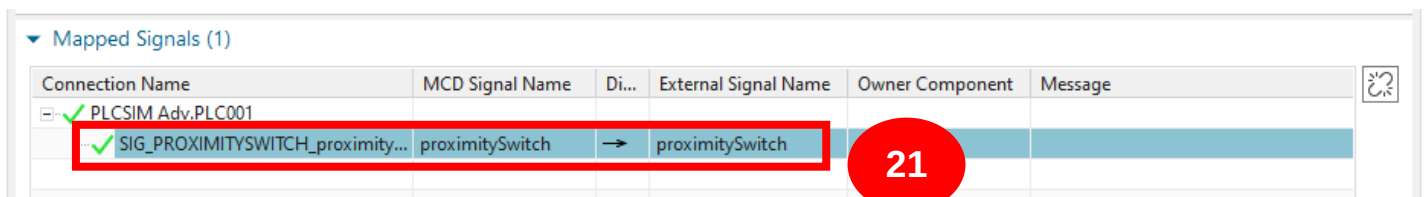
12. The **EXTERNAL SIGNAL CONFIGURATION** window will open again.
13. Select **IOM** from the **UPDATE OPTIONS – AREA** dropdown box.
14. Click **UPDATE TAGS**.
15. Click **SELECT ALL**. This will add the 3 available tags.
16. Click **OK**.



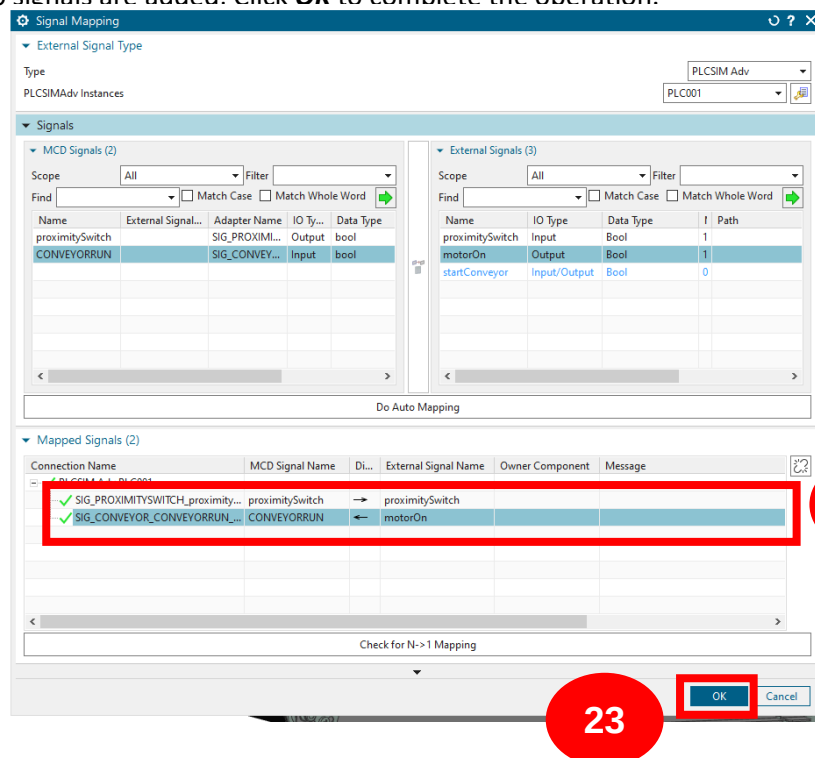
17. The **SIGNAL MAPPING** window will reappear.
18. In the **SIGNALS** section, **HIGHLIGHT** the proximitySwitch entry on the **MCD SIGNALS** section.
19. **HIGHLIGHT** the proximitySwitch entry in the **EXTERNAL SIGNALS** section.
20. Click the **MAP SIGNAL** icon the middle of the dialog window.



21. Steps 18 – 20 will add an entry to the **MAPPED SIGNALS** table.

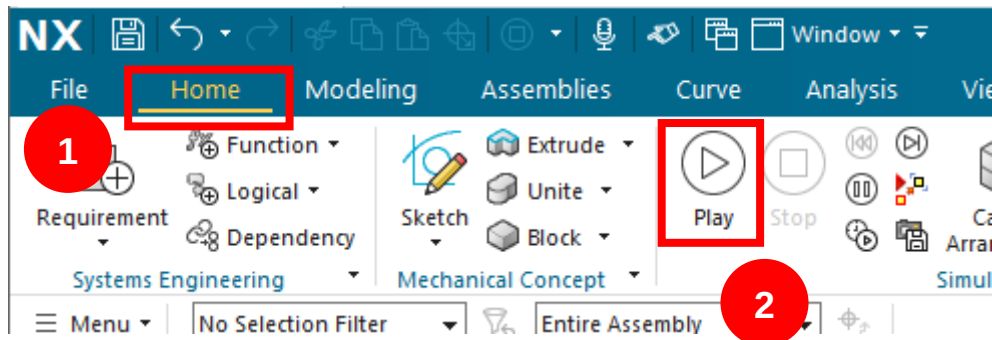


22. **REPEAT** steps 18 – 20, except this time use **CONVEYORRUN (MCD SIGNALS)** and **MOTORON (EXTERNAL SIGNALS)**.
23. Now that the two signals are added. Click **OK** to complete the operation.

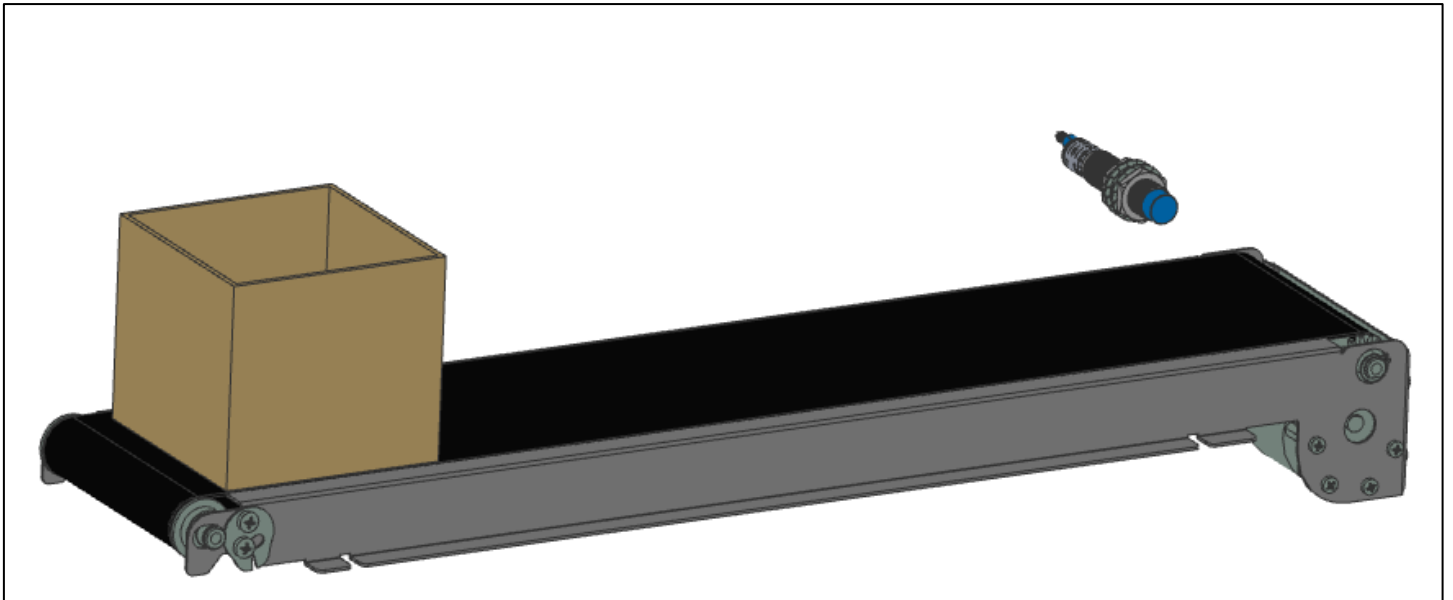


Starting the simulation in Siemens NX-MCD

1. Click on the **HOME** tab.
2. Click on the **PLAY** button in the **SIMULATE** area.



3. This will enable the workspace to react as per the signal inputs including things like gravity.
4. The Cardboard box should drop to the top of the conveyor.

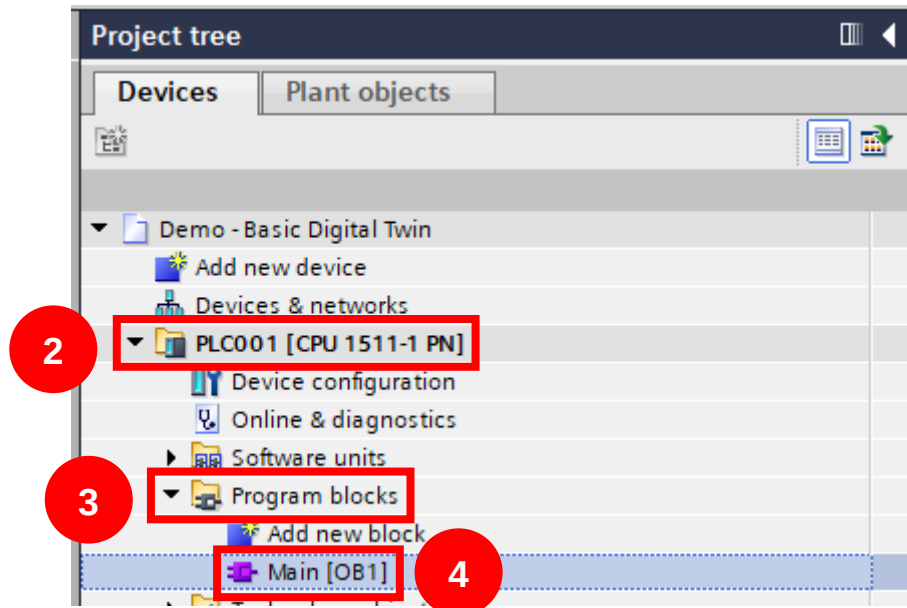


Running the conveyor via the PLC Online with TIA Portal V16.0

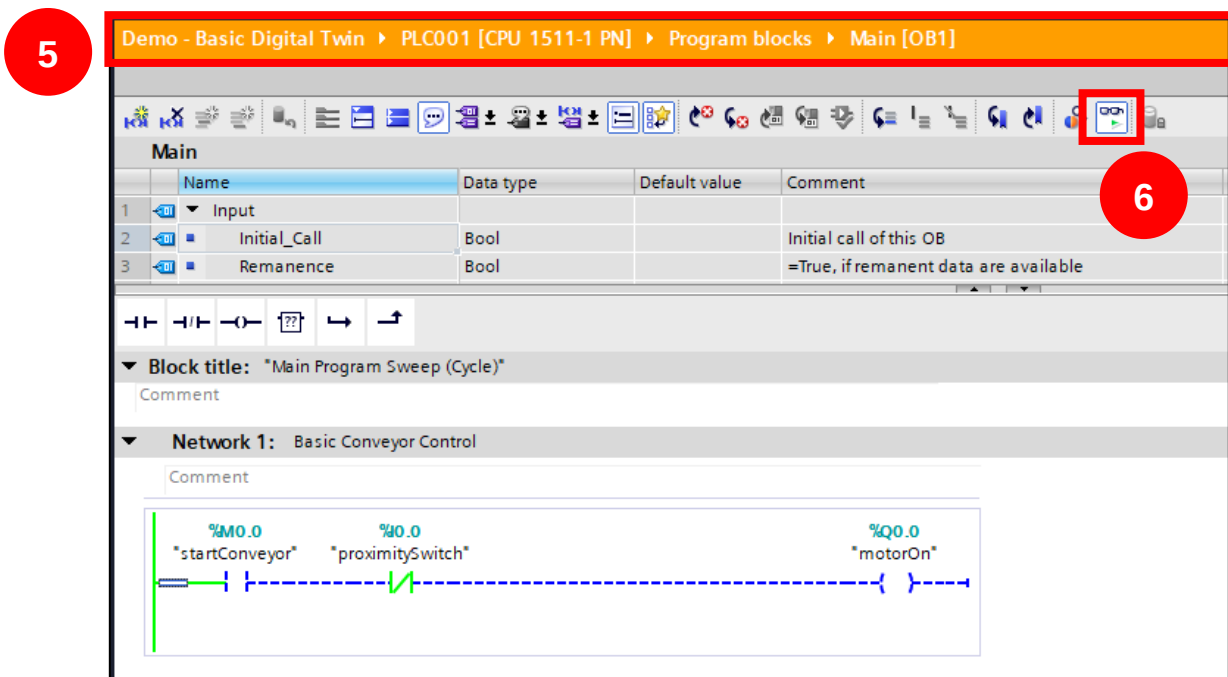
1. Open **TIA PORTAL V16.0** by clicking on the icon in the Windows taskbar as shown below.



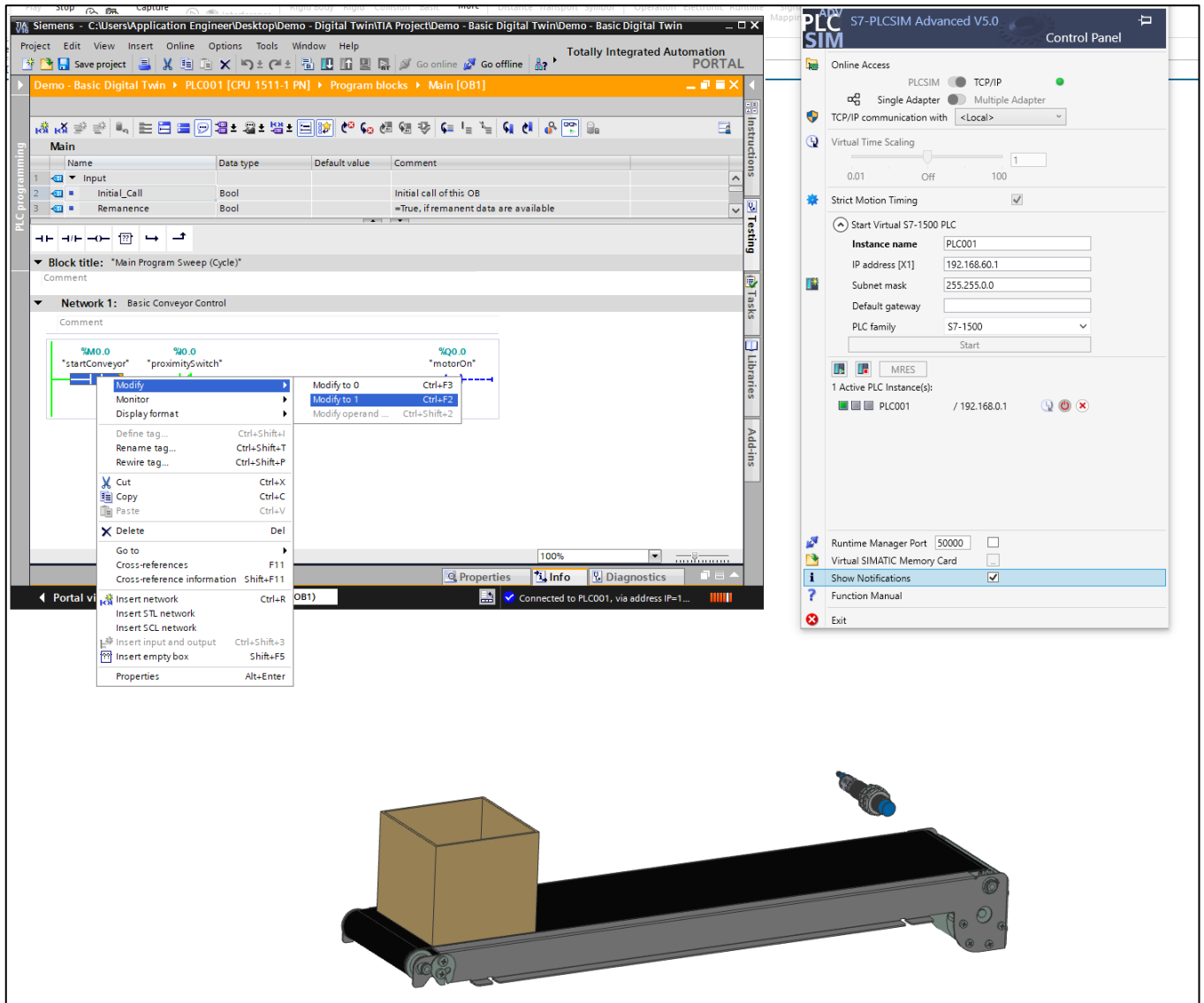
2. Expand the **PLC001** folder in the **PROJECT TREE**.
3. Expand the **PROGRAM BLOCKS**.
4. Double Click **MAIN (OB1)**.



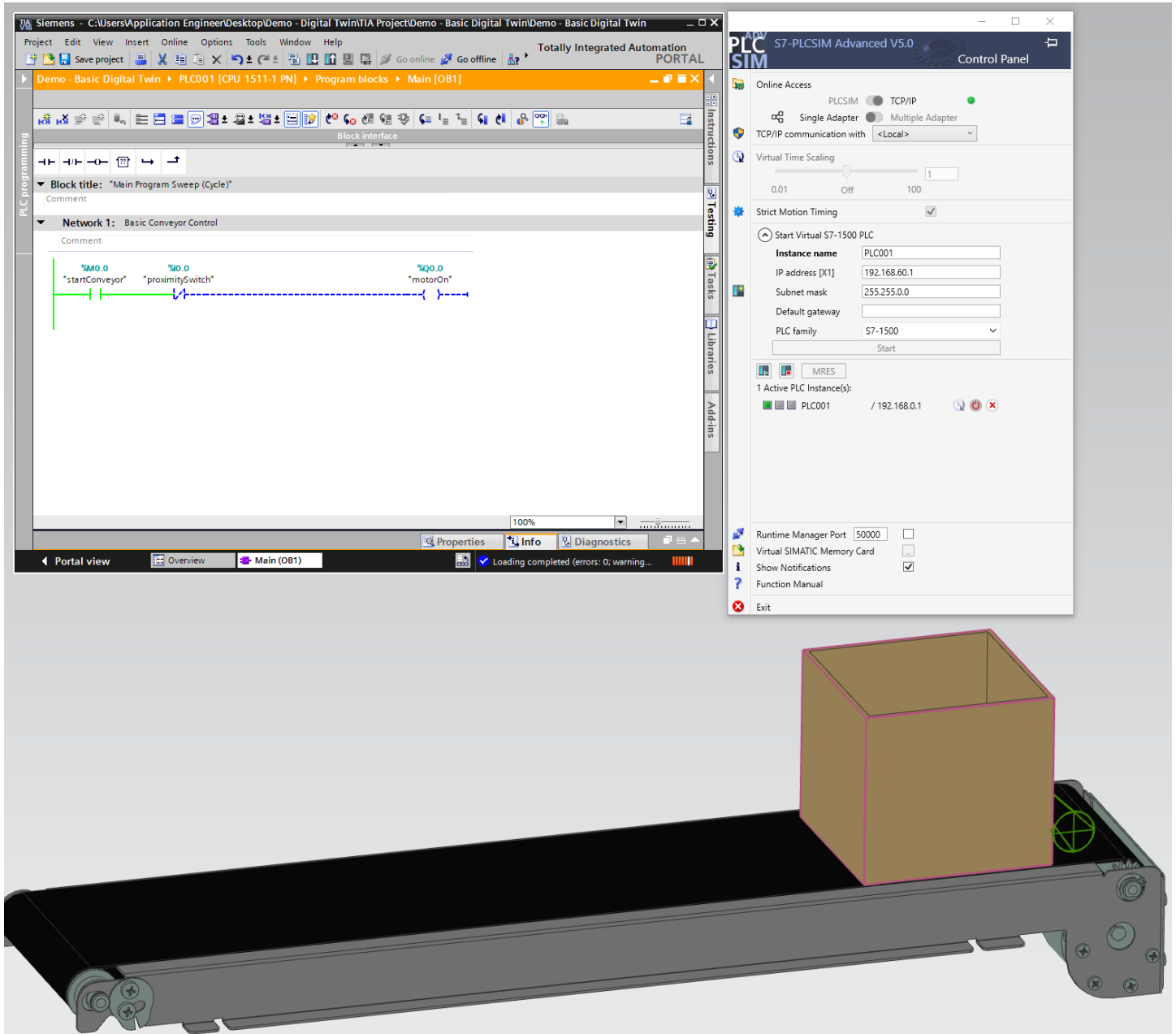
5. In the **PROGRAM WINDOW**, Click on the **MONITOR** icon. This will
6. The **WINDOW TITLE** should turn **ORANGE** to indicate that TIA Portal is monitoring **ONLINE**.



7. Minimise and move both the TIA Portal and Siemens NX-MCD windows around so that both are visible.
8. Right click on the normally open contact called **STARTCONVEYOR** in **NETWORK 1**.
9. Click on **MODIFY**.
10. Click on **MODIFY TO 1**.



11. The conveyor in Siemens NX-MCD should **MOVE** and **STOP** when it reaches the proximity switch.
12. The proximity switch will light up **GREEN** when the cardboard box is **DETECTED**.
13. The **PLC INPUT** will also change state, which will stop the motor.



Abbreviations/Definitions

MCD – Mechatronics Concept Designer

WCS – World Coordinate System

UCS – User Coordinate System

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